



Notes

Probability & statistics 1A (University of Bath)



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Y1 S1 Probability and Statistics

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Overview of Probability and Statistics 1A

Content & Learning outcomes

Content

This unit introduces the basic tools and principles of probability theory. We will talk about:

- *Foundations of probability theory*
Sets (sample spaces, events, intersections, unions, differences and complements) and the rules of probability (Kolmogorov's axioms).
- *Equally likely outcomes*
Sampling methods (with or without ordering and replacement).
- *Conditional probability and independence*
Partition theorem, Bayes' theorem.
- *Random variables*
Discrete and continuous, probability mass functions (PMFs), probability density functions (PDFs), cumulative distribution functions (CDFs), joint and marginal discrete distributions, independence of random variables.
- *Common distributions*
Discrete (Bernoulli trials, Binomial, Geometric and Poisson distribution) and continuous (Uniform, Exponential and Normal distribution).
- *Expectation and variance*
Definition and properties for discrete and continuous random variables, standard deviation, covariance and correlation. Sums of independent random variables and the law of large numbers.
- *Key application* (Random walks).

Learning outcomes

After taking this unit, you should be able to:

- Apply the basic laws of probability.
- Solve a variety of problems with probability, including the use of combinations, permutations and standard probability distributions.
- Perform common expectation calculations.
- Calculate marginal and conditional distributions of discrete random variables from joint distributions.
- Calculate and explore the behaviour of sums of independent random variables.

Organisation

Lectures

There are three weekly lectures:

- In-person session on Mondays at 17:15 in 2 West, University Hall.
- In-person session on Tuesdays at 13:15 in 2 West, University Hall.
- Online Zoom session on Wednesdays at 11:15.
 - Zoom meeting ID: 940 6197 2755
 - Passcode: 110246

Single Sign On (SSN) is necessary to access the online lectures. If you try to access using your personal email and not the University one, you will end up in the waiting room for the entirety of the session.

Tutorials

There is one tutorial per week. Tutorials are small groups guided by a tutor who will help you to work through the problem sheets – more details below. You are encouraged to use the tutorials as interactive sessions - ask questions and discuss problems, don't just expect to sit and listen.

Office hours

If you have a question that your tutor can't answer, I will be available on Tuesdays from 11:00 to 12:30 in my office, 4 West 1.13.

Recordings

Recordings of the lecture sessions will be made available on Panopto and the Moodle unit page after the session.

Lecture notes

A set of comprehensive lecture notes will be available through the unit's Moodle page during the course of the semester. The notes are available in different formats with identical content but different layouts.

Problem sheets

Each Wednesday, there will be a homework problem sheet released on the Moodle page of the unit. These do not form part of your final mark. There will be three types of questions, all clearly indicated in the problem sheet:

- Some of the problems are designed to start discussions in your small-group tutorials. They may also introduce topics that are needed for the marked problems below.
- Some of the problems are marked by your tutor to give you feedback on your learning and help you to understand whether you are doing the right things. This is purely for your own benefit and will not contribute to your final mark for the unit.
- Some of the problems are extra questions that provide additional insight into the course material.

Your solutions **to the marked questions** should be handed in at a time and date decided by your tutor. Your solutions will have to be put into the correct folder, identified by your tutorial group label, in the "Probability and Statistics 1A" pigeon hole at the bottom floor of 4W.

The problems will be discussed in the tutorials, and your tutors will mark the handed-in scripts and return them with comments at the tutorials that take place the following week.

Full solutions to *tutorial and marked* questions will be published on Moodle *after* the last hand-in deadline.

Assessment

Your assessment will be an exam in January, with marks from 0 to 60 and passing grade being 24/60. Past exams can be found on the Library website, under "Past exam papers", code MA10211. Alternatively, [click here](#).

The mark given to you will then be combined with all the other units that constitute MA12002, MA12005, or MA12012.

Probability and Statistics 1A will be 40% of the final mark for MA12002, MA12005 and MA12012.

Resources

Extra textbooks:

This unit is self-contained in the sense that you will not strictly need to consult textbooks. However, you may wish to consult extra books to support your learning and understanding. A list of suggested works available at the Library can be found on the Moodle page of the unit.

What if I have found an error in the lecture notes?

Send me an email at mir20@bath.ac.uk and I will correct it as soon as possible.

FAQ

What is the difference between probability and statistics?

The problems considered by probability and statistics can be thought of being complementary to each other: in probability theory, we consider a process which has some randomness and we build a model of what will happen and then study it, while in statistics, we observe something that has happened and try to infer what process could explain the observations.

- **Probability:** Assuming that a coin is fair, what is the probability that it takes me more than 10 tosses to see a head?
- **Statistics:** It takes me more than 10 tosses to see a head. Is the coin fair?

This is a simplification; the two subjects intersect and take inspiration from each other. But they are different subjects. In the first semester we will mainly study probability, whereas in the second semester you will build on the probability you have learnt and also learn some statistics. From the second year onwards, there will be separate probability units and statistics units.

1 Foundations of Probability

1.1 Sets

Example 1.1 (Coin tosses). I toss a coin 100 times. How many heads do I see?

In probability theory, we sometimes use the term “**experiment**” to describe a process whose outcome is not known in advance. We assign **probabilities** to possible outcomes, or combinations of such outcomes. Like,

- what is the probability I see exactly 38 heads?
- what is the probability I see more than 75 heads?

So before we can talk about probabilities, we need to talk about what are the possible outcomes of our experiment. We call this collection, or **set**, of possible outcomes the **sample space**.

1.1.1 Sample spaces and events

Definition 1.1 (Sample space). The set Ω of all possible outcomes of an experiment is called the sample space.

Example 1.2. Examples of sample spaces:

- Suppose that we toss a coin 100 times and count the number of heads obtained. The sample space is $\Omega = \{0, 1, \dots, 100\}$. (A *finite* sample space.)
- Suppose that we count the tosses of a coin until the first head is obtained. The sample space $\Omega = \{1, 2, 3, \dots\}$ contains an infinite number of elements. (A *countable* or *countably infinite* sample space.)
- Suppose I want to know how long it is between the start of this lecture and the first time someone in the room sneezes. We could use $\Omega = [0, \infty)$. (An *uncountable* sample space.)

We often write ω to represent a generic point in Ω , just like you might use x to represent a generic real number, or n to represent a generic natural number.

In the first example, if we just want to know the probability of seeing exactly 38 heads, we are interested in the probability of one point in the sample space. But if we want to know e.g. the probability of seeing more than 75 heads, we are interested in the probability that the outcome of our experiment is any one of a whole bunch of points - a **subset** of the sample space.

Definition 1.2 (Event). An event is a collection of possible outcomes of an experiment.

If the actual outcome of an experiment belongs to the event E , we say that the event E has **occurred**.

Example 1.3. Consider the experiment of rolling a (6-sided) dice. The possible outcomes, or sample points, are the scores 1, 2, 3, 4, 5 and 6 making the sample space $\Omega = \{1, 2, 3, 4, 5, 6\}$. Examples of events include:

- (a) $E_1 = \{\text{score is an even number}\} = \{2, 4, 6\}$.
- (b) $E_2 = \{\text{score is bigger than 4}\} = \{5, 6\}$.
- (c) $E_3 = \{\text{score is 1}\} = \{1\}$.

If the outcome of a roll is 6, then the event E_1 occurs (as $6 \in E_1$), the event E_2 occurs (as $6 \in E_2$), but the event E_3 does not occur (as $6 \notin E_3$).

1.1.2 Set relations

Definition 1.3 (Subset and equal sets). - A set (or event¹) E is a subset of F , written $E \subset F$, if whenever E occurs F also occurs.

Another way of saying this is that for every $\omega \in E$, we also have $\omega \in F$.

¹In probability theory, it is customary to consider the probability of an *event* rather than a *set*. We will often use the terms set and event interchangeably. Really events are just subsets of a sample space Ω .

- We say that sets (or events) E and F are equal, written $E = F$, if both $E \subset F$ and $F \subset E$.

Note that when we write $E \subset F$, we include the possibility that E is the whole of F (i.e. $E = F$)².

We can use Venn³ diagrams to provide a pictorial representation for events. For example, Figure 1 represents the case $E \subset F$.

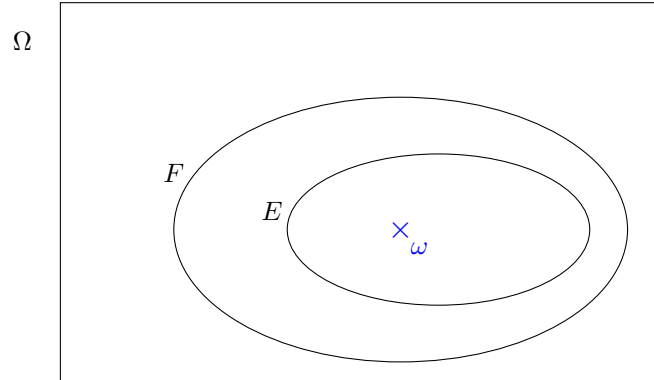


Figure 1: Event E is a subset of event F if, whenever E occurs F also occurs. Note that any event is a subset of Ω and so we have that $E \subset \Omega$ and $F \subset \Omega$.

We write $\emptyset$ for the **empty set**, the set (or event) containing no points (or outcomes).

Event $\emptyset$ contains no points and hence it is logically correct to say that each point belonging to $\emptyset$ also belongs to E , for any set E . Consequently $\emptyset \subset E \subset \Omega$ is true for any event E .

1.1.3 Operations of set theory

Definition 1.4 (Elementary set operations). For any two sets (or events) E and F we have the following elementary operations:

1. The **union** of E and F , written as $E \cup F$, is the event containing all outcomes that belong to either E or F or both so that $E \cup F = \{\omega : \omega \in E \text{ or } \omega \in F\}$. See Figure 2 below.
2. The **intersection** of E and F , written as $E \cap F$, is the event containing all outcomes that belong to both E and F so that $E \cap F = \{\omega : \omega \in E \text{ and } \omega \in F\}$. See Figure 3 below.
3. The **difference** of E and F , written as $E \setminus F$, is the event containing all outcomes that belong to E but not to F so that $E \setminus F = \{\omega : \omega \in E \text{ and } \omega \notin F\}$. See Figure 4 below.
4. The **complement** of E , written as E^c , is the event containing all outcomes that do not belong to E so that $E^c = \{\omega : \omega \notin E\}$. See Figure 5 below.

Remarks:

- Figure 2 shows the union of E and F . Note that, for any event E , $\Omega \cup E = \Omega$ and $\emptyset \cup E = E$.
- Figure 3 shows the intersection of E and F . For any event E , $\Omega \cap E = E$ and $\emptyset \cap E = \emptyset$.
- Figure 4 shows the difference of E and F . Note that $E \setminus F = E \cap F^c$ and $E = (E \cap F) \cup (E \cap F^c)$. More on this later!
- Figure 5 shows the complement of E . Note that, for any event E , we have $E^c = \Omega \setminus E$, i.e. all elements of Ω that are not in E . The following properties are easy to prove:

²In some areas of mathematics it is more customary to write $E \subseteq F$ to allow $E = F$, and $E \subset F$ only if $E \neq F$. In probability theory it is fairly common, but not universal, to write $E \subset F$ even if we allow $E = F$. This is the convention we adopt.

³John Venn (1834 – 1923)

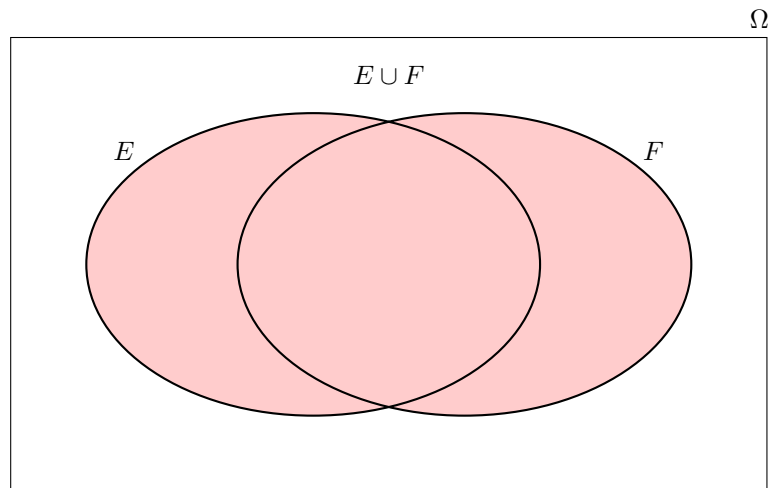


Figure 2: Union of E and F , $E \cup F$. The union $E \cup F$ occurs whenever at least one (and possibly both) of E and F occur.

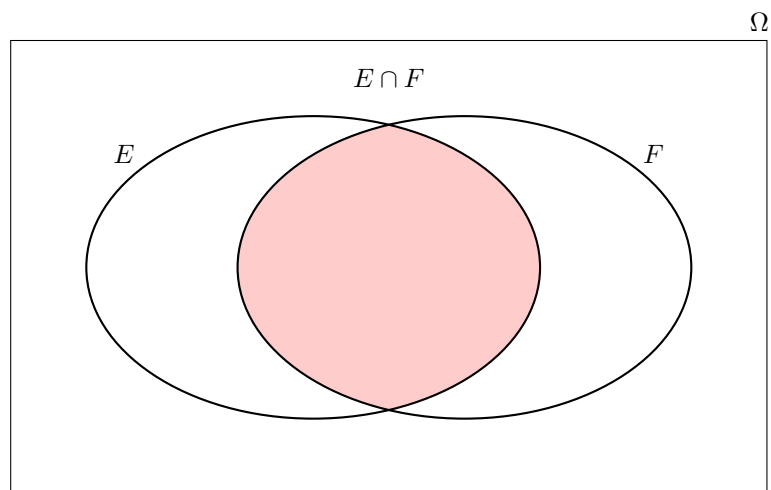


Figure 3: Intersection of E and F , $E \cap F$. The intersection $E \cap F$ occurs whenever both E and F occur simultaneously.

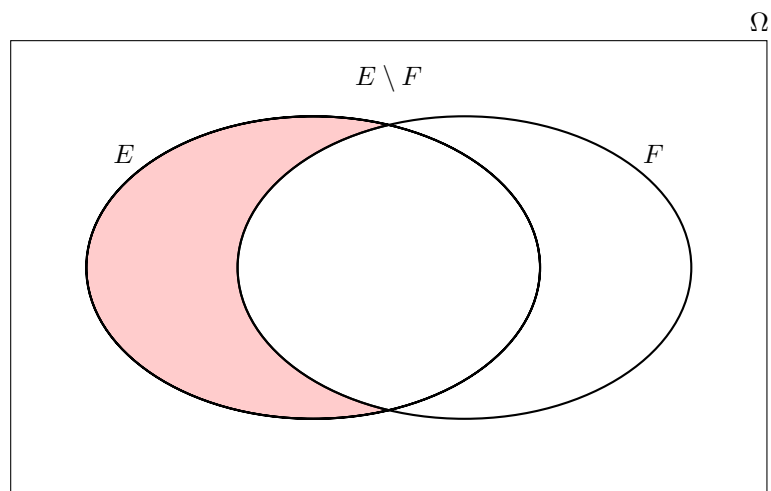


Figure 4: Difference of E and F , $E \setminus F$. The difference $E \setminus F$ occurs whenever E occurs but F does not occur.

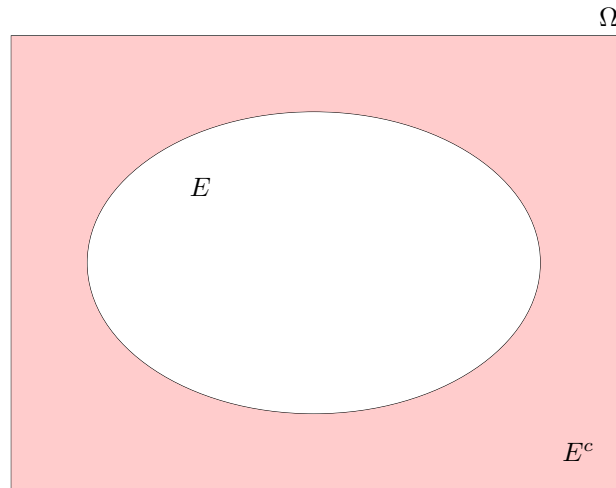


Figure 5: Complement of E , E^c . The complement E^c occurs precisely when E does not occur.

1. $(E^c)^c = E$ (the complement of the complement of an event is the event).
2. $\emptyset^c = \Omega$ and $\Omega^c = \emptyset$ (the complement of the empty set is the sample space and vice versa).
3. $E \cup E^c = \Omega$ (the sample space is the union of the events E and E^c).
4. $E \cap E^c = \emptyset$ (E and E^c have no outcomes in common).

Example 1.4 (Example 1.3 revisited). For the events E_1 , E_2 , and E_3 of Example 1.3 we have:

1. $E_1 \cup E_2 = \{2, 4, 5, 6\} = \{\text{score is an even number or bigger than 4}\}$.
2. $E_1 \cap E_2 = \{6\} = \{\text{score is an even number and bigger than 4}\}$.
3. $E_1^c = \{1, 3, 5\} = \{\text{score is not an even number}\} = \{\text{score is an odd number}\}$. From this we see that $E_1 \cup E_1^c = \Omega = \{\text{score is an even or an odd number}\}$ and $E_1 \cap E_1^c = \emptyset = \{\text{score is an even and an odd number}\}$.

Definition 1.5 (Disjoint events). Events E and F are disjoint or mutually exclusive if they cannot both occur at the same time, that is $E \cap F = \emptyset$.

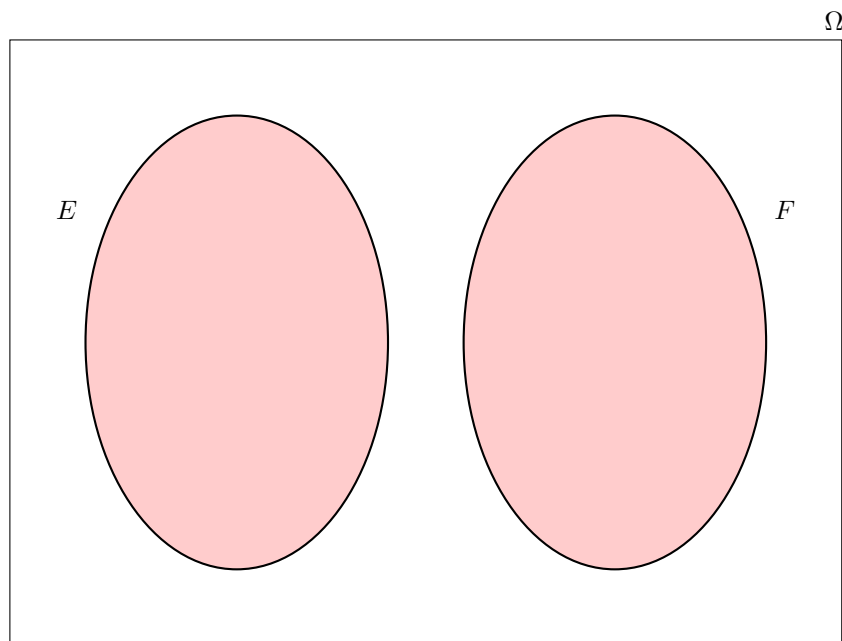


Figure 6: Events E and F are *disjoint* or *mutually exclusive* if they cannot both occur at the same time.

Figure 6 gives an example of disjoint events E and F . Note that an equivalent statement to $E \cap F = \emptyset$ for disjoint events is that $E \subset F^c$. One way of seeing this is to note that for any events E and F we can write $E = (E \cap F) \cup (E \cap F^c)$; but if $E \cap F = \emptyset$ then $E = E \cap F^c$.

Example 1.5 (Example 1.3 revisited). Events E_2 and E_3 are disjoint as $E_2 \cap E_3 = \{\} = \emptyset = \{\text{score is bigger than 4 and score is 1}\}$.

Disjoint events play an important role in calculating probabilities as we shall see in Section 1.2.2.

Sometimes it may be helpful to think of union ($\cup$) and intersection ($\cap$) as the “set analogues” of addition (+) and multiplication ($\times$) respectively. Be careful! This is not always the right intuition - but it can be useful for remembering the following relationships:

Theorem 1.1 (Laws of set theory). *For any sets E , F and G the following relationships hold:*

1. *Commutative Laws:* (a) $E \cup F = F \cup E$
(b) $E \cap F = F \cap E$
2. *Associative Laws:* (a) $E \cup (F \cup G) = (E \cup F) \cup G$
(b) $E \cap (F \cap G) = (E \cap F) \cap G$
3. *Distributive Laws:* (a) $(E \cap F) \cup G = (E \cup G) \cap (F \cup G)$
(b) $(E \cup F) \cap G = (E \cap G) \cup (F \cap G)$
4. *De Morgan’s Laws:* (a) $(E \cup F)^c = E^c \cap F^c$
(b) $(E \cap F)^c = E^c \cup F^c$

Remark: Rather than trying to remember these by rote, think about what they mean (say them out loud: for the second of De Morgan’s laws, “if you’re not in E and F , you must be either not in E or not in F ”), or imagine Venn diagrams.

Indeed, the best way to prove these statements is by drawing Venn diagrams. For example, to prove the first of De Morgan’s⁴ laws, note that the area covered in both wavy blue lines and diagonal red stripes in Figure 7 - which represents $E^c \cap F^c$ - is equal to the area hashed in green in Figure 8 - which represents $(E \cup F)^c$. Thus $E^c \cap F^c = (E \cup F)^c$.

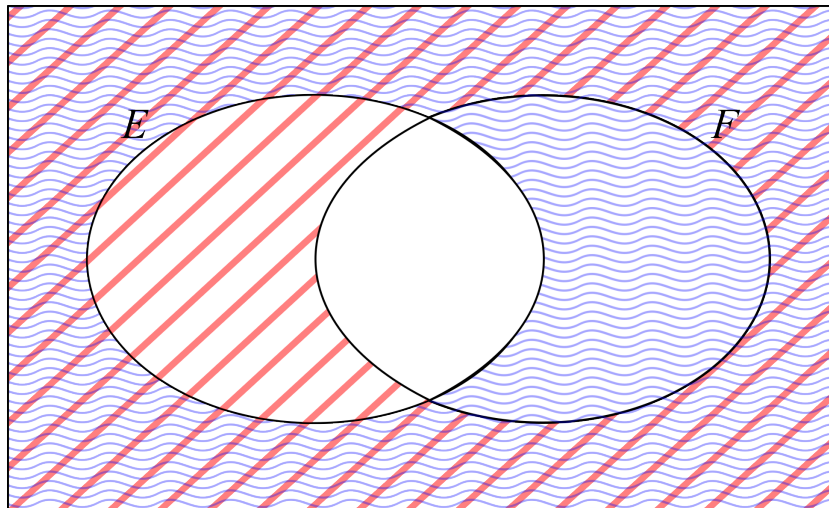


Figure 7: A Venn diagram showing E^c in wavy blue lines and F^c in diagonal red stripes. Thus $E^c \cap F^c$ is the region covered in both wavy blue lines and diagonal red stripes.

⁴ Augustus De Morgan (1806 – 1871).

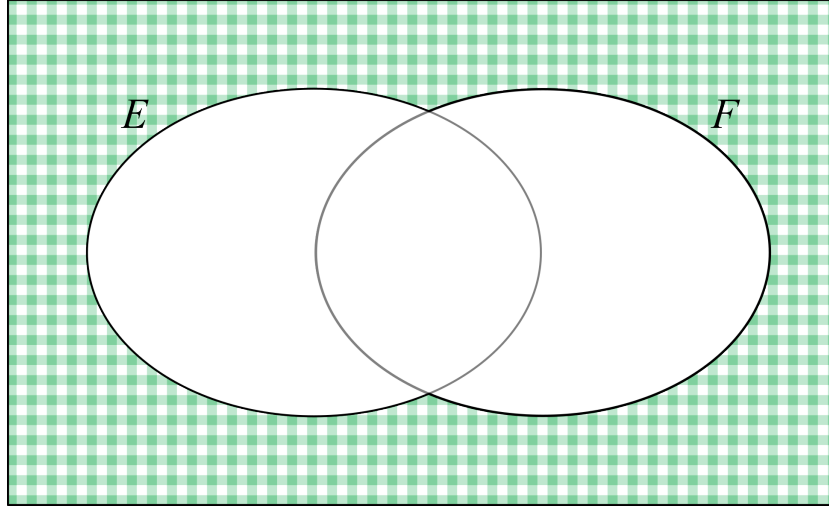


Figure 8: A Venn diagram showing $(E \cup F)^c$ hashed in green.

1.1.4 Extensions to collections of sets

By the Associative Laws in Theorem 1.1, $E \cup (F \cup G) = (E \cup F) \cup G$ and $E \cap (F \cap G) = (E \cap F) \cap G$ and therefore the brackets do not really matter. Thus, we can write $E \cup F \cup G$ for the event that at least one of these events occurs and $E \cap F \cap G$ for all three events occurring. We can extend this notion to both finite and countably infinite collections of events (or sets).

1.1.4.1 Unions and intersections of collections of sets

1. If $E_1, \dots, E_n$ is any finite collection of sets then we define

$$\bigcup_{i=1}^n E_i := E_1 \cup E_2 \cup \dots \cup E_n = \{\omega : \omega \in E_i \text{ for some } i \in \{1, 2, \dots, n\}\},$$

$$\bigcap_{i=1}^n E_i := E_1 \cap E_2 \cap \dots \cap E_n = \{\omega : \omega \in E_i \text{ for all } i \in \{1, 2, \dots, n\}\}.$$

2. If $E_1, E_2, \dots$, is any countably infinite collection of sets then we define

$$\bigcup_{i=1}^{\infty} E_i = E_1 \cup E_2 \cup \dots = \{\omega : \omega \in E_i \text{ for some } i \in \mathbb{N}\},$$

$$\bigcap_{i=1}^{\infty} E_i = E_1 \cap E_2 \cap \dots = \{\omega : \omega \in E_i \text{ for all } i \in \mathbb{N}\}.$$

In a similar way the Distributive Laws and De Morgan's Laws from Theorem 1.1 can be extended to many events. We only write out the latter here.

1.1.4.2 De Morgan's Laws for collections of sets

1. If $E_1, \dots, E_n$ is any finite collection of sets then

$$\left(\bigcup_{i=1}^n E_i \right)^c = \bigcap_{i=1}^n E_i^c, \quad \left(\bigcap_{i=1}^n E_i \right)^c = \bigcup_{i=1}^n E_i^c. \quad (1)$$

2. If $E_1, E_2, \dots$, is any countably infinite collection of events defined on Ω then

$$\left(\bigcup_{i=1}^{\infty} E_i \right)^c = \bigcap_{i=1}^{\infty} E_i^c, \quad \left(\bigcap_{i=1}^{\infty} E_i \right)^c = \bigcup_{i=1}^{\infty} E_i^c. \quad (2)$$

We will not prove these here.

Example 1.6. As in Example 1.2, suppose that we toss a coin repeatedly and count the number of tosses until the first head is obtained. Recall that our sample space is $\Omega = \{1, 2, 3, \dots\}$, where i represents the outcome that the first head occurs on the i th toss. Consider

$$E_{\text{even}} = \{\text{first head occurs on an even toss}\} = \{2\} \cup \{4\} \cup \{6\} \cup \dots = \bigcup_{i=1}^{\infty} \{2i\}.$$

Let $E_{\text{odd}} = \{\text{first head occurs on an odd toss}\}$ then $E_{\text{odd}} = E_{\text{even}}^c$ and so by using De Morgan's Laws, Equation (2),

$$E_{\text{odd}} = E_{\text{even}}^c = \left(\bigcup_{i=1}^{\infty} \{2i\} \right)^c = \bigcap_{i=1}^{\infty} \{2i\}^c,$$

i.e., the first head does not occur on the 2nd toss, and it does not occur on the 4th toss, and so on.

1.1.4.3 Pairwise disjoint events (extension of Definition 1.5 to pairwise disjoint events)

1. The events $E_1, E_2, \dots, E_n$ are said to be *pairwise disjoint* or *mutually exclusive* if no two can simultaneously occur, that is, $E_i \cap E_j = \emptyset$ for all $i, j = 1, \dots, n$, $i \neq j$.
2. Similarly, if we have countably many events $E_1, E_2, \dots$, they are pairwise disjoint if $E_i \cap E_j = \emptyset$ for all $i, j \in \mathbb{N}$, $i \neq j$.

Thus, for example E_1, E_2, E_3 are pairwise disjoint if and only if $E_1 \cap E_2 = \emptyset$, $E_1 \cap E_3 = \emptyset$ and $E_2 \cap E_3 = \emptyset$.

In Example 1.3, recall the events $E_1 = \{2, 4, 6\}$, $E_2 = \{5, 6\}$, $E_3 = \{1\}$. The events are *not* pairwise disjoint, since $E_1 \cap E_2 \neq \emptyset$. (Even though $E_1 \cap E_2 \cap E_3 = \emptyset$.)

Sometimes we might say “disjoint” when we mean “pairwise disjoint”, but it is good to be precise.

1.2 The rules of probability

If we just have a finite number of points in our sample space, then usually we can assign a probability to every possible event. For example, if we are tossing a coin 100 times and counting how many heads we see, we can ask for the probability that the number of heads is 5 or even or greater than 72.

There are only finitely many possible outcomes and therefore only finitely many *collections of outcomes* - or possible events, i.e. subsets of the sample space (there are 2^{100} in fact).

Definition 1.6 (Power set). For any set S , the power set of S , denoted by $\mathcal{P}(S)$, is the set of all subsets of S , including $\emptyset$ and S itself.

This semester, we will usually be able to assign a probability $\mathbb{P}(E)$ to every $E \in \mathcal{P}(\Omega)$, i.e. to every possible event.

But this is not true for all sample spaces; in fact it is not true for $\Omega = [0, 1]$ or $\Omega = \mathbb{R}$. (**Non-examinable!**) Suppose e.g. $\Omega = [0, 1]$. There are some subsets A of $[0, 1]$ where, if we had $\mathbb{P}(A) = 0$, then we must have $\mathbb{P}(\Omega) = 0$, and if we had $\mathbb{P}(A) > 0$, then we must have $\mathbb{P}(\Omega) = \infty$. Any probability measure should have $\mathbb{P}(\Omega) = 1$, so neither option is allowed. So we just can't allow ourselves to assign a probability $\mathbb{P}(A)$ to this kind of set.

So eventually, we have to be careful about what sets we can assign probabilities to. The following definition gives us three rules that any reasonable collection of sets has to satisfy. Thankfully, these are quite simple. (Unfortunately, the name we give to collections of sets satisfying the rules sounds complicated and even confusing - but it is just a name, and since everyone uses it, we might as well get used to it.)

1.2.1 Sigma-algebras

Definition 1.7 (Sigma-algebra). A collection $\mathcal{F}$ of subsets of Ω is called a σ -algebra (or sometimes a σ -field) if it satisfies

1. $\emptyset \in \mathcal{F}$ (the empty set is an element of the sigma-algebra).
2. if $E \in \mathcal{F}$, then $E^c \in \mathcal{F}$ (we say “the σ -algebra is closed under taking complements”).
3. if $E_1, E_2, \dots \in \mathcal{F}$ then $\bigcup_{i=1}^{\infty} E_i \in \mathcal{F}$ (we say “the σ -algebra is closed under countable unions”).

Remark: The sample space Ω is always an element of any σ -algebra $\mathcal{F}$, because $\emptyset \in \mathcal{F}$ and $\Omega = \emptyset^c$, so by rule 2, $\Omega \in \mathcal{F}$.

Lemma 1.1 (Sigma-algebras are closed under countable intersections). *Any σ -algebra is closed under countable intersections. That is, if $E_1, E_2, \dots \in \mathcal{F}$ then $\bigcap_{i=1}^{\infty} E_i \in \mathcal{F}$.*

Proof. Idea: Use De Morgan's law to turn intersection into union, and then use closure under countable unions.

If $E_1, E_2, \dots \in \mathcal{F}$ then by rule 2, $E_1^c, E_2^c, \dots \in \mathcal{F}$ and so by rule 3, $\bigcup_{i=1}^{\infty} E_i^c \in \mathcal{F}$. Thus by rule 2, $(\bigcup_{i=1}^{\infty} E_i^c)^c \in \mathcal{F}$ as well. By De Morgan's Laws, Equation (2),

$$\left(\bigcup_{i=1}^{\infty} E_i^c \right)^c = \bigcap_{i=1}^{\infty} (E_i^c)^c = \bigcap_{i=1}^{\infty} E_i$$

and therefore $\bigcap_{i=1}^{\infty} E_i \in \mathcal{F}$ too. □

The definition says that σ -algebras are closed under *countable* unions, and we have just seen that they are also closed under *countable* intersections. But what about *finite* unions and intersections? Fortunately they are closed under these too.

Lemma 1.2 (Sigma-algebras are closed under finite unions and intersections). *Any σ -algebra is closed under finite unions and intersections. That is, if $n \in \mathbb{N}$ and $E_1, E_2, \dots, E_n \in \mathcal{F}$ then $\bigcup_{i=1}^n E_i \in \mathcal{F}$ and $\bigcap_{i=1}^n E_i \in \mathcal{F}$.*

Proof. Idea 1 (finite unions): Turn a finite union into an infinite union by using a load of $\emptyset$'s.

Recall that for any event E we have $E \cup \emptyset = E$, and thus, if we let $E_{n+1} = \emptyset, E_{n+2} = \emptyset, \dots$ then we have

$$\bigcup_{i=1}^{\infty} E_i = E_1 \cup \dots \cup E_n \cup E_{n+1} \cup \dots = E_1 \cup \dots \cup E_n \cup \emptyset \cup \emptyset \cup \dots = \bigcup_{i=1}^n E_i,$$

so that if $E_1, \dots, E_n \in \mathcal{F}$, then $\bigcup_{i=1}^n E_i \in \mathcal{F}$ by rule 3 of the definition of σ -algebras.

Idea 2 (finite intersections): Turn a finite intersection into an infinite intersection by using a load of Ω 's.

Recall that for any event E we have $E \cap \Omega = E$, and thus, if we let $E_{n+1} = \Omega, E_{n+2} = \Omega, \dots$ then we have

$$\bigcap_{i=1}^{\infty} E_i = E_1 \cap \dots \cap E_n \cap E_{n+1} \cap \dots = E_1 \cap \dots \cap E_n \cap \Omega \cap \Omega \cap \dots = \bigcap_{i=1}^n E_i,$$

so that if $E_1, \dots, E_n \in \mathcal{F}$, then $\bigcap_{i=1}^n E_i \in \mathcal{F}$ by Lemma 1.1. □

Many different σ -algebras can be associated with a given sample space Ω .

Example 1.7. Some examples of σ -algebras:

1. The collection of the two sets $\{\emptyset, \Omega\}$ is always a σ -algebra (often called the trivial σ -algebra).
2. If $E \subset \Omega$ then $\{\emptyset, E, E^c, \Omega\}$ is a σ -algebra. (Exercise: check this!)
3. The power set of Ω , $\mathcal{P}(\Omega)$, is always a σ -algebra.

Non-examinable: As hinted above, if Ω is finite, then we can always take the σ -algebra $\mathcal{F}$ to be $\mathcal{P}(\Omega)$, the power set. But as we mentioned, for uncountable Ω , the situation is more complicated due to reasons that are beyond the scope of this course. Broadly speaking, for an uncountable Ω (e.g., $[0, 1]$ or $\mathbb{R}$), $\mathcal{P}(\Omega)$ is simply *too large* and will contain weird subsets that cannot be reasonably assigned probabilities. These sets are said to be non-measurable⁵ and they are a topic for courses in measure theory. For an uncountable Ω , such as $\Omega = \mathbb{R}$, a suitable σ -algebra is the smallest collection of events containing all sets of the form $[a, b]$ for any $a, b \in \mathbb{R}$ such that $a < b$.

⁵In 1905 Giuseppe Vitali (1875–1932) provided the first example of a non-measurable subset of real numbers nowadays known as Vitali sets.

1.2.2 Kolmogorov's axioms of probability

We now want to assign probabilities to each event in the σ -algebra. There are rules that need to be satisfied to make these consistent. These rules, or axioms, were formulated by Andrey Kolmogorov⁶ in the 1930s.

Definition 1.8 (Probability measure and probability space). Let Ω be a sample space and $\mathcal{F}$ a σ -algebra defined on Ω . A *probability measure* $\mathbb{P}$ on $(\Omega, \mathcal{F})$ must satisfy the following, sometimes known as Kolmogorov's axioms:

(A1) $\mathbb{P}(E) \geq 0$ for all $E \in \mathcal{F}$.

(A2) $\mathbb{P}(\Omega) = 1$.

(A3) If $E_1, E_2, \dots \in \mathcal{F}$ are pairwise disjoint, then

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(E_i) = \mathbb{P}(E_1) + \mathbb{P}(E_2) + \dots$$

The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is then called a *probability space*.

Remark: We call property (A3) *countable additivity*: if the E_i are pairwise disjoint then the probability that (at least) one of them occurs is the sum of their individual probabilities.

1.2.3 Properties of probability measures

We can use Kolmogorov's axioms (A1)-(A3) to build up properties of a probability measure that will be useful when calculating complicated probabilities.

Lemma 1.3 (The probability of the empty set is zero). *For any probability space $(\Omega, \mathcal{F}, \mathbb{P})$, we have $\mathbb{P}(\emptyset) = 0$.*

Proof. Idea: Write Ω as a union of Ω and a load of $\emptyset$'s, and use countable additivity.

Let $E_1 = \Omega$ and $E_2 = E_3 = \dots = \emptyset$. Thus

$$\bigcup_{i=1}^{\infty} E_i = \Omega \cup \emptyset \cup \emptyset \cup \dots = \Omega \tag{3}$$

and, since $\Omega \cap \emptyset = \emptyset$ and $\emptyset \cap \emptyset = \emptyset$, we have $E_i \cap E_j = \emptyset$ for all $i \neq j$ so that $E_1, E_2, \dots$ are pairwise disjoint. By (3),

$$\begin{aligned} \mathbb{P}(\Omega) &= \mathbb{P}\left(\bigcup_{i=1}^{\infty} E_i\right) \\ &= \sum_{i=1}^{\infty} \mathbb{P}(E_i) = \mathbb{P}(E_1) + \mathbb{P}(E_2) + \dots \quad [\text{by (A3), since } E_i \text{ are pairwise disjoint}] \\ &= \mathbb{P}(E_1) + \sum_{i=2}^{\infty} \mathbb{P}(E_i) \\ &= \mathbb{P}(\Omega) + \sum_{i=2}^{\infty} \mathbb{P}(\emptyset). \end{aligned} \tag{4}$$

Subtracting $\mathbb{P}(\Omega)$ from both sides of Equation (4) gives

$$\sum_{i=2}^{\infty} \mathbb{P}(\emptyset) = 0.$$

From (A1), we have $\mathbb{P}(\emptyset) \geq 0$, and if $\mathbb{P}(\emptyset) > 0$, then $\sum_{i=1}^{\infty} \mathbb{P}(\emptyset) = \infty \neq 0$. Thus we must have $\mathbb{P}(\emptyset) = 0$. $\square$

⁶Andrey Kolmogorov (1903-1987). His 1933 book *Foundations of the Theory of Probability* laid the framework for the axiomatic foundations of probability theory.

Theorem 1.2 (Finite additivity of $\mathbb{P}$ for pairwise disjoint events). *For any probability space $(\Omega, \mathcal{F}, \mathbb{P})$, if $E, F \in \mathcal{F}$ with $E \cap F = \emptyset$ then*

$$\mathbb{P}(E \cup F) = \mathbb{P}(E) + \mathbb{P}(F). \quad (5)$$

Similarly, if $E_1, E_2, \dots, E_n \in \mathcal{F}$ are pairwise disjoint then

$$\mathbb{P}\left(\bigcup_{i=1}^n E_i\right) = \sum_{i=1}^n \mathbb{P}(E_i) = \mathbb{P}(E_1) + \dots + \mathbb{P}(E_n). \quad (6)$$

Proof. We prove (6), since (5) is just the case $n = 2$.

Idea: Same trick as in Lemma 1.2. Turn a finite union into an infinite union by using a load of $\emptyset$'s.

Suppose that $E_1, \dots, E_n$ are pairwise disjoint and let $E_{n+1} = E_{n+2} = \dots = \emptyset$. Then, as $E \cap \emptyset = \emptyset$ for any event E , we know that $E_1, E_2, \dots$ are pairwise disjoint and

$$\bigcup_{i=1}^{\infty} E_i = E_1 \cup \dots \cup E_n \cup \emptyset \cup \emptyset \cup \dots = E_1 \cup \dots \cup E_n = \bigcup_{i=1}^n E_i.$$

Thus, by (A3) as the E_i s are pairwise disjoint,

$$\begin{aligned} \mathbb{P}\left(\bigcup_{i=1}^n E_i\right) &= \mathbb{P}\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(E_i) \\ &= \sum_{i=1}^n \mathbb{P}(E_i) + \mathbb{P}(E_{n+1}) + \mathbb{P}(E_{n+2}) + \dots \\ &= \sum_{i=1}^n \mathbb{P}(E_i) + \mathbb{P}(\emptyset) + \mathbb{P}(\emptyset) + \dots \end{aligned} \quad (7)$$

From Theorem 1.3, we have $\mathbb{P}(\emptyset) = 0$ and substituting this into (7) gives (6). $\square$

Just to check we haven't done anything totally unreasonable, we consider the following example.

Example 1.8. Consider an experiment where we toss a coin once. The sample space is $\Omega = \{H, T\}$ where $H = \{\text{toss is a head}\}$ and $T = \{\text{toss is a tail}\}$.

We choose the σ -algebra to be $\mathcal{F} = \mathcal{P}(\Omega) = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}$. From (A2) of Definition 1.8, we have $\mathbb{P}(\{H, T\}) = 1$, and by Theorem 1.3 we have $\mathbb{P}(\emptyset) = 0$. As $\{H\} \cap \{T\} = \emptyset$, we have by Theorem 1.2

$$\mathbb{P}(\{H\} \cup \{T\}) = \mathbb{P}(\{H\}) + \mathbb{P}(\{T\}). \quad (8)$$

As $\{H\} \cup \{T\} = \Omega$, then it follows from Equation (8) that

$$\mathbb{P}(\{H\}) + \mathbb{P}(\{T\}) = 1. \quad (9)$$

From (A1) of Definition 1.8, we have $\mathbb{P}(\{H\}) \geq 0$ and $\mathbb{P}(\{T\}) \geq 0$, so for any $p \in [0, 1]$ the specification $\mathbb{P}(\{H\}) = p$ and $\mathbb{P}(\{T\}) = 1 - p$ is a valid probability measure on $(\Omega, \mathcal{F})$. If we **additionally** want the coin to be *fair*, i.e., $\mathbb{P}(\{H\}) = \mathbb{P}(\{T\})$, then we must take $p = \frac{1}{2}$.

Corollary 1.1 (Probability of complements). *For any probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and any event $E \in \mathcal{F}$, we have $\mathbb{P}(E^c) = 1 - \mathbb{P}(E)$.*

Proof. Idea: Use the fact that $\Omega = E \cup E^c$ and apply Theorem 1.2.

As $E \in \mathcal{F}$, we have, by Definition 1.7, $E^c \in \mathcal{F}$, and $E \cup E^c = \Omega$. By (A2)

$$\mathbb{P}(E \cup E^c) = \mathbb{P}(\Omega) = 1. \quad (10)$$

Moreover, $E \cap E^c = \emptyset$, i.e., E, E^c are disjoint. Thus, by (5) of Theorem 1.2 and (10)

$$1 = \mathbb{P}(E \cup E^c) = \mathbb{P}(E) + \mathbb{P}(E^c) \quad (11)$$

Rearranging Equation (11) yields the claim. $\square$

Corollary 1.2 (Probabilities are between zero and one). *For any probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and any event $E \in \mathcal{F}$, we have $0 \leq \mathbb{P}(E) \leq 1$. Thus $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$.*

Proof. By (A1) and Corollary 1.1, $\mathbb{P}(E^c) = 1 - \mathbb{P}(E) \geq 0$, implying $\mathbb{P}(E) \leq 1$. $\square$

Corollary 1.3 (Partition rule). *For any probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and any events $E, F \in \mathcal{F}$, we have*

$$\mathbb{P}(F) = \mathbb{P}(F \cap E) + \mathbb{P}(F \cap E^c). \quad (12)$$

Proof. Problem Sheet Exercise. $\square$

Corollary 1.4 (Containment rule). *For any probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and events $E, F \in \mathcal{F}$ with $E \subset F$, we have*

$$\mathbb{P}(F) = \mathbb{P}(E) + \mathbb{P}(F \cap E^c). \quad (13)$$

so that

$$\mathbb{P}(F) \geq \mathbb{P}(E). \quad (14)$$

Proof. Since $E \subset F$ we have $F \cap E = E$: see Figure 9. Substituting this into (12) of Corollary 1.3 gives (13). By (A1), $\mathbb{P}(F \cap E^c) \geq 0$ and hence (14) follows from (13). $\square$

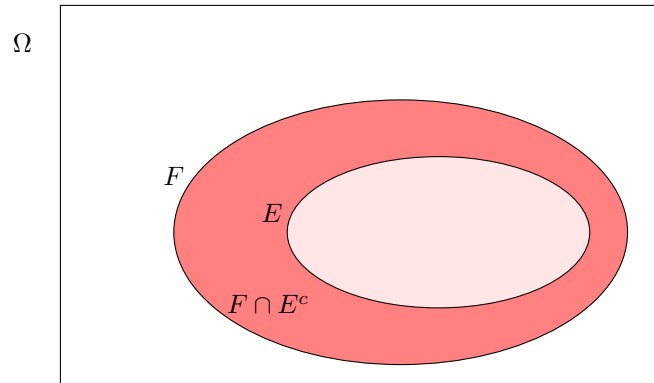


Figure 9: If $E \subset F$, then $F \cap E = E$.

Some of the results above will have been presented to you as facts at school (e.g. the fact that probabilities are between 0 and 1), and some may seem “intuitively obvious”. This can be a good thing, as your schooling and intuition can help you to remember useful properties of probability measures. But **be careful!** Sometimes your intuition can get carried away. Union is *not* the same as addition. For example, it is **not** always true that for two events E and F , we have $\mathbb{P}(E \cup F) = \mathbb{P}(E) + \mathbb{P}(F)$. Theorem 1.2 says this is true for *disjoint* events, but if our events are not disjoint, we need to use the following simple but important result.

Corollary 1.5 (Inclusion-exclusion for two sets). *For any probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and any events $E, F \in \mathcal{F}$, we have*

$$\mathbb{P}(E \cup F) = \mathbb{P}(E) + \mathbb{P}(F) - \mathbb{P}(E \cap F). \quad (15)$$

Proof. Problem Sheet Exercise. $\square$

Perhaps a useful rule is to “check your intuition with a Venn diagram”.

Example 1.9. In a game of Cluedo, a murder has been committed by a single person with a single weapon. We have narrowed the choice of murderer and the choice of weapon down to three possible choices each:

- **Murderer:** Colonel Mustard (M), Professor Plum (P), Miss Scarlett (S)
- **Weapon:** Candlestick (C), Lead Pipe (L), Rope (R)

Our sample space is thus

$$\Omega = \{(M, C), (M, L), (M, R), (P, C), (P, L), (P, R), (S, C), (S, L), (S, R)\}.$$

Each possible outcome has been assigned the following probabilities⁷:

	C	L	R
M	$\frac{4}{27}$	$\frac{4}{27}$	$\frac{4}{27}$
P	$\frac{4}{27}$	$\frac{4}{27}$	$\frac{4}{27}$
S	$\frac{1}{27}$	$\frac{1}{27}$	$\frac{1}{27}$

Thus, for example, $\mathbb{P}(\{(M, C)\}) = \frac{4}{27}$ and $\mathbb{P}(\{(S, C)\}) = \frac{1}{27}$.

1. **Find the probability of Colonel Mustard having committed the murder.** Event

$$M_{\text{Mustard}} = \{\text{Mustard committed the murder}\} = \{(M, C)\} \cup \{(M, L)\} \cup \{(M, R)\}$$

is a union of three pairwise disjoint events. Thus, from Theorem 1.2,

$$\begin{aligned}\mathbb{P}(M_{\text{Mustard}}) &= \mathbb{P}(\{(M, C)\}) + \mathbb{P}(\{(M, L)\}) + \mathbb{P}(\{(M, R)\}) \\ &= \frac{4}{27} + \frac{4}{27} + \frac{4}{27} = \frac{12}{27}.\end{aligned}\tag{16}$$

2. **Find the probability of Professor Plum or Colonel Mustard having committed the murder.** Let M_{Plum} denote the event that Professor Plum committed the murder. In a similar way to the previous part, using Theorem 1.2,

$$\begin{aligned}\mathbb{P}(M_{\text{Plum}}) &= \mathbb{P}(\{(P, C)\}) + \mathbb{P}(\{(P, L)\}) + \mathbb{P}(\{(P, R)\}) \\ &= \frac{4}{27} + \frac{4}{27} + \frac{4}{27} = \frac{12}{27}.\end{aligned}$$

Now, the murder has been committed by a single person and so M_{Mustard} and M_{Plum} are disjoint events. Once again by Theorem 1.2

$$\begin{aligned}\mathbb{P}(M_{\text{Mustard}} \cup M_{\text{Plum}}) &= \mathbb{P}(M_{\text{Mustard}}) + \mathbb{P}(M_{\text{Plum}}) \\ &= \frac{12}{27} + \frac{12}{27} = \frac{24}{27}.\end{aligned}$$

3. **Find the probability that Colonel Mustard is innocent.** Note that M_{Mustard}^c corresponds to Colonel Mustard being innocent. By Corollary 1.1

$$\mathbb{P}(\text{Mustard is innocent}) = \mathbb{P}(M_{\text{Mustard}}^c) = 1 - \mathbb{P}(M_{\text{Mustard}}) = 1 - \frac{12}{27} = \frac{15}{27}.$$

4. **Find the probability that Colonel Mustard is the murderer, or that rope was used.** Let W_{Rope} denote the event that the murder weapon was a rope. First we calculate

$$\begin{aligned}\mathbb{P}(W_{\text{Rope}}) &= \mathbb{P}(\{(M, R)\}) + \mathbb{P}(\{(P, R)\}) + \mathbb{P}(\{(S, R)\}) \\ &= \frac{4}{27} + \frac{4}{27} + \frac{1}{27} = \frac{9}{27}.\end{aligned}\tag{17}$$

As Mustard could have committed the murder using rope, these are **not** disjoint events and so Theorem 1.2 does not apply. We must instead use the inclusion-exclusion rule, Corollary 1.5, to find

$$\begin{aligned}\mathbb{P}(M_{\text{Mustard}} \cup W_{\text{Rope}}) &= \mathbb{P}(M_{\text{Mustard}}) + \mathbb{P}(W_{\text{Rope}}) - \mathbb{P}(M_{\text{Mustard}} \cap W_{\text{Rope}}) \\ &= \frac{12}{27} + \frac{9}{27} - \frac{4}{27} = \frac{17}{27}.\end{aligned}\tag{18}$$

⁷Note that these non-negative numbers sum to one. In Section 1.2.4 we will demonstrate that this means that we have a probability measure on $(\Omega, \mathcal{P}(\Omega))$.

Note that we could have calculated this probability in another way, by writing the event we want as the union of the five disjoint events:

$$M_{\text{Mustard}} \cup W_{\text{Rope}} = \{(M, C)\} \cup \{(M, L)\} \cup \{(M, R)\} \cup \{(P, R)\} \cup \{(S, R)\}.$$

If you think about this carefully, you can notice that if we had added $\mathbb{P}(M_{\text{Mustard}})$ to $\mathbb{P}(W_{\text{Rope}})$ together then, from Equations (16) and (17), we would have double-counted $\mathbb{P}(\{(M, R)\}) = \mathbb{P}(M_{\text{Mustard}} \cap W_{\text{Rope}})$. This is exactly the term that we subtract in Equation (18), so you can see why the inclusion-exclusion formula works.

1.2.4 Specifying probabilities

We now continue our mission to show that our definition of a probability measure is sensible. We check that, if we have n elements in our state space and n non-negative real numbers that sum to 1, we can use these to define a probability measure. The same holds if we have a countably infinite state space and non-negative real numbers $p_1, p_2, p_3 \dots$ that sum to 1.

Theorem 1.3 (Specifying probabilities for a countable sample space). *Suppose that $\Omega = \{\omega_1, \dots, \omega_n\}$ is a finite set and $\mathcal{F}$ is any σ -algebra on Ω . Suppose that $p_1, \dots, p_n$ are non-negative numbers with $\sum_{i=1}^n p_i = 1$. For any event $E \in \mathcal{F}$ define $\mathbb{P}(E)$ by*

$$\mathbb{P}(E) = \sum_{i: \omega_i \in E} p_i \quad (19)$$

where the sum over an empty set is defined to be zero. Then $\mathbb{P}$ is a probability measure on $(\Omega, \mathcal{F})$.

This remains true if $\Omega = \{\omega_1, \omega_2, \dots\}$ with corresponding non-negative numbers $p_1, p_2, \dots$ with $\sum_{i=1}^{\infty} p_i = 1$.

In words, if we specify $\mathbb{P}(\{\omega_i\}) = p_i$, then the probability of any event E can be found by adding the probabilities p_i of all outcomes contained in E .

Proof. Idea: We just need to check that Kolmogorov's axioms hold. The key observation is that if events $E_1, \dots, E_k$ are pairwise disjoint, then each element of the state space can be in at most one of them.

Suppose that $\Omega = \{\omega_1, \dots, \omega_n\}$ is finite. (The countably infinite case is similar.) As each p_i is non-negative then, by Equation (19), $\mathbb{P}(E) \geq 0$ and so (A1) holds. Now,

$$\mathbb{P}(\Omega) = \sum_{i: \omega_i \in \Omega} p_i = \sum_{i=1}^n p_i = 1,$$

so (A2) holds. For (A3), since Ω is finite we only need to consider finite unions. Let $E_1, \dots, E_k$ be pairwise disjoint events. Then for any $i \neq j$, $E_i \cap E_j = \emptyset$ so that if $\omega \in E_i$ then $\omega \notin E_j$; this means that each ω is in at most one of the E_i . We thus have that

$$\begin{aligned} \mathbb{P}\left(\bigcup_{i=1}^k E_i\right) &= \sum_{j: \omega_j \in \bigcup_{i=1}^k E_i} p_j \\ &= \sum_{i=1}^k \left(\sum_{j: \omega_j \in E_i} p_j \right) \quad [\text{as each } \omega_j \text{ is only in one of the } E_i] \\ &= \sum_{i=1}^k \mathbb{P}(E_i) \end{aligned}$$

where the first and last equalities hold by Equation (19). Thus (A3) holds. $\square$

Example 1.10. Again consider rolling one dice, so that $\Omega = \{1, 2, 3, 4, 5, 6\}$. Then we can set

$$p_i = \mathbb{P}(\{\text{score on the dice is } i\}) = \mathbb{P}(\{i\}) = \frac{1}{6}$$

for each $i = 1, 2, 3, 4, 5, 6$. For any event E in $\mathcal{P}(\Omega)$, set $\mathbb{P}(E)$ as in Equation (19). Then $\mathbb{P}$ is a probability measure on $(\Omega, \mathcal{P}(\Omega))$. For example,

$$\begin{aligned}\mathbb{P}(\{\text{score is even}\}) &= \mathbb{P}(\{2, 4, 6\}) = \mathbb{P}(\{2\}) + \mathbb{P}(\{4\}) + \mathbb{P}(\{6\}) = \frac{3}{6} = \frac{1}{2}; \\ \mathbb{P}(\{\text{score is less than 3}\}) &= \mathbb{P}(\{1, 2\}) = \mathbb{P}(\{1\}) + \mathbb{P}(\{2\}) = \frac{2}{6} = \frac{1}{3}.\end{aligned}$$

Notice that Theorem 1.3 only requires the p_i to be non-negative and to sum to one. It does not determine the actual values of the p_i . For example in the dice-rolling example of Example 1.10, the specification $p_1 = \frac{1}{12}$, $p_2 = \frac{1}{12}$, $p_3 = \frac{1}{12}$, $p_4 = \frac{1}{4}$, $p_5 = \frac{1}{4}$ and $p_6 = \frac{1}{4}$ would be a valid probability measure. In the next section we study one way of defining the p_i , known as the classical definition of probability, via the notion of equally likely outcomes.

2 The Classical Interpretation of Probability

2.1 Equally likely outcomes and the classical interpretation of probability

The classical interpretation of probability is based on the situation where the possible outcomes of an experiment are all considered to be **equally likely**.

Example 2.1. - A coin is tossed, with heads and tails equally likely.

- A (fair) dice is rolled.
- A card is drawn at random from a well-shuffled deck, so that each card is equally likely to be drawn.

Definition 2.1 (Classical interpretation of probability). For a finite sample space $\Omega = \{\omega_1, \dots, \omega_n\}$, so that $|\Omega| = n$, the classical interpretation of probability assumes that the outcomes $\omega_1, \dots, \omega_n$ are equally likely and defines the probability of each ω_i to be

$$\mathbb{P}(\{\omega_i\}) = \frac{1}{|\Omega|} = \frac{1}{n}.$$

For any event $E \subset \Omega$, the probability of E is then

$$\mathbb{P}(E) = \frac{|E|}{|\Omega|} = \frac{\text{number of ways } E \text{ can occur}}{\text{total number of outcomes}}. \quad (20)$$

We can use Theorem 1.3 to check that this defines a valid probability measure.

Indeed, we have $p_i = \frac{1}{|\Omega|} = \frac{1}{n} \geq 0$ for all i . Moreover,

$$\mathbb{P}(E) = \frac{|E|}{|\Omega|} = \sum_{i: \omega_i \in E} \frac{1}{|\Omega|} = \sum_{i: \omega_i \in E} p_i,$$

which Theorem 1.3 tells us is a valid probability measure.

Example 2.2. Suppose that we toss a fair coin three times. What is the probability of obtaining at least two heads?

We may write the sample space as

$$\Omega = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

where, for example, THT denotes that the first toss was a tail, the second a head, and the third a tail. There are thus $|\Omega| = 8$ possible outcomes. As the coin is stated to be fair, we proceed under the

assumption that all the outcomes are equally likely. Let E be the event that we obtain at least two heads then

$$E = \{HHH, HHT, HTH, THH\}$$

so that $|E| = 4$. Thus, using the classical interpretation of probability, Equation (20),

$$\mathbb{P}(E) = \frac{|E|}{|\Omega|} = \frac{4}{8} = \frac{1}{2}.$$

Note: We could have used a different sample space, $\Omega' = \{0, 1, 2, 3\}$, to count the number of heads observed over the three coin tosses. This would **not** be an example of the classical interpretation of probability, as then the four possibilities are *not* equally likely (you are more likely to see 1 head than 0 heads, for example - as we can see if we use Ω as above).

Example 2.3. Suppose that we roll two fair dice, a red one and a blue one. What is the probability of the total score of the two dice being 6?

Let (r, b) denote the score on the red dice being r and the score on the blue dice being b . Use the sample space

$$\Omega = \{(1, 1), (1, 2), (1, 3), \dots, (6, 4), (6, 5), (6, 6)\}$$

with $|\Omega| = 36$ ($= 6 \times 6$). As the dice are fair, we can assume that each outcome is equally likely. If we let $E = \{\text{total score is 6}\}$ then to find $\mathbb{P}(E)$ we need to find $|E|$, i.e. the total number of outcomes such that $r + b = 6$. We tabulate the possible outcomes and add the scores.

Red	Blue					
	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

We thus have $E = \{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}$, so that $|E| = 5$. By the classical interpretation of probability, (20), we have

$$\mathbb{P}(E) = \frac{|E|}{|\Omega|} = \frac{5}{36}.$$

Note: Again we could have chosen a different sample space, e.g. $\Omega' = \{2, 3, \dots, 12\}$ representing the sum of the two dice. But then the set of possible outcomes $2, 3, \dots, 12$ would not be equally likely. We emphasise that to use the classical interpretation of probability, the choice of sample space is crucial!

Example 2.4. From a sample of 400 adults, 300 cycle or swim (or both), 160 swim, 120 swim and cycle. What is the probability of an adult selected at random not cycling?

Let Ω be the set of all the sampled adults, so that $|\Omega| = 400$. Let S be set of the adults who swim and C the set of the adults who cycle. In this case

$$|S| = 160, \quad |S \cup C| = 300, \quad |S \cap C| = 120.$$

We can use a Venn diagram to easily summarise the number of adults who fall into the disjoint events of Ω as shown in Figure 10.

Choosing an adult at random corresponds to assuming that each adult is equally likely to be chosen. Then, using the classical interpretation of probability, (20),

$$\begin{aligned}\mathbb{P}(S) &= \frac{|S|}{|\Omega|} = \frac{160}{400} = \frac{2}{5} \\ \mathbb{P}(S \cup C) &= \frac{|S \cup C|}{|\Omega|} = \frac{300}{400} = \frac{3}{4} \\ \mathbb{P}(S \cap C) &= \frac{|S \cap C|}{|\Omega|} = \frac{120}{400} = \frac{3}{10}.\end{aligned}$$

Now, using the inclusion-exclusion rule, see Corollary 1.5, $\mathbb{P}(S \cup C) = \mathbb{P}(S) + \mathbb{P}(C) - \mathbb{P}(S \cap C)$ and by rearranging we have

$$\mathbb{P}(C) = \mathbb{P}(S \cup C) + \mathbb{P}(S \cap C) - \mathbb{P}(S) = \frac{3}{4} + \frac{3}{10} - \frac{2}{5} = \frac{13}{20}.$$

By using the probability of complements, Corollary 1.1, we have

$$\mathbb{P}(\{\text{adult doesn't cycle}\}) = \mathbb{P}(C^c) = 1 - \mathbb{P}(C) = \frac{7}{20}.$$

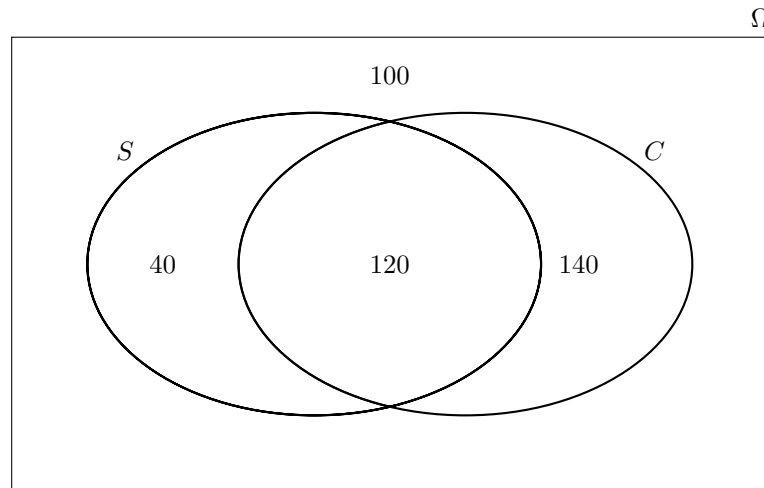


Figure 10: If $|\Omega| = 400$ with $|S| = 160$, $|S \cup C| = 300$ and $|S \cap C| = 120$ then we can calculate the size of each disjoint subset: $|S \cap C^c| = 40$, $|S^c \cap C| = 140$ (so that $|C| = 260$) and $|S^c \cap C^c| = 100$.

In the examples we have seen so far, it was relatively easy to count the number of outcomes and thus calculate the probabilities. In more complicated situations, we need to develop general counting techniques to find the number of outcomes. This is known as **combinatorics**.

2.2 Multiplication principle

Consider the problem where we have two finite sets, say A and B , and we wish to count the number of ways of choosing one element from A and one from B . How many ways are there of doing this?

Suppose that $A = \{a_1, \dots, a_n\}$ and $B = \{b_1, \dots, b_m\}$ so that $|A| = n$ and $|B| = m$. We can write an outcome of choosing one element from each set as a pair (a_i, b_j) and arrange them in an $n \times m$ table:

$$\begin{array}{ccccc}(a_1, b_1) & (a_1, b_2) & (a_1, b_3) & \dots & (a_1, b_m) \\ (a_2, b_1) & (a_2, b_2) & (a_2, b_3) & \dots & (a_2, b_m) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ (a_n, b_1) & (a_n, b_2) & (a_n, b_3) & \dots & (a_n, b_m)\end{array}$$

Thus there are $n \times m = nm$ ($= |A| \times |B|$) possible ways to choose one element from each set.

Theorem 2.1 (Multiplication principle). *Let $A_1, A_2, \dots, A_k$ be sets with $n_1, n_2, \dots, n_k$ elements, respectively. The number of ways of choosing one element from each of the k sets is*

$$\prod_{i=1}^k n_i = n_1 n_2 \cdots n_k.$$

Proof. Idea: Use induction. The number of ways of choosing from $i + 1$ sets is the number of ways of choosing from the first j , and then choosing from the $(i + 1)$ th.

The proof is by induction. The result is trivial for $k = 1$. Suppose it is true up to $k = i$. This means that if we let B_i be the set of all possible choices from sets $A_1, \dots, A_i$, i.e.

$$B_i = \{(a_1, a_2, \dots, a_i) : a_1 \in A_1, \dots, a_i \in A_i\},$$

then

$$|B_i| = |A_1| \times |A_2| \times \cdots \times |A_i| = n_1 n_2 \cdots n_i.$$

Then

$$\begin{aligned} |B_{i+1}| &= |\{(a_1, a_2, \dots, a_{i+1}) : a_1 \in A_1, \dots, a_{i+1} \in A_{i+1}\}| \\ &= |\{(a_1, a_2, \dots, a_{i+1}) : (a_1, \dots, a_i) \in B_i, a_{i+1} \in A_{i+1}\}|, \end{aligned}$$

i.e. the number of ways of choosing one element from B_i and one from A_{i+1} . We have already seen that the number of ways of choosing one element from a set of size n , and another from a set of size m , is $n \times m$. Here our two sets are B_i of size $n_1 n_2 \cdots n_i$, and A_{i+1} of size n_{i+1} , so we obtain that

$$|B_{i+1}| = n_1 n_2 \cdots n_i \times n_{i+1}.$$

Thus we have shown that if the result is true for $k = i$ then it is true for $k = i + 1$. Since it is true for $k = 1$, by induction it is true for all k . $\square$

Example 2.5. *Suppose that we toss a fair coin, roll a fair dice and pick a card at random from a full pack of cards. What is the probability of tossing a head, scoring 5 or 6 on the dice, and picking a red picture card (Jack, Queen, or King)?*

Let $\Omega = \{(t, d, c) : t \in \{H, T\}, d \in \{1, \dots, 6\}, c \in \{A \heartsuit, \dots, K \clubsuit\}\}$. In this case

$$|\Omega| = |\{H, T\}| \times |\{1, \dots, 6\}| \times |\{A \heartsuit, \dots, K \clubsuit\}| = 2 \times 6 \times 52 = 624.$$

Thus, there are 624 equally likely outcomes as the coin and the dice are fair and the card is picked at random.

Let E be the event of interest, i.e.,

$$E = \{(t, d, c) : t \in \{H\}, d \in \{5, 6\}, c \in \{J \heartsuit, Q \heartsuit, K \heartsuit, J \diamondsuit, Q \diamondsuit, K \diamondsuit\}\}$$

so that

$$|E| = |\{H\}| \times |\{5, 6\}| \times |\{J \heartsuit, Q \heartsuit, K \heartsuit, J \diamondsuit, Q \diamondsuit, K \diamondsuit\}| = 1 \times 2 \times 6 = 12.$$

Thus, E can occur in 12 different ways and by the classical interpretation of probability we have

$$\mathbb{P}(E) = \frac{|E|}{|\Omega|} = \frac{12}{624} = \frac{1}{52}.$$

2.3 Ordered choice: permutations

A **permutation** is an ordered arrangement of objects. Suppose that we have a set of n objects and we wish to choose r elements from the set and list them in order. How many ways are there to do this?

The answer depends on whether or not elements can be chosen more than once. If we are allowed to choose each element more than once, then we are **sampling with replacement**; otherwise we are **sampling without replacement**.

2.3.1 Sampling with replacement

Corollary 2.1 (Ordered choice: sampling with replacement). *The number of ways of choosing r elements with replacement in a specific order from a set of n elements is n^r .*

Proof. Follows from Theorem 2.1, the multiplication principle. Let A be the set of n elements, and then let $A_1 = A_2 = \dots = A_r$, with $n_i = n$ for each $i = 1, \dots, r$. Then choosing r elements with replacement in a specific order is the same as picking one element from A_1 , one from A_2 , and so on up to A_r . Theorem 2.1 says the number of ways of doing this is

$$\prod_{i=1}^r n_i = \prod_{i=1}^r n = n^r.$$

□

Example 2.6. *How many different 4 digit PIN codes are there?*

A 4 digit PIN code is a specific way of choosing $r = 4$ digits out of 10 digits (i.e., $0, 1, \dots, 9$). This is sampling with replacement as the same digit can appear more than once, and this is an ordered choice as the PIN codes 1234 and 4321 are different. By Corollary 2.1 there are thus $10^4 = 10000$ different codes.

What is the probability that my PIN contains only the digits 0 to 6 (assuming PINs are chosen at random and all 4-digit numbers are allowed)?

Let E be the event that my PIN contains only the digits 0 to 6. Again by 2.1 there are thus $7^4 = 2401$ possible 4-digit codes using the seven numbers 0, 1, 2, 3, 4, 5, 6, so $|E| = 2401$. Thus by the classical interpretation of probability

$$\mathbb{P}(E) = \frac{|E|}{|\Omega|} = \frac{2401}{10000} = 0.2401.$$

2.3.2 Sampling without replacement

We now consider the number of ways of choosing r **different** elements from a set of n elements in a specific order, i.e. sampling *without* replacement.

Choose the first item. There are n ways to do this. Once this item has been chosen, it is not replaced and so the number of remaining elements decreases by 1. Thus there are only $n - 1$ ways to choose the second item, and then $n - 2$ ways to choose the third item, and so on. Finally, once $(r - 1)$ items have been chosen without replacement, there are $n - (r - 1)$ ways to choose the r th item. Using the multiplication principle, Theorem 2.1, we thus have $n \times (n - 1) \times \dots \times (n - (r - 1))$ ways of choosing the r elements in a specific order without replacement.

Before we state this formally, it will be easier if we have a way of writing numbers of this form more concisely. We will need to recall the definition of *factorials*.

Definition 2.2 (Factorial). For any $m \in \{0, 1, \dots\}$, m factorial, written as $m!$, is defined as

$$m! = \begin{cases} m(m-1)(m-2)\dots 3 \times 2 \times 1 & \text{if } m = 1, 2, 3, \dots, \\ 1 & \text{if } m = 0. \end{cases}$$

Using this definition, the reasoning above can be formally stated as follows.

Corollary 2.2 (Ordered choice: sampling without replacement). *The number of permutations of length r taken from a set of size n , or equivalently, the number of ways to choose r different elements in a specific order from a set of n elements is*

$$\begin{aligned} n(n-1)(n-2)\dots(n-(r-1)) &= \frac{n(n-1)\dots(n-(r-1))(n-r)(n-(r+1))\dots 1}{(n-r)(n-(r+1))\dots 2 \cdot 1} \\ &= \frac{n!}{(n-r)!}. \end{aligned}$$

Example 2.7. Five cards from a shuffled pack are dealt face up in a row. What is the probability the first three are picture cards and the last two are not?

Since the order of the cards is important, we need our sample space Ω to be the set of ordered sequences of five cards drawn from a deck of 52 cards. By Corollary 2.2,

$$|\Omega| = 52 \times 51 \times 50 \times 49 \times 48.$$

Let E be the event that we get three picture cards and then two non-picture cards. As there are 12 picture cards in total (four suits, King, Queen and Jack in each suit), the number of ways of choosing the first three cards to all be picture cards is (sampling without replacement) $12 \times 11 \times 10$. The number of ways of choosing the last two cards to both be non-picture cards is similarly 40×39 . By the multiplication principle, we have

$$|E| = 12 \times 11 \times 10 \times 40 \times 39,$$

and so

$$\mathbb{P}(E) = \frac{|E|}{|\Omega|} = \frac{12 \times 11 \times 10 \times 40 \times 39}{52 \times 51 \times 50 \times 49 \times 48} = \frac{11}{1666} \approx 0.0066.$$

2.4 Unordered choice: combinations

A **combination** is an unordered arrangement of objects. Suppose that we have a set of n *distinct* elements and wish to choose r elements from E so that the order of the elements is irrelevant. In how many ways can we do this?

Similar to the case with permutations, the answer depends on whether the same element can be chosen more than once.

2.4.1 Sampling without replacement

Let $C(n, r)$ denote the number of ways of choosing r **different** elements from a set of n distinct elements when **the order does not matter**. What is $C(n, r)$? From Corollary 2.2 we know that the number of *ordered* samples (or permutations) is $n!/(n-r)!$. If the order does not matter, then we can re-order (or *permute*) the r chosen elements in any way we like and still get the same sample.

Lemma 2.1 (Number of permutations of k objects). *The number of permutations of k objects, i.e., the number of ways k elements can be permuted/ordered/arranged, is $k!$.*

Proof. We apply Corollary 2.2 with $n = r = k$, i.e. the number of ways to choose k different elements in a specific order from a set of k elements. This gives $k!/0! = k!$. $\square$

Therefore in the $n!/(n-r)!$ permutations, each combination appears $r!$ times, and thus

$$\frac{n!}{(n-r)!} = r! C(n, r).$$

We deduce that

$$C(n, r) = \frac{n!}{r!(n-r)!},$$

which we formally state as the following corollary:

Corollary 2.3 (Unordered choice: sampling without replacement, combinations). *The number of combinations of length r taken from a set of n distinct elements, i.e., the number of ways of choosing r different elements from a set of n distinct elements when the order does not matter is*

$$\binom{n}{r} := \frac{n!}{r!(n-r)!}. \quad (21)$$

Remarks:

1. $\binom{n}{r}$ is read as “ n choose r ”.
2. From Equation (21), we have immediately that for $r = 0, 1, \dots, n$

$$\binom{n}{r} = \binom{n}{n-r}.$$

Example 2.8. *The national lottery chooses 6 balls at random (without replacement) from a set of 59 balls labelled from 1 to 59. If you pick 6 numbers, what is the probability that your numbers match those drawn?*

The sample space is the set of choices of 6 distinct elements from a set of 59, i.e.

$$\Omega = \{(a_1, a_2, a_3, a_4, a_5, a_6) : \text{no two } a_i \text{ are equal, } a_i \in \{1, \dots, 59\} \text{ for all } i.\}$$

From Corollary 2.3 we know that

$$|\Omega| = \binom{59}{6} = \frac{59!}{53!6!} = \frac{59 \times 58 \times 57 \times 56 \times 55 \times 54}{6 \times 5 \times 4 \times 3 \times 2 \times 1} = 45,057,474.$$

For us to win, we have to match exactly the one chosen sequence, so $\mathbb{P}(\text{we win}) = 1/|\Omega| \approx 0.000000022194$ or about one in 45 million.

Example 2.9. *Five cards are dealt from a shuffled pack of 52 cards. What is the probability of a full house, i.e., three of one rank and a pair of another rank? (The rank of a card is just whether it is Ace, 2, 3, ..., 10, Jack, Queen or King. So a full house could be e.g. three 9s and two Jacks.)*

Let $\Omega = \{\text{choices of 5 cards}\}$. Then $|\Omega| = \binom{52}{5}$. We now count how many possible ways there are of obtaining a full house. There are 13 ways of choosing the rank of the card that is in the triple. Once the rank is chosen, there are $\binom{4}{3}$ ways to choose the three suits. Having chosen the rank for the triple, there are only 12 ways (the remaining ranks) of choosing the rank for the pair and $\binom{4}{2}$ ways of choosing the suits for the pair. If we let $E = \{\text{get a full house}\}$ then

$$|E| = 13 \times \binom{4}{3} \times 12 \times \binom{4}{2}.$$

As all hands are equally likely,

$$\begin{aligned} \mathbb{P}(E) &= \frac{|E|}{|\Omega|} \\ &= \frac{13 \times 12 \times \binom{4}{3} \times \binom{4}{2}}{\binom{52}{5}} \\ &= 13 \times 12 \times 4 \times \frac{4 \times 3}{2 \times 1} \times \frac{5 \times 4 \times 3 \times 2 \times 1}{52 \times 51 \times 50 \times 49 \times 48} \\ &= \frac{6}{4165} = 0.0014 \text{ (4dp)}. \end{aligned}$$

2.4.2 Sampling with replacement

Consider choosing r elements from a set $A = \{a_1, \dots, a_n\}$ of n distinct elements, with replacement, when the order of the r elements does not matter. This means that a_1 could appear any number of times from 0 up to r ; so could a_2 , and so on. Say that a_1 appears m_1 times, a_2 appears m_2 times, and so on. Then we need each m_i to be a non-negative integer, and we also need $m_1 + \dots + m_n = r$, so that the total number of chosen elements is r .

The ice cream vending machine

Imagine there is a vending machine with n different flavours of ice cream set out in a row. You put enough money in for r scoops of ice cream. (You are allowed more than one scoop of each flavour.) You have to tell the machine which flavours you want, and also get the machine to the end so that it resets for the next customer. There are only two buttons: “scoop” (S) and “move” (M). The machine starts from flavour 1, so to get it to the end you will need to press the “move” button $n - 1$ times. And to get your helping of ice cream you will need to press the “scoop” button r times. Apart from these conditions you are free to press the two buttons in any order: if you want r scoops of flavour 1, you press “scoop” r times followed by “move” $n - 1$ times. Another example, if $n = 10$ and $r = 5$, is

MMSMMSSMMSMMSM

which would get you one scoop of flavour 3, two scoops of flavour 5, one scoop of flavour 7 and one scoop of flavour 9.

Another way of looking at this is that we have to press $n - 1 + r$ buttons, and we can choose any r of them to be S . This is the same as if the national lottery had $n - 1 + r$ balls and r of them were chosen as the winning numbers. So the number of ways of making r choices from a set of n distinct elements, with replacement, is exactly the same as the number of ways of choosing r out of $n - 1 + r$ distinct elements, *without* replacement. But we already know that this is exactly $\binom{n-1+r}{r}$.

(This explanation is based on one from mathsisfun.com. See e.g. the numberphile episode “stars and bars (and bagels)” for another explanation.)

Corollary 2.4 (Unordered choice: sampling with replacement). *The number of ways of choosing r elements from a set of n elements, with replacement, so that the order does not matter is*

$$\binom{n-1+r}{r} = \binom{n-1+r}{n-1} = \frac{(n-1+r)!}{r!(n-1)!}.$$

Example 2.10. *A doughnut shop offers four kinds of doughnut: ring, chocolate, jam, and lemon. You want to buy 12 doughnuts. How many different possible choices are there?*

We are choosing $r = 12$ doughnuts out of $n = 4$ possible kinds. (Obviously each kind can be chosen more than once, otherwise we would struggle to buy 12!) Thus the answer is, by Corollary 2.4,

$$\binom{4-1+12}{12} = \binom{15}{12} = \binom{15}{3} = \frac{15!}{12!3!} = \frac{15 \times 14 \times 13}{3 \times 2 \times 1} = 455.$$

To summarise all the different scenarios, we can refer to the following table.

	Without replacement	With replacement
Ordered	$\frac{n!}{(n-r)!}$	n^r
Unordered	$\binom{n}{r}$	$\binom{n-1+r}{r}$

It might help to remember a key example for each:

	Without replacement	With replacement
Ordered	podium in a race	PIN code
Unordered	lotto	buying doughnuts

2.5 Examples

Example 2.11. (a) How many distinct permutations of the word *METHODS* are there?

There are seven distinct letters so there are $7! = 5040$ permutations.

(b) How many distinct permutations of the word *ALGEBRA* are there?

Notice that there are two As, so we need to be careful: Corollary 2.3 insists that the n elements are *distinct*, which is not the case here.

If we label the As as A_1, A_2 and treat these as different letters, then there are $7!$ permutations of the letters $A_1 A_2 L G E B R A_2$. Consider any one of these permutations, say $A_1 A_2 L G E B R$. This is equivalent to $A_2 A_1 L G E B R$: there are $2!$ ways of permuting the As whilst keeping the remaining five letters fixed. Thus, there are $7!/2! = 2520$ distinct permutations of *ALGEBRA*.

Example 2.12. I turn over 5 cards one by one from a well-shuffled deck. What is the probability that they are A, 2, 3, 4, 5 (in that order)?

The order matters, and we aren't replacing cards, so with $\Omega = \{\text{ordered samples of 5 cards without replacement}\}$ we have

$$|\Omega| = \frac{52!}{(52-5)!} = 52 \times 51 \times 50 \times 49 \times 48 = 311,875,200.$$

Now, if

$$E = \{\text{first card is A, second is 2, \dots, fifth is 5}\},$$

then I have 4 options for my first card (since there are 4 aces), 4 for the second, 4 for the third, 4 for the fourth and 4 for the fifth. Thus

$$|E| = 4^5 = 1024,$$

and

$$\mathbb{P}(E) = \frac{|E|}{|\Omega|} = \frac{1024}{311,875,200} = \frac{8}{2,436,525} \approx 0.000003.$$

Example 2.13. The national lottery chooses 6 balls at random (without replacement) from a set of 59 balls labelled from 1 to 59. If you pick 6 numbers, what is the probability that at least 5 of your numbers match those drawn?

We already know that if

$$\Omega = \{\text{set of 6 distinct choices without replacement from 59 possibilities}\},$$

$$|\Omega| = \binom{59}{6} = 45,057,474,$$

and therefore

$$\mathbb{P}(\text{all 6 of my numbers match}) = \frac{1}{45,057,474}.$$

Now, since

$$\mathbb{P}(\text{at least 5 of my numbers match}) = \mathbb{P}(\text{all 6 match}) + \mathbb{P}(\text{exactly 5 match})$$

and the two events on the right are disjoint, it remains to calculate $\mathbb{P}(\text{exactly 5 match})$.

Now, to match exactly 5 numbers, I need to choose which chosen ball not to match (6 choices) and then I have 53 choices (the balls not chosen) for that number. Apart from this, I have no flexibility: my remaining 5 numbers must match exactly the ones on the remaining 5 balls. So

$$\mathbb{P}(\text{exactly 5 match}) = \frac{6 \times 53}{45,057,474} = \frac{318}{45,057,474} \approx 0.000007,$$

and

$$\mathbb{P}(\text{at least 5 match}) = \frac{319}{45,057,474} \approx 0.000007.$$

Example 2.14. *An urn contains r red balls and b blue balls. You remove balls one by one from the urn at random, without replacing them. What is the probability that the first time you remove a red ball is on the k th pick?*

The sample space is the set of sequences of $r + b$ balls, exactly b of which must be blue. This is the same as if we have $r + b$ distinct elements (the numbers 1 up to $r + b$) and we need to choose b of them to colour blue; the order doesn't matter (if I choose number 3 and then number 7, that's the same as choosing number 7 and number 3; they both just get coloured blue) and we are working without replacement (once I have chosen number 3 and coloured it blue, I can't choose number 3 again). Therefore

$$|\Omega| = \binom{r+b}{b}.$$

If $E = \{\text{first red ball is the } k\text{th}\}$, then our first k balls are fixed (they must be $k - 1$ blue balls followed by a red); and the remaining $r + b - k$ must consist of $r - 1$ red balls (and thus $b - k + 1$ blue balls). Or in other words, we have already coloured the numbers 1 up to k , and we now have $r + b - k$ numbers, $r - 1$ of which need to be chosen (to colour red). Thus

$$|E| = \binom{r+b-k}{r-1},$$

and therefore

$$\mathbb{P}(E) = \frac{|E|}{|\Omega|} = \frac{\binom{r+b-k}{r-1}}{\binom{r+b}{b}}.$$

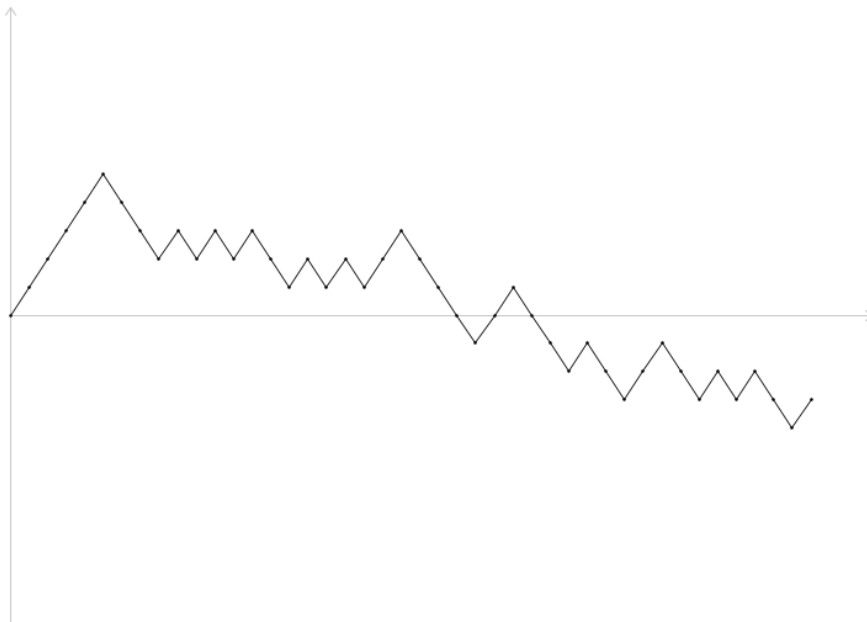


Figure 11: The first 43 steps of a random walk.

Example 2.15 (Random walk). *In the first lecture we briefly saw a key example, a **random walk** that at each time step moves up by 1 with probability $1/2$, and down by 1 with probability $1/2$. (See figures 11 and 12.) What is the probability that the random walk is at 0 after n steps?*

In order to be at 0, the walk must have taken the same number of up-steps as down-steps.

If n is odd, then this is impossible, so the probability the walk is at 0 is zero.

What about if n is even? Let

$$\Omega = \{\text{ordered sequences of } +1 \text{ and } -1 \text{ of length } n\},$$

and

$$E = \{\text{ordered sequences of } +1 \text{ and } -1 \text{ of length } n \text{ such that exactly } n/2 \text{ are } +1\}.$$

Then (ordered sampling of n elements from a set of 2 with replacement)

$$|\Omega| = 2^n$$

and (unordered sampling of $n/2$ elements from a set of n without replacement)

$$|E| = \binom{n}{n/2}.$$

Each outcome in Ω is equally likely, and thus, for even n ,

$$\mathbb{P}(E) = \frac{|E|}{|\Omega|} = \frac{\binom{n}{n/2}}{2^n} = \frac{n!}{2^n (n/2)! (n/2)!}.$$

(Is this formula correct for odd n ? Is $\mathbb{P}(E) = 0$ for odd n ? Why / why not?)

This tells us, implicitly, the probability that our random walk is at 0 after n steps. But in this form, it's not very helpful - how big is $\frac{n!}{2^n (n/2)! (n/2)!}$ when n is large? To answer this question we need (the remarkable) *Stirling's formula*, which is non-examinable, to tell us how big $n!$ is.

Stirling's formula (non-examinable):

$$n! \approx \frac{n^n}{e^n} \sqrt{2\pi n}.$$

Using Stirling's formula, we see that

$$\mathbb{P}(E) \approx \frac{n^n e^{-n} \sqrt{2\pi n}}{2^n (n/2)^{n/2} e^{-n/2} \sqrt{\pi n} (n/2)^{n/2} e^{-n/2} \sqrt{\pi n}} = \sqrt{\frac{2}{\pi n}}.$$

You might like to ask yourself: does the random walk hit 0 infinitely often? How long do we have to wait for it to hit 0?

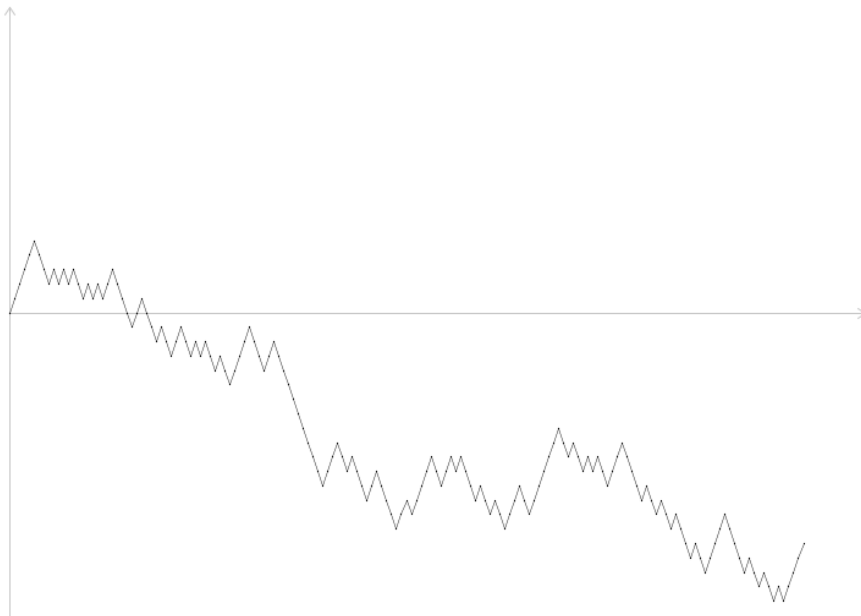


Figure 12: The first 200 steps of a random walk.

3 Conditional probability and independence

3.1 Conditional probability

Suppose that in an experiment, we learn that the event F has occurred. Does this information change the probability of some other event, say E , occurring?

Definition 3.1 (Conditional probability). For a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ if $E, F \in \mathcal{F}$ such that $\mathbb{P}(F) > 0$, then the conditional probability of E given F , written as $\mathbb{P}(E | F)$, is

$$\mathbb{P}(E | F) = \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(F)}. \quad (22)$$

Example 3.1. Your friend tossed a coin three times, but you do not know the outcomes. What is the probability of observing three tails, when you know that at least two tails are observed?

Let $\Omega = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$ as before.

If we set $E = \{\text{Three tails}\}$ and $F = \{\text{At least two tails}\}$, then $E = \{TTT\}$, $F = \{TTT, TTH, THT, HTT\}$ and $E \cap F = \{TTT\}$. If we assume that all sample points of Ω are equally likely then

$$\mathbb{P}(E) = \frac{|E|}{|\Omega|} = \frac{1}{8}, \quad \text{and} \quad \mathbb{P}(F) = \frac{|F|}{|\Omega|} = \frac{4}{8} = \frac{1}{2}, \quad \text{and} \quad \mathbb{P}(E \cap F) = \frac{|E \cap F|}{|\Omega|} = \frac{1}{8}.$$

Using Equation (22), the probability of three tails given that there are at least two tails is

$$\mathbb{P}(E | F) = \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(F)} = \frac{1/8}{1/2} = \frac{1}{4}.$$

Notice that $\mathbb{P}(E | F) > \mathbb{P}(E)$ which is not a surprise: learning that there are at least two tails increases the probability that all three are tails.

You might be surprised that the probability of seeing three tails given at least two tails is not $1/2$, since once I have two tails, the probability that the third coin is tails is $1/2$. Indeed, if I tell you that **the first two tosses are tails**, then the conditional probability that all three are tails is $1/2$:

$$\mathbb{P}(\text{three tails} | \text{first two tosses are tails}) = \frac{\mathbb{P}(\{TTT\})}{\mathbb{P}(\{TTT, TTH\})} = \frac{1/8}{1/4} = \frac{1}{2}.$$

Think about the difference between these two conditionings.

Remarks: Return to Definition 3.1. Suppose that $\mathbb{P}(F) > 0$.

1. Essentially, F becomes the new sample space for the conditional probability $\mathbb{P}(\cdot | F)$, i.e., $\mathbb{P}(F | F) = 1$. All probabilities are then re-normalised by $\mathbb{P}(F)$.
2. It is straightforward to check that $\mathbb{P}(\cdot | F)$ satisfies Kolmogorov's axioms (A1)–(A3). Hence, all implications of these axioms, i.e. the properties in Section 1.2.3, remain valid for conditional probability as well. For example, the inclusion-exclusion rule of Corollary 1.5 for conditional probability is that for any events E and G we have:

$$\mathbb{P}(E \cup G | F) = \mathbb{P}(E | F) + \mathbb{P}(G | F) - \mathbb{P}(E \cap G | F).$$

3. If E and F are disjoint, that is $E \cap F = \emptyset$ (Definition 1.5), then $\mathbb{P}(E \cap F) = 0$ so that $\mathbb{P}(E | F) = 0$.

We may re-arrange Equation (22) to obtain

$$\mathbb{P}(E \cap F) = \mathbb{P}(F)\mathbb{P}(E | F), \quad (23)$$

which can be a useful way of calculating intersection probabilities, as in some cases conditional probabilities may be easy to calculate directly.

Example 3.2. Two cards are drawn at random from a deck of 52 cards. What is the probability of both being aces?

Let $E_1 = \{\text{first card is an ace}\}$ and $E_2 = \{\text{second card is an ace}\}$. Then

$$\mathbb{P}(\{\text{Two aces}\}) = \mathbb{P}(E_1 \cap E_2) = \mathbb{P}(E_1)\mathbb{P}(E_2 | E_1) = \frac{4}{52} \times \frac{3}{51} = \frac{1}{221}.$$

We can extend Equation (23) to intersections of n events.

Theorem 3.1. Suppose that $E_1, \dots, E_n$ are any events such that $\mathbb{P}(E_1 \cap \dots \cap E_{n-1}) > 0$. Then

$$\mathbb{P}(E_1 \cap \dots \cap E_n) = \mathbb{P}(E_1)\mathbb{P}(E_2 | E_1)\mathbb{P}(E_3 | E_1 \cap E_2) \cdots \mathbb{P}(E_n | E_1 \cap \dots \cap E_{n-1}). \quad (24)$$

Proof. Idea: Just write out the definition of the right-hand side and cancel lots of terms!

First we need to check that the right-hand side is well-defined. For any $k \in \{1, \dots, n-1\}$, $\bigcap_{i=1}^{n-1} E_i \subset \bigcap_{i=1}^k E_i$ and so, by the containment rule (Corollary 1.4, if $E \subset F$, then $\mathbb{P}(F) \geq \mathbb{P}(E)$), and so

$$\mathbb{P}\left(\bigcap_{i=1}^k E_i\right) \geq \mathbb{P}\left(\bigcap_{i=1}^{n-1} E_i\right) > 0.$$

Thus, all conditional probabilities in Equation (24) are well defined. Now, starting from the right-hand side of (24),

$$\begin{aligned} & \mathbb{P}(E_1)\mathbb{P}(E_2 | E_1)\mathbb{P}(E_3 | E_1 \cap E_2) \cdots \mathbb{P}(E_n | E_1 \cap \dots \cap E_{n-1}) \\ &= \mathbb{P}(E_1) \frac{\mathbb{P}(E_1 \cap E_2)}{\mathbb{P}(E_1)} \frac{\mathbb{P}(E_1 \cap E_2 \cap E_3)}{\mathbb{P}(E_1 \cap E_2)} \cdots \frac{\mathbb{P}(E_1 \cap \dots \cap E_n)}{\mathbb{P}(E_1 \cap \dots \cap E_{n-1})} \\ &= \mathbb{P}(E_1 \cap \dots \cap E_n) \end{aligned}$$

yielding the claim. □

Example 3.3. An electrician's toolbox contains five good and two bad fuses. Fuses are selected for testing at random, without replacement.

1. Find the probability that the first two tested fuses are defective.

Let D_i denote the event that the i th fuse is defective and G_i that it is good, so that $G_i^c = D_i$. Then

$$\mathbb{P}(\{\text{First two defective}\}) = \mathbb{P}(D_1 \cap D_2) = \mathbb{P}(D_1)\mathbb{P}(D_2 | D_1) = \frac{2}{7} \times \frac{1}{6} = \frac{1}{21}.$$

2. Find the probability of the second defective fuse being encountered on the third test.

Let E be the event that the second defective fuse is found on third test, i.e.

$$E = (G_1 \cap D_2 \cap D_3) \cup (D_1 \cap G_2 \cap D_3).$$

Thus

$$\begin{aligned} \mathbb{P}(E) &= \mathbb{P}(G_1 \cap D_2 \cap D_3) + \mathbb{P}(D_1 \cap G_2 \cap D_3) && [\text{since disjoint events}] \\ &= \mathbb{P}(G_1)\mathbb{P}(D_2 | G_1)\mathbb{P}(D_3 | G_1 \cap D_2) \\ &\quad + \mathbb{P}(D_1)\mathbb{P}(G_2 | D_1)\mathbb{P}(D_3 | D_1 \cap G_2) && [\text{by theorem above}] \\ &= \left(\frac{5}{7} \times \frac{2}{6} \times \frac{1}{5}\right) + \left(\frac{2}{7} \times \frac{5}{6} \times \frac{1}{5}\right) = \frac{2}{21}. \end{aligned}$$

3.2 The law of total probability

Definition 3.2 (Partition). If $E_1, \dots, E_n$ is a collection of pairwise disjoint non-empty events, i.e., $E_i \neq \emptyset$ and $E_i \cap E_j = \emptyset$ for all $i, j \in \{1, \dots, n\}$, $i \neq j$, and if $\bigcup_{i=1}^n E_i = \Omega$, then $E_1, \dots, E_n$ form a *partition* of Ω .

Partitions are useful as they divide the sample space Ω into disjoint events whose probabilities can be added. That is, for a partition $\{E_1, \dots, E_n\}$ we have

$$\mathbb{P}\left(\bigcup_{i=1}^n E_i\right) = \sum_{i=1}^n \mathbb{P}(E_i) = 1.$$

Example 3.4. For any event E with $E \neq \emptyset$ and $E \neq \Omega$, $\{E, E^c\}$ is a partition of Ω , since $E \cap E^c = \emptyset$ and $E \cup E^c = \Omega$.

Recall the partition rule (Corollary 1.3, Section 1.2.3,

$$\mathbb{P}(F) = \mathbb{P}(F \cap E) + \mathbb{P}(F \cap E^c). \quad (25)$$

Now, if $\mathbb{P}(E) > 0$ and $\mathbb{P}(E^c) > 0$ so that $\mathbb{P}(F | E) = \mathbb{P}(F \cap E) / \mathbb{P}(E)$ and $\mathbb{P}(F | E^c) = \mathbb{P}(F \cap E^c) / \mathbb{P}(E^c)$ are well defined, then

$$\mathbb{P}(F \cap E) = \mathbb{P}(E) \mathbb{P}(F | E), \quad (26)$$

$$\mathbb{P}(F \cap E^c) = \mathbb{P}(E^c) \mathbb{P}(F | E^c). \quad (27)$$

Substituting Equations (26) and (27) into Equation (25) gives

$$\mathbb{P}(F) = \mathbb{P}(E) \mathbb{P}(F | E) + \mathbb{P}(E^c) \mathbb{P}(F | E^c).$$

This result can be extended into the following theorem.

Theorem 3.2 (The Law of Total Probability). Suppose that $\{E_1, \dots, E_n\}$ form a partition of Ω and $\mathbb{P}(E_i) > 0$ for all $i \in \{1, \dots, n\}$. Then for any event F ,

$$\mathbb{P}(F) = \sum_{i=1}^n \mathbb{P}(E_i) \mathbb{P}(F | E_i).$$

Proof. Idea: Write F as a union of its intersections with the E_i , and use the definition of conditional probability.

Note that $F = F \cap \Omega$ and $\bigcup_{i=1}^n E_i = \Omega$. Thus by the Distributive Law (Theorem 1.1)

$$F = F \cap \Omega = F \cap \left(\bigcup_{i=1}^n E_i\right) = \bigcup_{i=1}^n (F \cap E_i).$$

For any $i \neq j$,

$$(F \cap E_i) \cap (F \cap E_j) = F \cap (E_i \cap E_j) = F \cap \emptyset = \emptyset,$$

so that $F \cap E_1, \dots, F \cap E_n$ are pairwise disjoint. Thus

$$\begin{aligned} \mathbb{P}(F) &= \mathbb{P}\left(\bigcup_{i=1}^n (F \cap E_i)\right) \\ &= \sum_{i=1}^n \mathbb{P}(F \cap E_i) && \text{[disjoint events]} \\ &= \sum_{i=1}^n \mathbb{P}(E_i) \mathbb{P}(F | E_i) && \text{[conditional probability]} \end{aligned}$$

□

Remarks:

1. We can extend Definition 3.2 to countably infinite partitions $E_1, E_2, \dots$ of Ω . In this case, Theorem 3.2 extends naturally to

$$\mathbb{P}(F) = \sum_{i=1}^{\infty} \mathbb{P}(E_i) \mathbb{P}(F | E_i).$$

2. Theorem 3.2 also holds for disjoint events $E_1, \dots, E_n$ such that $F \subset \bigcup_{i=1}^n E_i \neq \Omega$.

Example 3.5. In a bank's trading office, Buster, Rich, and Owen made 30%, 50% and 20% of all the deals last year, respectively. Moreover, $1/2$, $1/3$ and $1/4$ of their deals, respectively, made in excess of £1 million in profit. What is the probability that the profit of a randomly chosen deal from last year was over £1 million?

Let Ω be collection of last year's deals and let B, R, W denote the events that the chosen deal was made by Buster, Rich, and Owen, respectively. Events B, R , and W are disjoint and $B \cup R \cup W = \Omega$, so $\{B, R, W\}$ is a partition of Ω . Let us write $E = \{\text{the chosen deal made over £1 million in profit}\}$, and so, by the law of total probability,

$$\begin{aligned} \mathbb{P}(E) &= \mathbb{P}(B) \mathbb{P}(E | B) + \mathbb{P}(R) \mathbb{P}(E | R) + \mathbb{P}(W) \mathbb{P}(E | W) \\ &= \left(\frac{3}{10} \times \frac{1}{2} \right) + \left(\frac{5}{10} \times \frac{1}{3} \right) + \left(\frac{2}{10} \times \frac{1}{4} \right) = \frac{11}{30}. \end{aligned}$$

3.3 Bayes' theorem

Suppose that $\mathbb{P}(E) > 0$ and $\mathbb{P}(F) > 0$. Then we have

$$\mathbb{P}(E \cap F) = \mathbb{P}(F) \mathbb{P}(E | F) = \mathbb{P}(E) \mathbb{P}(F | E)$$

so that

$$\mathbb{P}(E | F) = \frac{\mathbb{P}(E) \mathbb{P}(F | E)}{\mathbb{P}(F)}, \quad (28)$$

which gives a formula for reversing the conditioning. This is useful if $\mathbb{P}(F | E)$ is straightforward to obtain but we are interested in $\mathbb{P}(E | F)$. For example, as a patient visiting the doctor we are interested in $\mathbb{P}(\text{disease} | \text{symptoms})$ whilst medical evidence typically knows $\mathbb{P}(\text{symptoms} | \text{disease})$. Equation (28) is often called Bayes' theorem⁸, but there is a more general form which applies to partitions of Ω .

Theorem 3.3 (Bayes' theorem). Suppose that $\{E_1, \dots, E_n\}$ is a partition of Ω and $\mathbb{P}(E_i) > 0$ for all $i \in \{1, \dots, n\}$. Then for any $j \in \{1, \dots, n\}$, and any event F with $\mathbb{P}(F) > 0$,

$$\mathbb{P}(E_j | F) = \frac{\mathbb{P}(E_j) \mathbb{P}(F | E_j)}{\sum_{i=1}^n \mathbb{P}(E_i) \mathbb{P}(F | E_i)}. \quad (29)$$

Proof. Equation (29) follows from Equation (28) by taking $E = E_j$ and then using the law of total probability, Theorem 3.2, for $\mathbb{P}(F)$. $\square$

Remarks:

1. Recall that $\{E, E^c\}$ is a partition for any E , so that

$$\mathbb{P}(E | F) = \frac{\mathbb{P}(E) \mathbb{P}(F | E)}{\mathbb{P}(E) \mathbb{P}(F | E) + \mathbb{P}(E^c) \mathbb{P}(F | E^c)}.$$

2. As with the Law of Total Probability, the result can be extended to the case where $E_1, E_2, \dots$ is a countably infinite partition, and also to the case where $E_1, \dots, E_n$ are disjoint and $F \subset \bigcup_{i=1}^n E_i \neq \Omega$.

⁸Thomas Bayes (1701–1761).

Example 3.6 (Example 3.5 revisited). Suppose that $1/4$, $1/8$ and $1/16$ of the deals made by Buster, Rich and Owen, respectively, made a loss. What is the probability that a randomly chosen loss-making deal was made by Buster?

Let L be the event that the deal made a loss. Then we know

$$\begin{array}{lll} \text{Buster:} & \mathbb{P}(L | B) = 1/4, & \mathbb{P}(B) = 3/10 \\ \text{Rich:} & \mathbb{P}(L | R) = 1/8, & \mathbb{P}(R) = 5/10 \\ \text{Owen:} & \mathbb{P}(L | W) = 1/16, & \mathbb{P}(W) = 2/10 \end{array}$$

and so, by Bayes' Theorem,

$$\begin{aligned} \mathbb{P}(B | L) &= \frac{\mathbb{P}(B)\mathbb{P}(L | B)}{\mathbb{P}(B)\mathbb{P}(L | B) + \mathbb{P}(R)\mathbb{P}(L | R) + \mathbb{P}(W)\mathbb{P}(L | W)} \\ &= \frac{\left(\frac{3}{10} \times \frac{1}{4}\right)}{\left(\frac{3}{10} \times \frac{1}{4}\right) + \left(\frac{5}{10} \times \frac{1}{8}\right) + \left(\frac{2}{10} \times \frac{1}{16}\right)} = \frac{6}{6 + 5 + 1} = \frac{1}{2}. \end{aligned}$$

3.4 Independence

Definition 3.3 (Independence). Events E and F are independent if

$$\mathbb{P}(E \cap F) = \mathbb{P}(E)\mathbb{P}(F). \quad (30)$$

If two events are not independent, they are dependent.

Suppose that $\mathbb{P}(E) > 0$ and $\mathbb{P}(F) > 0$ and that E and F are independent. By (22) and (30) we have

$$\mathbb{P}(E | F) = \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(F)} = \frac{\mathbb{P}(E)\mathbb{P}(F)}{\mathbb{P}(F)} = \mathbb{P}(E),$$

and similarly,

$$\mathbb{P}(F | E) = \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(E)} = \frac{\mathbb{P}(E)\mathbb{P}(F)}{\mathbb{P}(E)} = \mathbb{P}(F).$$

Thus, Definition 3.3 of independence captures our intuition whereby E and F are independent if knowing one gives no information about the other: the conditional probability is equal to the unconditional probability, i.e., $\mathbb{P}(E | F) = \mathbb{P}(E)$ and $\mathbb{P}(F | E) = \mathbb{P}(F)$.

Remarks:

1. Assuming that $\mathbb{P}(E) > 0$ and $\mathbb{P}(F) > 0$, we could have equivalently defined independence by either $\mathbb{P}(E | F) = \mathbb{P}(E)$ or $\mathbb{P}(F | E) = \mathbb{P}(F)$, but Equation (30) is used as it treats events symmetrically and generalises naturally to many events as we shall see shortly.
2. Definition 3.3 allows us to consider the case when an event has zero probability. If $\mathbb{P}(E) = 0$ then E is independent of every event: $E \cap F \subset E$ so that $\mathbb{P}(E \cap F)\mathbb{P}(E) = 0$ and thus $\mathbb{P}(E \cap F) = 0 = \mathbb{P}(E)\mathbb{P}(F)$.

Example 3.7. A fair coin is tossed twice. Let E and F be the event of getting a head on the first and second toss, respectively.

Intuitively, if we know the coin to be fair, the outcome of the first toss has no influence on the second, $\mathbb{P}(F | E) = \mathbb{P}(F)$, so that $\mathbb{P}(E \cap F) = \mathbb{P}(E)\mathbb{P}(F)$.

This can be demonstrated formally. Let $\Omega = \{HH, HT, TH, TT\}$ be the sample space of equally likely outcomes. Then,

$$\mathbb{P}(E) = \frac{1}{2}, \quad \mathbb{P}(F) = \frac{1}{2}, \quad \text{and} \quad \mathbb{P}(E \cap F) = \frac{1}{4}.$$

Thus, $\mathbb{P}(E \cap F) = \frac{1}{4} = \mathbb{P}(E)\mathbb{P}(F)$ so that E and F are independent.

Example 3.8. Draw a card from a shuffled deck of cards. Let $E = \{\text{the card is an ace}\}$ and $F = \{\text{the suit of the card is hearts}\}$. Unlike in Example 3.7, here the two events are physically related (the same card). Are they independent?

Using equally likely events we have

$$\mathbb{P}(E) = \frac{4}{52} = \frac{1}{13}, \quad \mathbb{P}(F) = \frac{1}{4}, \quad \text{and} \quad \mathbb{P}(E \cap F) = \frac{1}{52}.$$

Thus, $\mathbb{P}(E \cap F) = \frac{1}{52} = \mathbb{P}(E)\mathbb{P}(F)$ and so E and F are independent. Note that $\mathbb{P}(E | F) = \mathbb{P}(E)$, so that learning the suit of a card does not give you any information about the rank.

Independence of events extends to the complementary events as well.

Theorem 3.4 (Independence of events extends to their complements). *If E and F are independent, then*

- (a) E and F^c are independent,
- (b) E^c and F are independent,
- (c) E^c and F^c are independent.

Proof. Idea: For part (a), use the partition rule to change a question about F^c into a question about F .

From the partition rule (Corollary 1.3), we have $\mathbb{P}(E) = \mathbb{P}(E \cap F) + \mathbb{P}(E \cap F^c)$, so that

$$\begin{aligned} \mathbb{P}(E \cap F^c) &= \mathbb{P}(E) - \mathbb{P}(E \cap F) \\ &= \mathbb{P}(E) - \mathbb{P}(E)\mathbb{P}(F) && \text{[by independence of } E \text{ and } F] \\ &= \mathbb{P}(E)(1 - \mathbb{P}(F)) = \mathbb{P}(E)\mathbb{P}(F^c). && \text{[probability of complements]} \end{aligned}$$

Hence we have (a). By symmetry we get (b), and (c) follows by applying (a) to (b). $\square$

Disjoint events are typically *not* independent!

If E and F are disjoint events, that is $E \cap F = \emptyset$, and both $\mathbb{P}(E) > 0$ and $\mathbb{P}(F) > 0$, then $\mathbb{P}(E)\mathbb{P}(F) > 0$. However, $\mathbb{P}(E \cap F) = 0$ so that E and F are dependent events.

Intuitively, if E and F are disjoint, then knowing that F has occurred tells us that E cannot have occurred, i.e., $\mathbb{P}(E | F) = 0 \neq \mathbb{P}(E) > 0$. Learning F changes the probability of E . Although this reasoning is quite intuitive, it is a common mistake to confuse independent and disjoint events.

3.5 Independence of many events

Be careful! If n events $E_1, \dots, E_n$ are independent then it is natural to expect that

$$\mathbb{P}(E_1 \cap \dots \cap E_n) = \mathbb{P}(E_1) \times \dots \times \mathbb{P}(E_n). \quad (31)$$

But also, if $E_1, \dots, E_n$ are independent, then it is also natural to expect that E_1 and E_2 are independent, and therefore

$$\mathbb{P}(E_1 \cap E_2) = \mathbb{P}(E_1)\mathbb{P}(E_2),$$

and this is **not** a consequence of (31). In the same way, we expect E_2 and E_3 to be independent, and E_1 , E_7 and E_{n-1} to be independent. In fact any combination of any number of the E_i for $i = 1, \dots, n$ should be independent, and each of these gives a different equation that needs to be satisfied.

This leads us to the following definition.

Definition 3.4 (Independence of many events). Events $E_1, \dots, E_n$ are said to be *independent* if for any $k \in \{2, 3, \dots, n\}$, and any $1 \leq i_1 < i_2 < \dots < i_k \leq n$, we have

$$\mathbb{P}(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_k}) = \mathbb{P}(E_{i_1}) \times \mathbb{P}(E_{i_2}) \times \dots \times \mathbb{P}(E_{i_k}).$$

Otherwise they are said to be *dependent*.

Similarly, events $E_1, E_2, \dots$ are said to be *independent* if for any $k \in \{2, 3, \dots\}$, and any $1 \leq i_1 < i_2 < \dots < i_k$, we have

$$\mathbb{P}(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_k}) = \mathbb{P}(E_{i_1}) \times \mathbb{P}(E_{i_2}) \times \dots \times \mathbb{P}(E_{i_k}).$$

Otherwise they are said to be *dependent*.

In other words, if we have countably many events $E_1, E_2, \dots$, they are independent if every finite sub-collection is independent.

Example 3.9. Events E_1, E_2, E_3 are independent if **all** of the following equalities hold:

$$\begin{array}{ll} \mathbb{P}(E_1 \cap E_2) = \mathbb{P}(E_1)\mathbb{P}(E_2) & [k = 2, i_1 = 1, i_2 = 2] \\ \mathbb{P}(E_1 \cap E_3) = \mathbb{P}(E_1)\mathbb{P}(E_3) & [k = 2, i_1 = 1, i_2 = 3] \\ \mathbb{P}(E_2 \cap E_3) = \mathbb{P}(E_2)\mathbb{P}(E_3) & [k = 2, i_1 = 2, i_2 = 3] \\ \mathbb{P}(E_1 \cap E_2 \cap E_3) = \mathbb{P}(E_1)\mathbb{P}(E_2)\mathbb{P}(E_3) & [k = 3, i_1 = 1, i_2 = 2, i_3 = 3] \end{array}$$

It is **not** enough to check that $\mathbb{P}(E_1 \cap E_2 \cap E_3) = \mathbb{P}(E_1)\mathbb{P}(E_2)\mathbb{P}(E_3)$, and it is also **not** enough to check pairwise independence (the first three equalities above). We absolutely need to check that **all four** of the equalities above hold! We will see this in the next two examples.

Example 3.10. Pick a card at random from a well-shuffled deck. Let $E_1 = \{\text{pick a red card}\}$, $E_2 = \{\text{pick a red ace or a black non-ace}\}$ and $E_3 = \{\text{pick the } A, J, Q \text{ or } K \text{ of diamonds}\}$. Then

$$\mathbb{P}(E_1) = \frac{26}{52} = \frac{1}{2}, \quad \mathbb{P}(E_2) = \frac{26}{52} = \frac{1}{2}, \quad \mathbb{P}(E_3) = \frac{4}{52} = \frac{1}{13}.$$

Also

$$\mathbb{P}(E_1 \cap E_2 \cap E_3) = \mathbb{P}(\text{pick } A \text{ of diamonds}) = \frac{1}{52} = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{13} = \mathbb{P}(E_1)\mathbb{P}(E_2)\mathbb{P}(E_3),$$

but the events are clearly not independent; for example,

$$\mathbb{P}(E_1 \cap E_3) = \mathbb{P}(E_3) = \frac{1}{13} \neq \mathbb{P}(E_1)\mathbb{P}(E_3),$$

i.e. knowing that E_3 holds tells you that E_1 holds, so in fact there is very strong dependence!

Example 3.11 (Example 3.7 revisited). Earlier we saw that if we toss a fair coin twice, $E = \{\text{head on first toss}\}$ and $F = \{\text{head on second toss}\}$ are independent. Consider a third event $G = \{\text{both tosses show the same side}\}$. In this case,

$$\mathbb{P}(G) = \mathbb{P}(HH) + \mathbb{P}(TT) = 1/2,$$

$$\mathbb{P}(E \cap G) = \mathbb{P}(HH) = 1/4$$

and

$$\mathbb{P}(F \cap G) = \mathbb{P}(HH) = 1/4.$$

Thus,

$$\mathbb{P}(E \cap G) = \mathbb{P}(E)\mathbb{P}(G), \quad \mathbb{P}(F \cap G) = \mathbb{P}(F)\mathbb{P}(G),$$

so that E and G are also independent as are F and G , i.e., E, F , and G are *pairwise* independent. However,

$$\mathbb{P}(E \cap F \cap G) = \mathbb{P}(HH) = \frac{1}{4} \neq \frac{1}{8} = \mathbb{P}(E)\mathbb{P}(F)\mathbb{P}(G),$$

so that E, F and G are not independent. Notice that $\mathbb{P}(G | E \cap F) = 1$, i.e., the events G and $E \cap F$ are not independent: knowing the result of each toss tells us the pair, so again we have very strong dependence!

Theorem 3.5 (Independence of complements, intersections and unions). *If $E_1, \dots, E_n$ are independent events, then*

- (a) $E_1^c, E_2, \dots, E_n$ are independent,
- (b) $E_1 \cap E_2, E_3, \dots, E_n$ are independent,
- (c) $E_1 \cup E_2, E_3, \dots, E_n$ are independent.

Proof. In each case we must show that the probability of the intersection of any sub-collection of the events is the product of the probabilities for each event. This follows immediately for any sub-collection that, respectively, does not include the events E_1^c , $E_1 \cap E_2$, $E_1 \cup E_2$ and so we only need to demonstrate this for the sub-collections including these events.

For part (a), for all $i_1 = 1 < i_2 < \dots < i_k \leq n$ we have

$$\begin{aligned}\mathbb{P}(E_1^c \cap E_{i_2} \cap \dots \cap E_{i_k}) &= \mathbb{P}(E_{i_2} \cap \dots \cap E_{i_k}) - \mathbb{P}(E_1 \cap E_{i_2} \cap \dots \cap E_{i_k}) \\ &= \mathbb{P}(E_{i_2}) \dots \mathbb{P}(E_{i_k}) - \mathbb{P}(E_1) \mathbb{P}(E_{i_2}) \dots \mathbb{P}(E_{i_k}) \\ &= (1 - \mathbb{P}(E_1)) \mathbb{P}(E_{i_2}) \dots \mathbb{P}(E_{i_k}) \\ &= \mathbb{P}(E_1^c) \mathbb{P}(E_{i_2}) \dots \mathbb{P}(E_{i_k})\end{aligned}$$

proving (a).

For part (b), for all $2 < i_2 < \dots < i_k \leq n$ we have

$$\begin{aligned}\mathbb{P}((E_1 \cap E_2) \cap E_{i_2} \cap \dots \cap E_{i_k}) &= \mathbb{P}(E_1) \mathbb{P}(E_2) \mathbb{P}(E_{i_2}) \dots \mathbb{P}(E_{i_k}) \\ &= \mathbb{P}(E_1 \cap E_2) \mathbb{P}(E_{i_2}) \dots \mathbb{P}(E_{i_k})\end{aligned}$$

proving (b).

For part (c), by two applications of (a), we see that $E_1^c, E_2^c, E_3, \dots, E_n$ are independent. Thus, by (b), $E_1^c \cap E_2^c, E_3, \dots, E_n$ are independent. From De Morgan's Laws (Theorem 1.1), $E_1^c \cap E_2^c = (E_1 \cup E_2)^c$ so that $(E_1 \cup E_2)^c, E_3, \dots, E_n$ are independent and thus, by another application of (a), $E_1 \cup E_2, E_3, \dots, E_n$ are independent. $\square$

Remarks:

1. Theorem 3.5 can be used repeatedly. For example, if $E_1, \dots, E_n$ are independent then, for example $E_1^c, E_2^c, \dots, E_n^c$ are independent as are $(E_1 \cup E_2) \cap E_3, E_4 \cup E_5, E_6^c, \dots, E_n$.
2. Theorem 3.5 also holds for countably infinite collections of events, and the proof is the same, since by Definition 3.4 we only need to check independence of any finite sub-collection.

Example 3.12. Three missiles are fired at a target which they hit independently with probabilities 0.7, 0.8, 0.9, respectively. What is the probability of the target being hit?

If $H_i = \{\text{the } i\text{th missile hits the target}\}$, then H_1, H_2, H_3 are independent and

$$\mathbb{P}(H_1) = 0.7, \quad \mathbb{P}(H_2) = 0.8, \quad \text{and} \quad \mathbb{P}(H_3) = 0.9.$$

If T is the event that the target is hit by at least one missile, then T^c is the event that all missiles miss. In this case,

$$\begin{aligned}\mathbb{P}(T) &= 1 - \mathbb{P}(T^c) \\ &= 1 - \mathbb{P}(H_1^c \cap H_2^c \cap H_3^c) \\ &= 1 - \mathbb{P}(H_1^c) \mathbb{P}(H_2^c) \mathbb{P}(H_3^c) && [\text{as } H_1^c, H_2^c, H_3^c \text{ are independent}] \\ &= 1 - 0.3 \times 0.2 \times 0.1 && [\text{as } \mathbb{P}(H_i^c) = 1 - \mathbb{P}(H_i)] \\ &= 0.994.\end{aligned}$$

Example 3.13. The electrical circuit in Figure 13 is made up of switches which are independently closed or open with probability p and $1 - p$, respectively. A signal is fed into the input. What is the probability of it being transmitted to the output?

Let $E_i = \{\text{the switch at location } i \text{ is closed.}\}$ where $i = 1, 2, 3, 4$. Then

$$\begin{aligned}\mathbb{P}(\{\text{transmitted}\}) &= \mathbb{P}((E_1 \cap E_2) \cup (E_3 \cap E_4)) \\ &= \mathbb{P}(E_1 \cap E_2) + \mathbb{P}(E_3 \cap E_4) - \mathbb{P}((E_1 \cap E_2) \cap (E_3 \cap E_4))\end{aligned} \tag{32}$$

$$\begin{aligned}&= \mathbb{P}(E_1) \mathbb{P}(E_2) + \mathbb{P}(E_3) \mathbb{P}(E_4) - \mathbb{P}(E_1) \mathbb{P}(E_2) \mathbb{P}(E_3) \mathbb{P}(E_4) \\ &= p^2 + p^2 - p^4 = p^2(2 - p^2),\end{aligned} \tag{33}$$

where Equation (33) follows from Equation (32) by the independence of E_1, E_2, E_3 and E_4 .

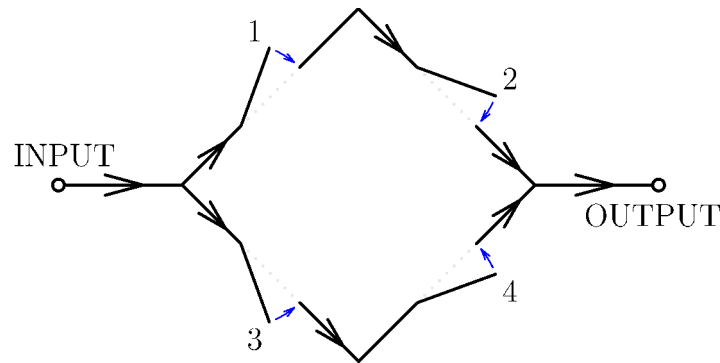


Figure 13: An electrical circuit: switches are independently closed with probability p .

3.6 Examples

3.6.1 Conditional probability animated

There is an animation showing conditional probability via balls landing on shelves at the setosa.io/conditional website.

3.6.2 Random walk

Recall the random walk from 2.15, which at each time step moves up by 1 with probability $1/2$, or down by 1 with probability $1/2$.

The following discussion is **non-examinable**.

Does the random walk return to 0 infinitely often? If so, we say it is *recurrent*.

Suppose the random walk is at position $x \in \mathbb{Z}$ after some time. We want to show that it definitely returns to 0 at some point in the future. What is the probability that it hits 0 before $2x$? It must be $1/2$ because the walk is symmetric, and can't stay trapped strictly between 0 and $2x$ forever. (Why not?)

If it hits 0 first then we are done. But maybe it hits $2x$ first. Then what is the probability it hits 0 before $4x$? Again, it must be $1/2$.

Again, if it hits 0 first then we are done, but maybe it hits $4x$ first. Then what is the probability it hits 0 before $8x$? Again, it must be $1/2$.

We continue in this way, getting *independent* (why?) trials each with probability $1/2$. The probability that the first k of these are all unsuccessful is $1/2^k$. For any k , we have

$$\mathbb{P}(\text{random walk never hits 0 starting from } x) \leq \mathbb{P}(\text{first } k \text{ trials are all unsuccessful}) = \frac{1}{2^k}.$$

Since this is true for any k , and $1/2^k$ is converging to 0 as $k \rightarrow \infty$, we must have

$$\mathbb{P}(\text{random walk never hits 0 starting from } x) = 0.$$

We deduce that **the random walk will hit 0 infinitely often, i.e. it is recurrent** - because wherever it ends up, it always has probability 1 of returning to 0.

Note that this doesn't tell us much about how long it takes to return to zero - each of the independent trials above might take a (very) long time to complete.

3.6.3 Prosecutor's fallacy: the costs of computing the wrong conditional probability (non-examinable)

See also what is the prosecutor's fallacy from Univeristy of Oxford and the Wikipedia page of the Sally Clark case.

A court needs to assess the guilt G or otherwise of a suspect of a crime, given evidence E . Then it is possible to compute the probability of the suspect being guilty or otherwise given the evidence as

$$\mathbb{P}(G | E) = \frac{\mathbb{P}(G \cap E)}{\mathbb{P}(E)} = \frac{\mathbb{P}(G)\mathbb{P}(E | G)}{\mathbb{P}(E)}$$

and

$$\mathbb{P}(G^c | E) = \frac{\mathbb{P}(G^c \cap E)}{\mathbb{P}(E)} = \frac{\mathbb{P}(G^c)\mathbb{P}(E | G^c)}{\mathbb{P}(E)}.$$

What could happen if $\mathbb{P}(E | G)$ is wrongly interpreted as $\mathbb{P}(G | E)$?

3.6.3.1 Sally Clark case (UK, 1999) Sally Clark's first son died in December 1996 within a few weeks of his birth of unexplained causes (E_1), and her second son died in similar circumstances in January 1998 (E_2). A month later, Clark was arrested and tried for both deaths.

Her defence said that both children died of sudden infant death syndrome (SIDS), while a paediatrician, as expert witness, testified that the chance of one child from an affluent family suffering SIDS was 1 in 8500 and concluded that the chance of two children from an affluent family suffering SIDS was 1 in 73 million, i.e.

$$\mathbb{P}(E_1 \cap E_2 | G^c) = \mathbb{P}(E_1 | G^c) \mathbb{P}(E_2 | G^c) = \left(\frac{1}{8500}\right)^2 \approx 1 \text{ in } 73 \text{ million.}$$

Also, the jury mistook $\mathbb{P}(E_1 \cap E_2 | G^c)$ with the probability that Clark was innocent, $\mathbb{P}(G^c | E_1 \cap E_2)$, and concluded that

$$\mathbb{P}(G | E_1 \cap E_2) = 1 - \mathbb{P}(G^c | E_1 \cap E_2) \approx 1 - \frac{1}{73,000,000} \approx 1,$$

so Clark was guilty “beyond reasonable doubt” and convicted for murder in November 2022.

There were both wrong assumptions and probabilistic errors involved in this case.

The first error was made by the paediatrician: medical literature showed that, if a first child dies of SIDS then a second child from the same family is significantly more likely to die of SIDS. In other words, susceptibility to SIDS is inherited, and therefore

$$\mathbb{P}(E_1 \cap E_2 | G^c) \neq \mathbb{P}(E_1 | G^c) \mathbb{P}(E_2 | G^c)$$

- we do not have independent events. If we suppose that the probability of the second child dying of SIDS given the first child died by SIDS, $\mathbb{P}(E_2 | E_1 \cap G^c)$, is almost 0.1, then

$$\mathbb{P}(E_1 \cap E_2 | G^c) = \mathbb{P}(E_1 | E_2 \cap G^c) \mathbb{P}(E_1 | G^c) \approx \frac{1}{10} \times \frac{1}{8500} = \frac{1}{85,000} \gg \frac{1}{73,000,000}.$$

The second error was made by the jury confusing $\mathbb{P}(E_1 \cap E_2 | G^c)$ with $\mathbb{P}(G^c | E_1 \cap E_2)$ – they should have used Bayes' rule to compute the latter probability.

What would a proper analysis look like?

The first step is to understand how likely the suspect is to be guilty based on just the first unexplained death. One statistic, obtained by surveying death certificates in the UK, is that of children who die in ways unexplained by medicine, fewer than 1 in 11 are subsequently discovered to have been murdered. Therefore we might assume that $\mathbb{P}(G | E_1) \leq \frac{1}{11}$.

Recall that we also believe that $\mathbb{P}(E_2 | E_1 \cap G^c) \approx \frac{1}{10}$.

We can then try to estimate $\mathbb{P}(G | E_1 \cap E_2)$. By Bayes' theorem (using the partition $\{G, G^c\}$, and working conditional on E_1 throughout),

$$\mathbb{P}(G | E_1 \cap E_2) = \frac{\mathbb{P}(G | E_1)\mathbb{P}(E_2 | G \cap E_1)}{\mathbb{P}(G | E_1)\mathbb{P}(E_2 | G \cap E_1) + \mathbb{P}(G^c | E_1)\mathbb{P}(E_2 | G^c \cap E_1)}.$$

Dividing through by the numerator, we get $\mathbb{P}(G | E_1 \cap E_2) = \frac{1}{1+F}$, where

$$F = \frac{\mathbb{P}(G^c | E_1)\mathbb{P}(E_2 | G^c \cap E_1)}{\mathbb{P}(G | E_1)\mathbb{P}(E_2 | G \cap E_1)}.$$

If $\mathbb{P}(G | E_1) \leq 1/11$ then $\mathbb{P}(G^c | E_1) \geq 10/11$ so

$$\frac{\mathbb{P}(G^c | E_1)}{\mathbb{P}(G | E_1)} \geq 10,$$

and if $\mathbb{P}(E_2 | E_1 \cap G^c) \gtrsim \frac{1}{10}$ then, since certainly $\mathbb{P}(E_2 | G \cap E_1) \leq 1$ (probabilities are always between 0 and 1!),

$$\frac{\mathbb{P}(E_2 | E_1 \cap G^c)}{\mathbb{P}(E_2 | E_1 \cap G)} \gtrsim \frac{1}{10}.$$

Thus $F \gtrsim 1$ and therefore

$$\mathbb{P}(G | E_1 \cap E_2) \lesssim \frac{1}{2}.$$

In the face of this evidence, Clark was not guilty beyond reasonable doubt. We made some assumptions here; with different opinions on the true values of $\mathbb{P}(G | E_1)$ and $\mathbb{P}(E_2 | E_1 \cap G^c)$ we would have to change our calculation, but any reasonable values would still give a value of $\mathbb{P}(G | E_1 \cap E_2)$ that is much, much larger than 1 in 73 million.

4 Discrete random variables

4.1 Real-valued random variables

Consider tossing a coin n times. In order to capture all the information from the n tosses, the sample space must have 2^n elements. However, we might only be interested in some aspects of the information, for example the number of heads obtained over the n tosses. We can encode this information in a *random variable*.

Despite the name, a random variable is neither random nor a variable. It is a function from the sample space to (usually) a subset of the real numbers. That is, $X : \Omega \rightarrow S$ for some $S \subset \mathbb{R}$. For example, X could be the number of heads obtained over n tosses, as mentioned above; then $X : \Omega \rightarrow \{0, 1, 2, \dots, n\}$.

Definition 4.1 (Random variable). A (real-valued) *random variable* (rv) X is a function that assigns a real-valued number to each possible outcome of an experiment, that is X is a mapping $X : \Omega \rightarrow S \subset \mathbb{R}$.

The set S is called the *support*⁹ of X .

Remarks:

1. We typically denote random variables by capital letters such as X , Y , and Z . Prior to our experiment, we imagine that the input of the function, i.e. the point ω in our sample space, is “unknown” (the result of n coin tosses for example). This means that the output $X(\omega)$ is also unknown. Lower case letters, such as x , y and z , are used for possible numerical outcomes, the real numbers in S .
2. If we have a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, then for a random variable X , we can define events of interest such as

$$\begin{aligned} \{X = x\} &= \{\omega \in \Omega : X(\omega) = x\}, \\ \{X \leq x\} &= \{\omega \in \Omega : X(\omega) \leq x\}, \\ \{X \in A\} &= \{\omega \in \Omega : X(\omega) \in A\}. \end{aligned}$$

We use the more compact notation on the left hand side frequently throughout the course, even though the right hand side is really what we mean. Moreover, we often drop the braces when referring to the probability of such events, i.e., write $\mathbb{P}(X \in A)$ instead of $\mathbb{P}(\{X \in A\})$ or even $\mathbb{P}(\{\omega \in \Omega : X(\omega) \in A\})$. These all mean the same thing.

Example 4.1. Consider tossing a fair coin three times. The sample space is

$$\Omega = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}.$$

⁹In fact, the support S^I of X is the smallest set such that $\mathbb{P}(X \in S^I) = 1$, but in practice in this course we will have $S^I = S$.

Let X denote the number of heads obtained in the three tosses. For each $\omega \in \Omega$ we can find $X(\omega)$ as follows:

ω	HHH	HHT	HTH	THH	HTT	THT	TTH	TTT
$X(\omega)$	3	2	2	2	1	1	1	0

Moreover,

$$\mathbb{P}(X = 1) = \mathbb{P}(\{\omega \in \Omega : X(\omega) = 1\}) = \mathbb{P}(\{HTT, THT, TTH\}) = 3/8,$$

and

$$\mathbb{P}(X \geq 2) = \mathbb{P}(\{\omega \in \Omega : X(\omega) \geq 2\}) = \mathbb{P}(\{HHH, HHT, HTH, THH\}) = 4/8 = 1/2.$$

We can define many random variables on Ω . For example, we could let Y denote the number of tails obtained in the three tosses (i.e., $Y = 3 - X$) or let Z denote the number of heads obtained on the last two tosses, or let W denote the number of times we see either a head followed by a tail or a tail followed by a head. In this case X , Y , Z and W are all random variables defined on the same probability space.

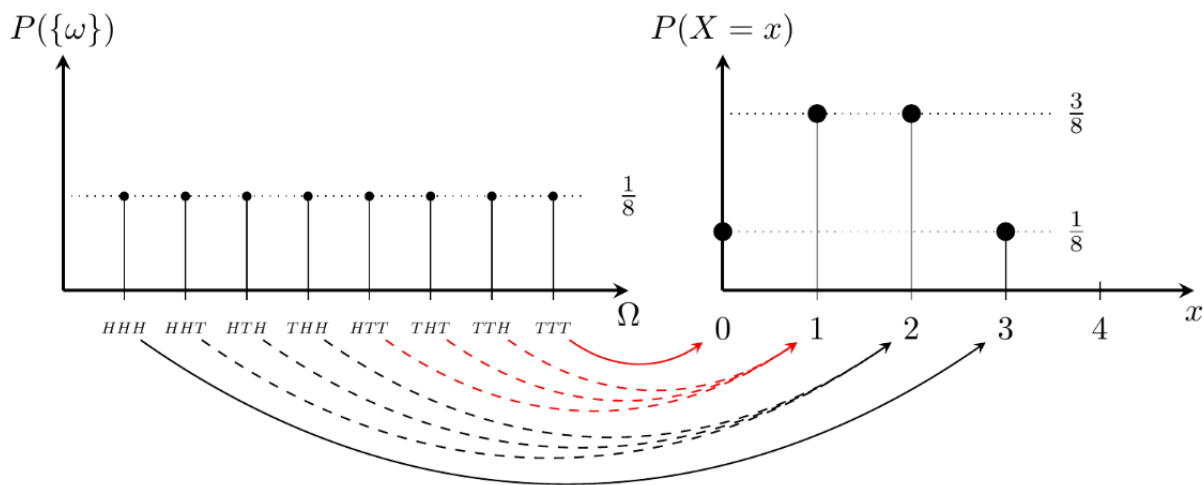


Figure 14: Random variable counting the number of heads when we toss a coin three times, as a function from the sample space to the real numbers.

4.1.1 Types of random variables: discrete and continuous

Definition 4.2 (Discrete RVs). A random variable X is discrete if it has a finite or countably infinite number of possible values, i.e. either

- (a) the support S is finite, so $S = \{x_1, x_2, \dots, x_n\}$ for some $x_1, x_2, \dots, x_n \in \mathbb{R}$, or
- (b) the support S is countably infinite so $S = \{x_1, x_2, \dots\}$ for some $x_1, x_2, \dots \in \mathbb{R}$.

Example 4.2. Examples of discrete random variables:

1. The number of heads obtained in three tosses of a fair coin described in Example 4.1 is a discrete random variable, with $S = \{0, 1, 2, 3\}$ finite.
2. The number of meteorites greater than 1 meter diameter that strike Earth in a year is a discrete random variable, with $S = \mathbb{Z}^+ = \{0, 1, 2, 3, \dots\}$.

Definition 4.3 (Continuous RVs). A random variable X is continuous if it has an uncountably infinite number of possible values, i.e., S is an un-countably infinite subset of $\mathbb{R}$. In particular, $S \subset \mathbb{R}$ could be an interval such as $(0, +\infty)$, $(-\infty, +\infty)$, or $(a, b]$ for some $a < b \in \mathbb{R}$.

Example 4.3. Examples of continuous random variables:

1. The time until the next person in the lecture hall sneezes could be modelled as a continuous random variable taking values in $S = [0, \infty)$.
2. The percentage of time that Nottingham Forest have possession of the ball in their next match could be modelled as a continuous random variable taking values in $S = [0, 100]$.

4.1.2 Probability mass function for discrete random variables

We saw in Theorem 1.3 that to describe a finite or countable probability space, it was enough to specify a collection $(p_1, p_2, \dots, p_n)$ or $(p_1, p_2, \dots)$ of non-negative numbers that sum to 1. By the same logic, to describe a discrete random variable X , it is enough to specify the non-negative numbers $\mathbb{P}(X = x)$ for each $x \in S$. These numbers are encoded in what we call the *probability mass function* of the random variable.

Definition 4.4 (Probability mass function). The *probability mass function* (pmf) $f_X : \mathbb{R} \rightarrow [0, 1]$ of a discrete random variable $X : \Omega \rightarrow S$ is defined for all $x \in \mathbb{R}$ as

$$f_X(x) = \mathbb{P}(X = x).$$

Thus $\sum_{x \in S} \mathbb{P}(X = x) = 1$ and $f_X(x) = 0$ for all $x \notin S$.

Example 4.4 (Example 4.1 revisited). We saw in Example 4.1 how X maps the equally probable sample points of Ω to $S = \{0, 1, 2, 3\} \subset \mathbb{R}$. These points are not equally likely but have the probabilities calculated in Example 4.1.

The pmf of X is

$$f_X(x) = \begin{cases} 1/8 & \text{if } x = 0, \\ 3/8 & \text{if } x = 1, \\ 3/8 & \text{if } x = 2, \\ 1/8 & \text{if } x = 3, \\ 0 & \text{if } x \notin \{0, 1, 2, 3\}. \end{cases}$$

We can check this satisfies the properties of a pmf:

1. $f_X(x) \geq 0$ for all $x \in \mathbb{R}$, since $f_X(x) = 0$ if $x \notin S = \{0, 1, 2, 3\}$ and $f_X(x) > 0$ if $x \in S = \{0, 1, 2, 3\}$.
2. $\sum_{x \in S} \mathbb{P}(X = x) = 1$, since

$$\begin{aligned} \sum_{x \in S} \mathbb{P}(X = x) &= \sum_{x \in \{0, 1, 2, 3\}} \mathbb{P}(X = x) \\ &= \sum_{i=0}^3 \mathbb{P}(X = i) \\ &= \mathbb{P}(X = 0) + \mathbb{P}(X = 1) + \mathbb{P}(X = 2) + \mathbb{P}(X = 3) \\ &= \frac{1}{8} + \frac{3}{8} + \frac{3}{8} + \frac{1}{8} = 1 \end{aligned}$$

Remark:

We can in fact think of the pmf f_X as *inducing* a new probability measure, $\mathbb{P}_X$, with sample space S and σ -algebra $\mathcal{F} = \mathcal{P}(S)$. We know this new probability measure will satisfy Kolmogorov's axioms by Theorem 1.3. We call $\mathbb{P}_X$ the *distribution* or *law* of X ; for any $A \subset S$ we have

$$\mathbb{P}_X(A) = \mathbb{P}(X \in A). \quad (34)$$

This is more complicated for continuous random variables as we can't define a (useful) probability mass function, since usually we will have $\mathbb{P}(X = x) = 0$ for all $x \in S$. But it is still possible to induce a probability measure $\mathbb{P}_X$, using (34) as the definition. (This is beyond the scope of this course since we don't have an equivalent of Theorem 1.3 for uncountable state spaces.) The point is that instead of looking at the probability that X is a certain value, we look at the probability that it is in a certain *set*.

Indeed, for *any* random variable, even if $\mathbb{P}(X = x)$ isn't useful, the quantity $\mathbb{P}(X \leq x)$ is always useful (and corresponds to choosing the set $A = (-\infty, x]$ in (34)). This is the idea behind the *cumulative distribution function* or cdf.

4.2 Cumulative distribution functions

The distribution of any random variable can be characterised by assigning probabilities to the events $\mathbb{P}(X \leq x)$, for all $x \in \mathbb{R}$.

Definition 4.5 (Cumulative distribution function). The *cumulative distribution function* (cdf) $F : \mathbb{R} \rightarrow [0, 1]$ is defined for any real valued random variable $X : \Omega \rightarrow \mathbb{R}$ as

$$F_X(x) = \mathbb{P}(X \leq x).$$

Sometimes people drop the word “cumulative” and just call this the *distribution function*.

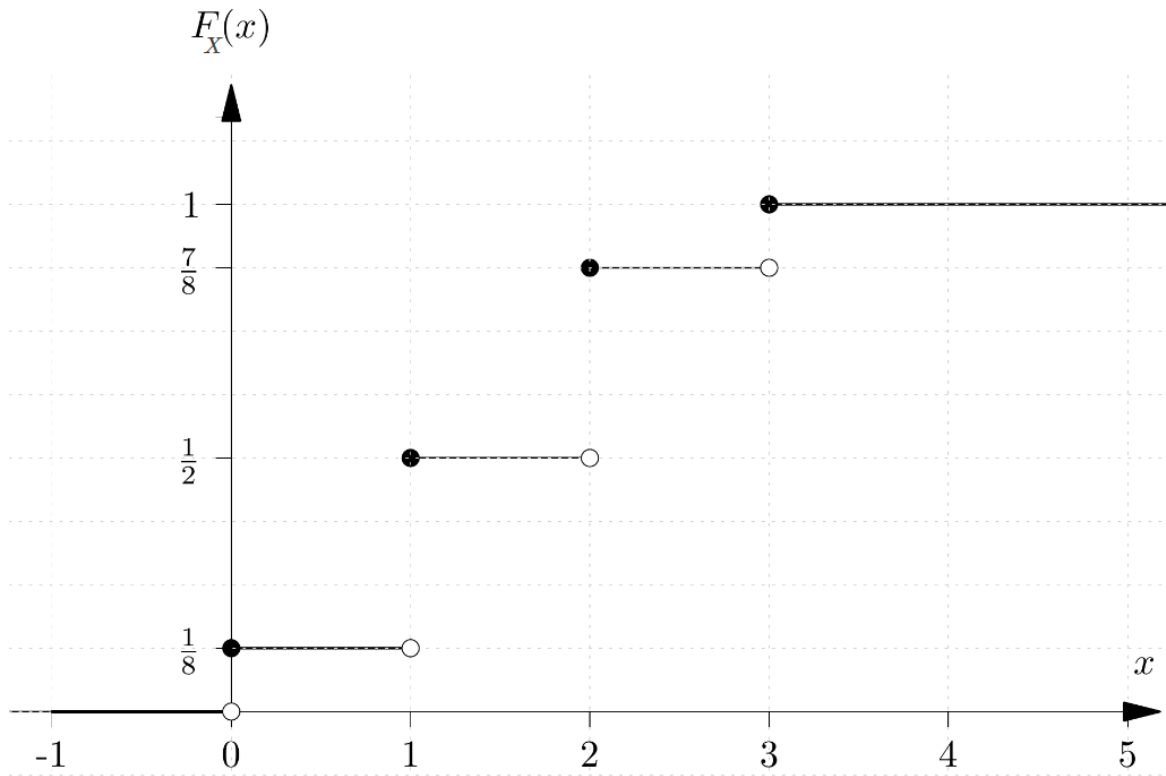


Figure 15: The cdf of the random variable X counting the number of heads in 3 tosses of a fair coin.

Example 4.5 (Example 4.4 revisited). Figure 15 shows the distribution function of X , where X is the random variable counting the number of heads in 3 tosses of a fair coin from Example 4.1. For example,

$$\begin{aligned}
 F_X(2) &= \mathbb{P}(X \leq 2) \\
 &= \mathbb{P}(\{X = 0\} \cup \{X = 1\} \cup \{X = 2\}) \\
 &= \mathbb{P}(X = 0) + \mathbb{P}(X = 1) + \mathbb{P}(X = 2) && [\text{disjoint events}] \\
 &= \frac{1}{8} + \frac{3}{8} + \frac{3}{8} = \frac{7}{8}.
 \end{aligned}$$

By Definition 4.5, if $x \leq y$, then $\{X \leq x\} \subset \{X \leq y\}$, and by the containment rule

$$F_X(x) = \mathbb{P}(X \leq x) \leq \mathbb{P}(X \leq y) = F_X(y),$$

i.e., F_X is always **non-decreasing**. Suppose that $X : \Omega \rightarrow S$ is a discrete random variable with $S = \{x_1, x_2, \dots, x_n\}$, with $x_1 < x_2 < \dots < x_n$. If $x \in \mathbb{R}$ is such that $x_k \leq x < x_{k+1}$ for some $k \in \{1, \dots, n\}$,

then

$$\begin{aligned}
F_X(x) &= \mathbb{P}(X \leq x) = \mathbb{P}(X \leq x_k) \\
&= \mathbb{P}(\{X = x_1\} \cup \{X = x_2\} \cup \dots \cup \{X = x_k\}) \\
&= \mathbb{P}(X = x_1) + \mathbb{P}(X = x_2) + \dots + \mathbb{P}(X = x_k) \quad [\text{disjoint events}] \\
&= \sum_{i=1}^k \mathbb{P}(X = x_i).
\end{aligned}$$

If $x > x_n$, then we have similarly

$$F_X(x) = \sum_{i=1}^n \mathbb{P}(X = x_i) = \sum_{x \in S} \mathbb{P}(X = x) = 1,$$

and if $x < x_1$ we have $\{X \leq x\} \subset \{X < x_1\}$, and thus

$$F_X(x) = \mathbb{P}(X \leq x) \leq \mathbb{P}(X < x_1) = 1 - \mathbb{P}(X \geq x_1) = 1 - \sum_{i=1}^n \mathbb{P}(X = x_i) = 1 - 1 = 0.$$

We collect these observations into a single theorem statement which is universally true for **all** random variables. A rigorous proof is omitted as, although it is not too difficult, it would require the analysis of convergent sequences, which you will not have learnt yet in your Analysis course.

Theorem 4.1 (Properties of cdfs). *For any random variable X , its distribution function F_X satisfies:*

1. For all $x < y$, $F_X(x) \leq F_X(y)$, i.e., F_X is non-decreasing.
 2. As x increases to $+\infty$, $F_X(x)$ approaches 1, i.e., $\lim_{x \rightarrow +\infty} F_X(x) = F_X(+\infty) = 1$.
 3. As x decreases to $-\infty$, $F_X(x)$ approaches 0, i.e., $\lim_{x \rightarrow -\infty} F_X(x) = F_X(-\infty) = 0$.
 4. As x approaches $x_0 \in \mathbb{R}$ from the right, $F_X(x)$ approaches $F_X(x_0)$, i.e., F_X is right-continuous.
- (Non-examinable)

Remark:

We do not have the mathematical tools to discuss property 4 rigorously (yet). For this course we can simply observe that in the case of a discrete random variable X with $S = \{x_1, \dots, x_n\}$, where $x_1 < x_2 < \dots < x_n$ are in increasing order,

$$F_X(x) = \mathbb{P}(X \leq x) = \mathbb{P}(X \leq x_k)$$

holds for all $x_k \leq x < x_{k+1}$, i.e. all the way up to x_{k+1} **but not for** $x = x_{k+1}$ itself. In Figure 15, this is highlighted by the black bullet markers.

Theorem 4.2 (Probabilities of intervals). *Let X be a real-valued random variable with cdf F_X . Then for any $a, b \in \mathbb{R}$ such that $a \leq b$, we have*

$$\mathbb{P}(X \in (a, b]) = F_X(b) - F_X(a),$$

(where $(a, a] = \emptyset$).

Proof. By using the cdf F_X , we can obtain the probabilities for the event that X takes a value in the interval $(a, b]$ as

$$\begin{aligned}
\mathbb{P}(X \in (a, b]) &= \mathbb{P}(\{X > a\} \cap \{X \leq b\}) \\
&= 1 - \mathbb{P}(\{X \leq a\} \cup \{X > b\}) && [\text{complement + De Morgan}] \\
&= 1 - \mathbb{P}(\{X \leq a\}) - \mathbb{P}(\{X > b\}) && [\text{disjoint}] \\
&= \mathbb{P}(X \leq b) - \mathbb{P}(X \leq a) && [\text{complement}] \\
&= F_X(b) - F_X(a),
\end{aligned}$$

completing the proof. □

Before concrete examples of well-known distributions, we need to state one more fundamental result which essentially provides a method for defining continuous random variables, and distributions, rigorously.

Theorem 4.3 (Defining cdfs). *For any F which has the properties 1–4 of Theorem 4.1, there exists a random variable X whose cdf is F , i.e., for all $x \in \mathbb{R}$, $\mathbb{P}(X \leq x) = F_X(x)$.*

Proof. Omitted. □

Thus we can take any function having properties 1–4 of Theorem 4.1, and be assured that it will be the distribution function of some random variable.

Example 4.6. *Take the cdf*

$$F_X(x) = \begin{cases} 0 & x < 0, \\ \frac{1}{8} & 0 \leq x < 1, \\ \frac{1}{2} & 1 \leq x < 2, \\ \frac{7}{8} & 2 \leq x < 3, \\ 1 & x \geq 3, \end{cases}$$

illustrated in Figure 15. How can we obtain the pmf from the cdf?

We see that for any $x < 0$, $F_X(x) = \mathbb{P}(X \leq x) = 0$, from which we conclude that $\mathbb{P}(X < 0) = 0$, but $F_X(0) = \mathbb{P}(X \leq 0) = \frac{1}{8}$, so by the containment rule, as $\{X = 0\} \subset \{X \leq 0\}$, we have

$$\begin{aligned} \mathbb{P}(X = 0) &= \mathbb{P}(X \leq 0) - \mathbb{P}(\{X \leq 0\} \cap \{X \neq 0\}) \\ &= \mathbb{P}(X \leq 0) - \mathbb{P}(X < 0) = \frac{1}{8} - 0 = \frac{1}{8}. \end{aligned}$$

Similarly we can conclude that

$$\begin{aligned} \mathbb{P}(X = 1) &= \mathbb{P}(X \leq 1) - \mathbb{P}(\{X \leq 1\} \cap \{X \neq 1\}) \\ &= \mathbb{P}(X \leq 1) - \mathbb{P}(X < 1) \\ &= \frac{1}{2} - \frac{1}{8} = \frac{3}{8}, \end{aligned}$$

and then similarly that $\mathbb{P}(X = 2) = \frac{3}{8}$, $\mathbb{P}(X = 3) = \frac{1}{8}$, and that for any $x \notin \{0, 1, 2, 3\}$, $\mathbb{P}(X = x) = 0$, which gives back the pmf in Example 4.4.

4.3 Common discrete random variables

We introduce some common discrete probability distributions. (Note that this can be done by giving the probability mass function, without any mention of the underlying probability space $(\Omega, \mathcal{F}, \mathbb{P})$.)

4.3.1 Bernoulli random variables

A particularly simple experiment is one in which there are only two possible outcomes. Examples of such pairs of random outcomes are heads or tails, success or failure, defective or not defective, and so on. It is convenient to label the two possible outcomes as 1 (usually the success) and 0 (usually the failure). Such an experiment is called a **Bernoulli**¹⁰ **trial**.

Definition 4.6 (Bernoulli distribution). A discrete random variable X is said to have the *Bernoulli* distribution with parameter $p \in [0, 1]$, written as $X \sim \text{Ber}(p)$, if its pmf is given by

$$f_X(x) = \mathbb{P}(X = x) = \begin{cases} 1 - p & \text{for } x = 0, \\ p & \text{for } x = 1, \\ 0 & \text{otherwise.} \end{cases} \quad (35)$$

That is, $\mathbb{P}(X = 1) = p$ and $\mathbb{P}(X = 0) = 1 - p$. The statement $X \sim \text{Ber}(p)$ should be read as “the random variable X has the Bernoulli distribution with parameter p ” so that “ $\sim$ ” corresponds to “has the distribution”.

¹⁰Jacob Bernoulli (1654 - 1705).

Notice that if $Y = 1 - X$ then

$$\begin{aligned}\mathbb{P}(Y = 1) &= \mathbb{P}(1 - X = 1) = \mathbb{P}(X = 0) = 1 - p \\ \mathbb{P}(Y = 0) &= \mathbb{P}(1 - X = 0) = \mathbb{P}(X = 1) = p,\end{aligned}$$

implying $Y \sim \text{Ber}(1 - p)$.

It is straightforward to compute the cumulative distribution function of a Bernoulli distribution: given $X \sim \text{Ber}(p)$,

$$F_X(x) = \mathbb{P}(X \leq x) = \begin{cases} 0 & \text{for } x < 0, \\ 1 - p & \text{for } 0 \leq x < 1, \\ 1 & \text{for } x \geq 1. \end{cases} \quad (36)$$

4.3.2 Binomial random variables

The Binomial distribution models the number of successes in n independent Bernoulli trials, each trial having the same success probability $p \in [0, 1]$. Examples of such collections of random outcomes are the number of heads (or tails) in n tosses of a coin, the number of successes or failures out of n trials, the number of defective (or non-defective) items out of n total items, and so on.

Definition 4.7 (Binomial distribution). A discrete random variable X is said to have the *binomial* distribution with parameters $n \in \mathbb{N}$ and $p \in [0, 1]$, written as $X \sim \text{Bin}(n, p)$, if its pmf is given by

$$f_X(x) = \mathbb{P}(X = x) = \begin{cases} \binom{n}{x} p^x (1 - p)^{n-x} & \text{for } x \in \{0, 1, \dots, n\}, \\ 0 & \text{otherwise.} \end{cases} \quad (37)$$

Probability mass functions of $\text{Bin}(n, p)$ for $n = 100$ and $p = 1/10, 1/2$ and $9/10$ are shown in Figure 16.

Example 4.7. Three components are made with each component independently being defective with probability $1/4$. What is the probability of at least two of them being defective?

There are $n = 3$ independent Bernoulli trials and each trial is a success (i.e., “defective”) with probability $p = 1/4$. If $X = \{\text{Number of defective components}\}$, then we have the model $X \sim \text{Bin}(3, 1/4)$. Hence,

$$\begin{aligned}\mathbb{P}(X \geq 2) &= \mathbb{P}(\{X = 2\} \cup \{X = 3\}) \\ &= \mathbb{P}(X = 2) + \mathbb{P}(X = 3) && \text{[disjoint events]} \\ &= \binom{3}{2} \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^1 + \binom{3}{3} \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^0 && \text{[} X \sim \text{Bin}(3, 1/4) \text{]} \\ &= 3 \times \frac{1}{16} \times \frac{3}{4} + \frac{1}{64} = \frac{5}{32}.\end{aligned}$$

Remarks:

1. From the binomial expansion, i.e.,

$$(a + b)^n = \sum_{x=0}^n \binom{n}{x} a^x b^{n-x},$$

by setting $a = p$ and $b = 1 - p$, we have

$$\sum_{x=0}^n \mathbb{P}(X = x) = \sum_{x=0}^n \binom{n}{x} p^x (1 - p)^{n-x} = (p + (1 - p))^n = 1.$$

2. If $n = 1$, we have a single Bernoulli trial, i.e., $X \sim \text{Bin}(1, p)$ is equivalent to $X \sim \text{Ber}(p)$.
3. If $Y_1, \dots, Y_n$ are independent and $Y_i \sim \text{Bin}(1, p)$ for all $i \in \{1, \dots, n\}$, then

$$X = \sum_{i=1}^n Y_i \sim \text{Bin}(n, p).$$

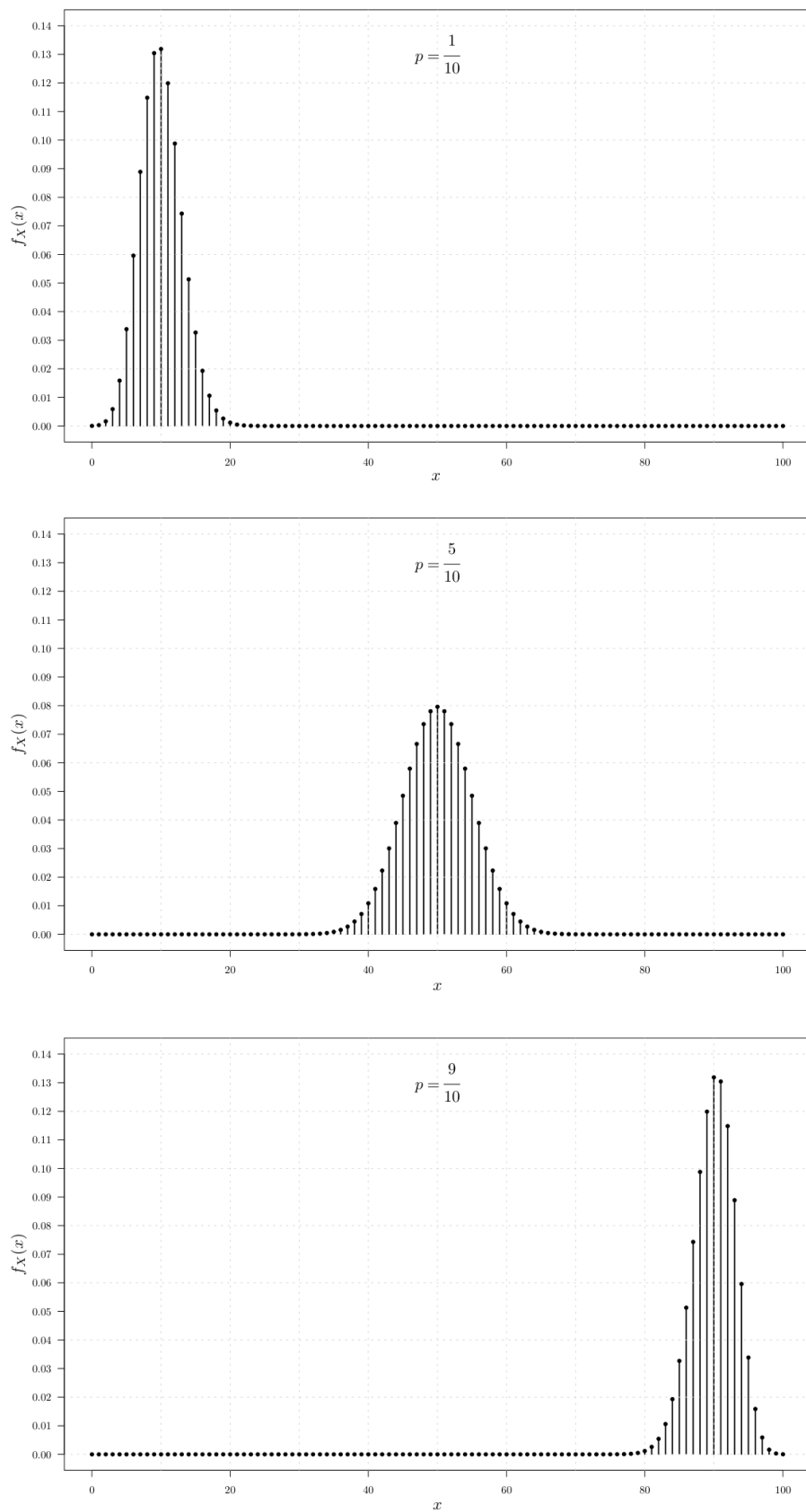


Figure 16: The pmf of a $\text{Bin}(n, p)$ with $n = 100$ and $p = 1/10, 5/10, 9/10$.

4. If the event $\{X = x\}$ occurs, then there must have been x successes and $n - x$ failures in the n trials. As we have seen earlier, there are $\binom{n}{x}$ ways of choosing x out of n trials to be the successful ones (the order in which we choose them does not matter). Once we have chosen which x of the trials should be successful, and thus which $n - x$ should be unsuccessful, the probability of this occurring is $p^x(1 - p)^{n-x}$.
5. If $X \sim \text{Bin}(n, p)$ and $Y = n - X$ then $Y \sim \text{Bin}(n, 1 - p)$: X is the number of successes in n trials and Y is the number of failures in n trials with $X + Y = n$.
6. Knowing the formula for the pmf, it is straightforward to compute also the cdf of $X \sim \text{Bin}(n, p)$: for $x \in \mathbb{R}$, if we write $\lfloor x \rfloor$ for the largest integer smaller than or equal to x , i.e. x “rounded down” to the nearest integer, then

$$F_X(x) = \mathbb{P}(X \leq x) = \begin{cases} 0 & \text{for } x < 0, \\ \sum_{i=0}^{\lfloor x \rfloor} \binom{n}{i} p^i (1 - p)^{n-i} & \text{for } 0 \leq x < n, \\ 1 & \text{for } x \geq n. \end{cases} \quad (38)$$

4.3.3 Geometric random variables

The geometric distribution models the number of independent Bernoulli trials **up to and including** the first success. Examples of such collections of random outcomes are the number of coin tosses until the first head, the number of tested items until you find the first defective one, and so on.

Definition 4.8 (Geometric distribution). A discrete random variable X is said to have the *geometric* distribution with parameter $p \in (0, 1)$, written $X \sim \text{Geom}(p)$, if its pmf is given by

$$f_X(x) = \mathbb{P}(X = x) = \begin{cases} (1 - p)^{x-1} p & \text{for } x \in \{1, 2, 3, \dots\} \\ 0 & \text{otherwise.} \end{cases} \quad (39)$$

Probability mass functions of $\text{Geom}(p)$ for $p = 1/10$, $1/2$ and $9/10$ are illustrated in Figure 17.

Remarks:

1. If the event $\{X = x\}$ occurs, then the first $x - 1$ trials must have been failures, each independently with probability $1 - p$, and the x th trial is a success, independently with probability p .
2. There are alternative formulations of the geometric distribution. For example, we might let Y be the number of independent Bernoulli trials **strictly before** the first success. In this case

$$\mathbb{P}(Y = y) = (1 - p)^y p$$

for each $y \in \{0, 1, 2, \dots\}$. Notice that if $X \sim \text{Geom}(p)$, then $Y = X - 1$.

Unlike for the Bernoulli or Binomial distribution, the set of possible values for a Geometric distribution is countably infinite. This makes the verification of $\sum_{x=1}^{\infty} \mathbb{P}(X = x) = 1$ a bit more complicated. To this end, we need the following geometric series theorem (which you may have met before).

Theorem 4.4 (Geometric series Theorem). *For any $r \neq 1$ we have*

$$\sum_{i=0}^n r^i = \frac{1 - r^{n+1}}{1 - r}, \quad (40)$$

and if $|r| < 1$, then

$$\sum_{i=0}^{\infty} r^i = 1 + r + r^2 + r^3 + \dots = \frac{1}{1 - r}. \quad (41)$$

Proof. The proof is omitted. □

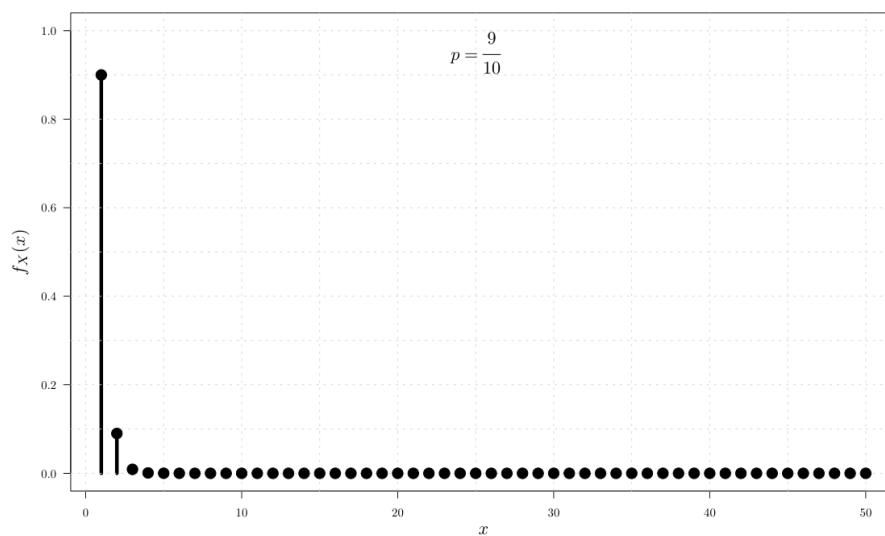
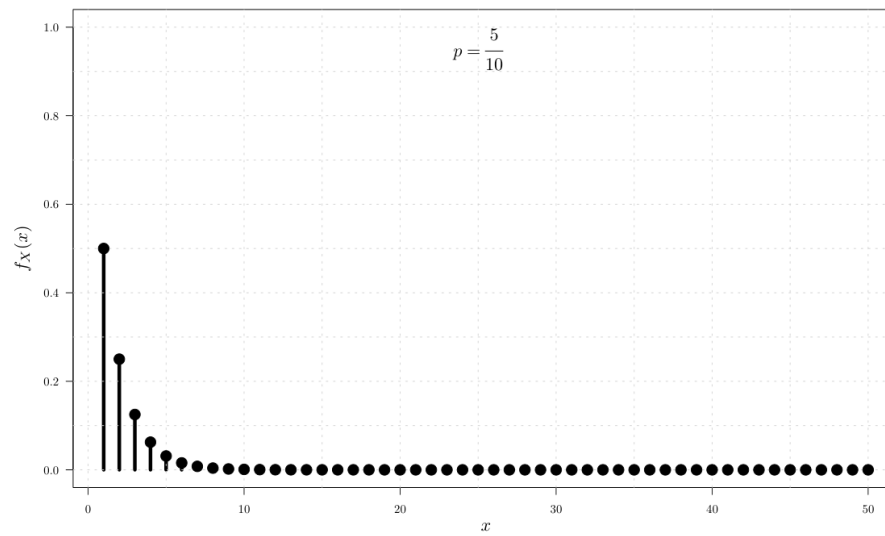
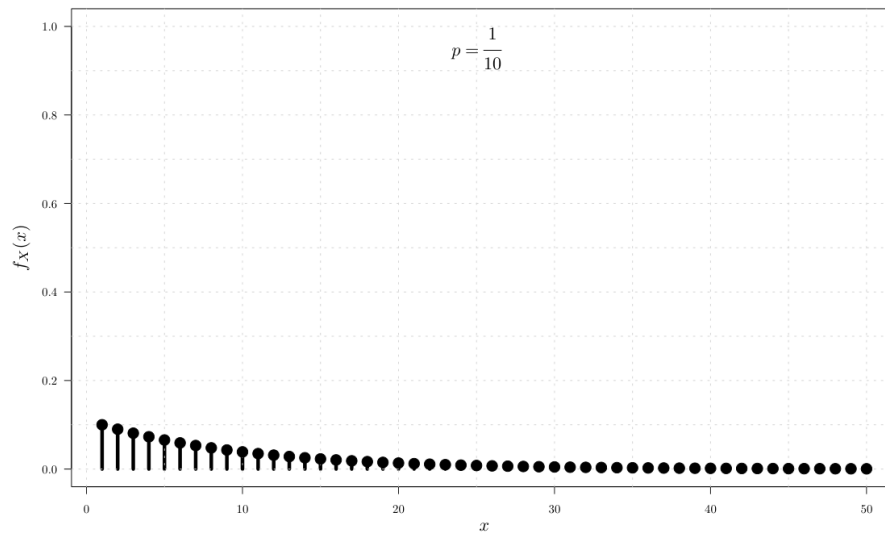


Figure 17: The pmf of a $\text{Geom}(p)$ with $p = 1/10, 5/10, 9/10$.

Example 4.8. Suppose that $X \sim \text{Geom}(p)$. Then

$$\begin{aligned}
 \sum_{k=1}^{\infty} \mathbb{P}(X = k) &= \sum_{k=1}^{\infty} (1-p)^{k-1} p && [\text{pmf of } \text{Geom}(p)] \\
 &= p \sum_{k=1}^{\infty} (1-p)^{k-1} \\
 &= p \sum_{j=0}^{\infty} (1-p)^j && [\text{let } j = k-1] \\
 &= p \left(\frac{1}{1-(1-p)} \right) = 1. && [\text{take } r = 1-p \in (0,1) \text{ for Geometric Series}]
 \end{aligned}$$

Thus the sum of probabilities in the pmf of a Geometric distribution is 1, as it should be.

Example 4.9. Suppose that $X \sim \text{Geom}(1/4)$.

1. Find $\mathbb{P}(X > 2)$.

As $X \sim \text{Geom}(1/4)$, we have (Definition 4.8)

$$\mathbb{P}(X = x) = \left(\frac{3}{4}\right)^{x-1} \left(\frac{1}{4}\right)$$

for each $x \in \{1, 2, 3, \dots\}$. By the probability of complements, we have $\mathbb{P}(E) = 1 - \mathbb{P}(E^c)$, and if we take $E = \{X > 2\}$ then $E^c = \{X \leq 2\}$ yielding

$$\begin{aligned}
 \mathbb{P}(X > 2) &= 1 - \mathbb{P}(X \leq 2) \\
 &= 1 - \mathbb{P}(\{X = 1\} \cup \{X = 2\}) && [\{X \leq 2\} = \{X = 1\} \cup \{X = 2\}] \\
 &= 1 - (\mathbb{P}(X = 1) + \mathbb{P}(X = 2)) && [\text{disjoint events}] \\
 &= 1 - \left[\left(\frac{1}{4}\right) + \left(\frac{3}{4}\right) \left(\frac{1}{4}\right) \right] && [\text{pmf of } \text{Geom}(1/4) \text{ for } x = 1, 2] \\
 &= 1 - \frac{1}{4} \times \frac{7}{4} = \frac{9}{16}.
 \end{aligned}$$

Notice that $\frac{9}{16} = \left(\frac{3}{4}\right)^2$, i.e., the squared probability of failure.

2. Find $\mathbb{P}(X > 3)$.

With the same reasoning as in point 1.,

$$\begin{aligned}
 \mathbb{P}(X > 3) &= 1 - \mathbb{P}(X \leq 3) \\
 &= 1 - \mathbb{P}(\{X = 1\} \cup \{X = 2\} \cup \{X = 3\}) \\
 &= \dots \\
 &= 1 - \left[\left(\frac{1}{4}\right) + \left(\frac{3}{4}\right) \left(\frac{1}{4}\right) + \left(\frac{3}{4}\right)^2 \left(\frac{1}{4}\right) \right] \\
 &= 1 - \frac{1}{4} \times \frac{37}{16} = \frac{27}{64} = \left(\frac{3}{4}\right)^3,
 \end{aligned}$$

i.e., the cubed probability of failure.

Example 4.10 (Generalisation of Example 4.9). Let n be a positive integer and suppose that $X \sim \text{Geom}(p)$. Find $\mathbb{P}(X \leq n)$ and $\mathbb{P}(X > n)$.

As we noticed in the previous example, $\{X > n\}$ is exactly equivalent to obtaining no successes in the first n trials, so we can immediately say that $\mathbb{P}(X > n) = (1-p)^n$ and therefore $\mathbb{P}(X \leq n) = 1 - (1-p)^n$.

However, to practice calculating sums and check that our logic is sound, we can do things step by step again. Notice that $\{X \leq n\} = \bigcup_{x=1}^n \{X = x\}$ which is a union of disjoint events so that

$$\begin{aligned}
 \mathbb{P}(X \leq n) &= \sum_{x=1}^n \mathbb{P}(X = x) \\
 &= \sum_{x=1}^n (1-p)^{x-1} p && [\text{as } X \sim \text{Geom}(p)] \\
 &= p \sum_{x=1}^n (1-p)^{x-1} \\
 &= p \sum_{i=0}^{n-1} (1-p)^i && [\text{set } i = x-1]
 \end{aligned} \tag{42}$$

The sum in Equation (42) is of the form $S_{n-1} = \sum_{i=0}^{n-1} r^i$ where $r = 1-p$. Thus, by Equation (40) and Theorem 4.4,

$$\mathbb{P}(X \leq n) = p \left(\frac{1 - (1-p)^n}{1 - (1-p)} \right) = 1 - (1-p)^n. \tag{43}$$

As $\{X > n\} = \{X \leq n\}^c$, we have by Equation (43),

$$\mathbb{P}(X > n) = (1-p)^n.$$

Both results match the answers we expected.

We have thus proved the formula the cdf of $X \sim \text{Geom}(p)$: for $x \in \mathbb{R}$, writing $\lfloor x \rfloor$ for the largest integer smaller than or equal to x , i.e. x “rounded down” to the nearest integer, we have

$$F_X(x) = \mathbb{P}(X \leq x) = \begin{cases} 0 & \text{for } x < 1, \\ 1 - (1-p)^{\lfloor x \rfloor} & \text{for } x \geq 1. \end{cases} \tag{44}$$

Example 4.11. Billy Forgetful independently remembers to attend each lecture with probability $p = 0.3$.

Let X be the number of lectures up to and including his first attendance.

In this case $X \sim \text{Geom}(0.3)$ and

$$\begin{aligned}
 \mathbb{P}(X > 5) &= (1 - 0.3)^5 = 0.16807 \quad (5 \text{ dp}), \\
 \mathbb{P}(X \leq 33) &= 1 - (1 - 0.3)^{33} = 1 - (0.7)^{33} = 0.99999 \quad (5 \text{ dp}).
 \end{aligned}$$

4.3.4 Poisson random variables

The Poisson distribution is used in situations where events are happening at a certain rate over a time period. For example, we might be counting the number of people joining a queue, or the number of radioactive decay events during a fixed time period.

Definition 4.9 (Poisson distribution). A discrete random variable X is said to have a Poisson distribution with parameter $\lambda \in (0, +\infty)$, written as $X \sim \text{Pois}(\lambda)$, if its pmf is given by

$$f_X(x) = \mathbb{P}(X = x) = \begin{cases} \frac{\lambda^x}{x!} e^{-\lambda} & \text{for } x \in \{0, 1, 2, \dots\} \\ 0 & \text{otherwise.} \end{cases} \tag{45}$$

Figure 18 shows examples of the pmfs of Poisson distributions with different values of λ .

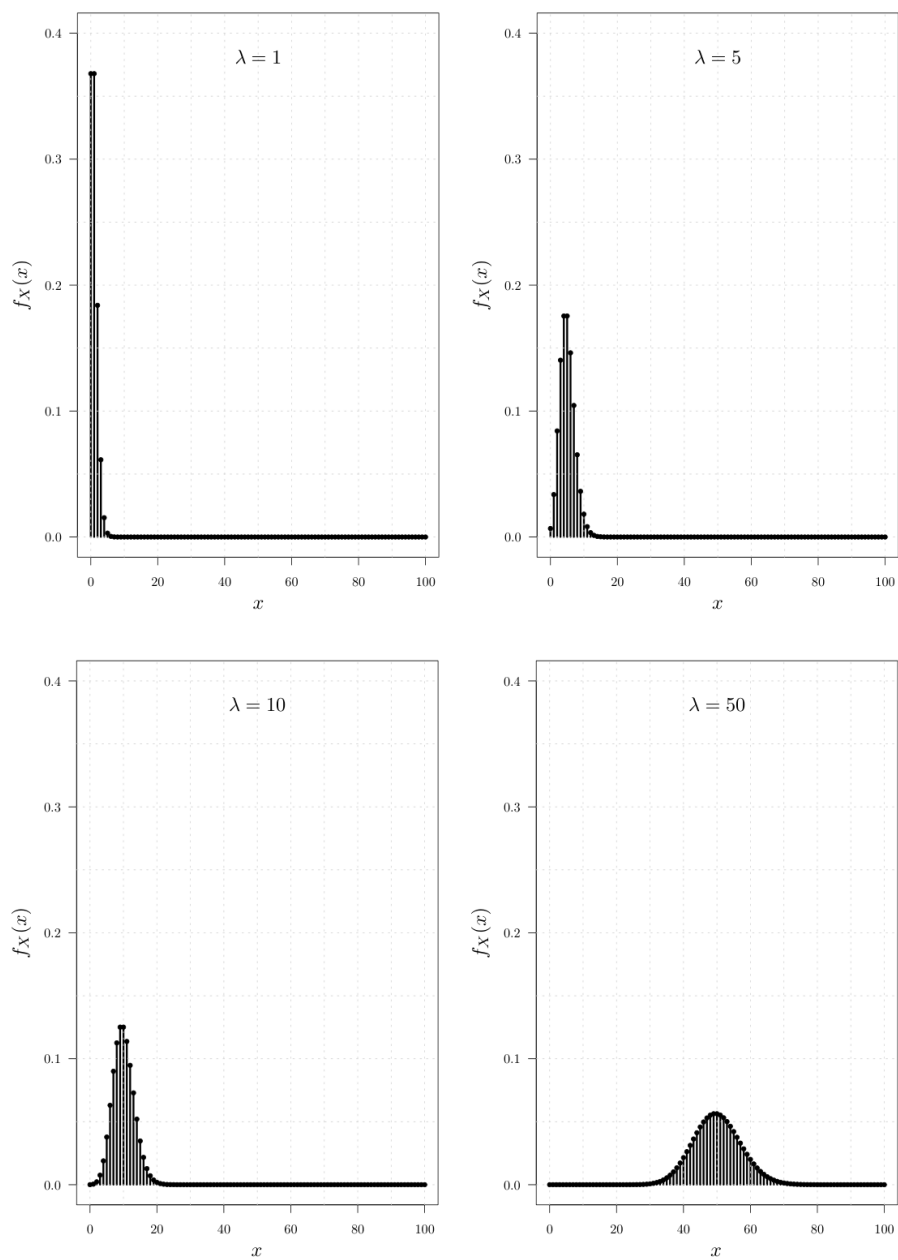


Figure 18: The pmf of a $\text{Pois}(\lambda)$ with $\lambda = 1, 5, 10, 50$.

To see that Definition 4.9 gives a valid pmf, we check that

$$\begin{aligned}
 \sum_{x=0}^{\infty} \mathbb{P}(X = x) &= \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} e^{-\lambda} && \text{[Poisson pmf]} \\
 &= e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} \\
 &= e^{-\lambda} e^{\lambda} && \text{[series expansion of } e^{\lambda}] \\
 &= 1
 \end{aligned}$$

Example 4.12. *Freddie has freshers' flu, and sneezes at a constant rate of 1.2 sneezes per minute. He needs 20 sneeze-less minutes to fall asleep. He goes to bed at 11pm. What is the probability that he is asleep by 11.20pm?*

We can model the number of Freddie's sneezes between 11pm and 11.20pm by a Poisson random variable X of parameter $20 \times 1.2 = 24$. Then

$$\mathbb{P}(\text{Freddie is asleep by 11.20pm}) = \mathbb{P}(X = 0) = f_X(0) = e^{-24}$$

(a very small number). Poor Freddie.

Remark:

Imagine that over a fixed time interval, we have a large number n of evenly spread Bernoulli trials all with a small probability of success p . Then we expect approximately np successful events over the time interval, which will occur at roughly a constant rate. Indeed it can be shown that for large n and small p , $\text{Bin}(n, p)$ and $\text{Pois}(np)$ are almost identical.

Example 4.13. *A typesetter, on average, makes one error in every 500 words typeset. A typical page contains 300 words. What is the probability that there will be no more than two errors in five pages?*

If we assume that setting a word is a Bernoulli trial with success probability $p = 1/500$ (notice that we are labeling an error as a "success") and that the trials are independent, then $X = \{\text{number of errors in 5 pages}\}$ is distributed according to a Binomial distribution with parameters $n = 1500$ (5 pages corresponds to $5 \times 300 = 1500$ words) and $p = 1/500$. Thus

$$\begin{aligned}
 \mathbb{P}(\text{no more than two errors}) &= \mathbb{P}(X \leq 2) \\
 &= \sum_{x=0}^2 \binom{1500}{x} \left(\frac{1}{500}\right)^x \left(\frac{499}{500}\right)^{1500-x} \\
 &= 0.4230 \quad (4 \text{ dp}),
 \end{aligned}$$

which is a fairly cumbersome calculation. If we use the Poisson approximation with $\lambda = np = 1500/500 = 3$, we have

$$\mathbb{P}(X \leq 2) \approx \sum_{x=0}^2 e^{-3} \frac{3^x}{x!} = e^{-3} \left(1 + 3 + \frac{9}{2}\right) = 0.4232 \quad (4 \text{ dp}).$$

The distribution function of a Poisson distribution can be written as follows: for $X \sim \text{Pois}(\lambda)$, with $\lfloor x \rfloor$ again representing x rounded down,

$$F_X(x) = \mathbb{P}(X \leq x) = \begin{cases} 0 & \text{for } x < 0, \\ \sum_{i=0}^{\lfloor x \rfloor} \frac{\lambda^i}{i!} e^{-\lambda} & \text{for } x \geq 0. \end{cases} \quad (46)$$

Remark:

We often think of λ as the average number of successes over a time interval, but time could be replaced, e.g., by area or volume. For example, in 1946, R. D. Clarke showed¹¹ that in Germany's flying-bomb

¹¹Accessed on 27 Aug 2021: <https://www.actuaries.org.uk/system/files/documents/pdf/0481.pdf>

attack on London in World War II, the number of bomb hits per $\frac{1}{4}$ square mile areas in south London were close to a Poisson random variable with parameter $\lambda = \frac{536}{576}$, suggesting that the hits were not clustered, but rather randomly scattered.

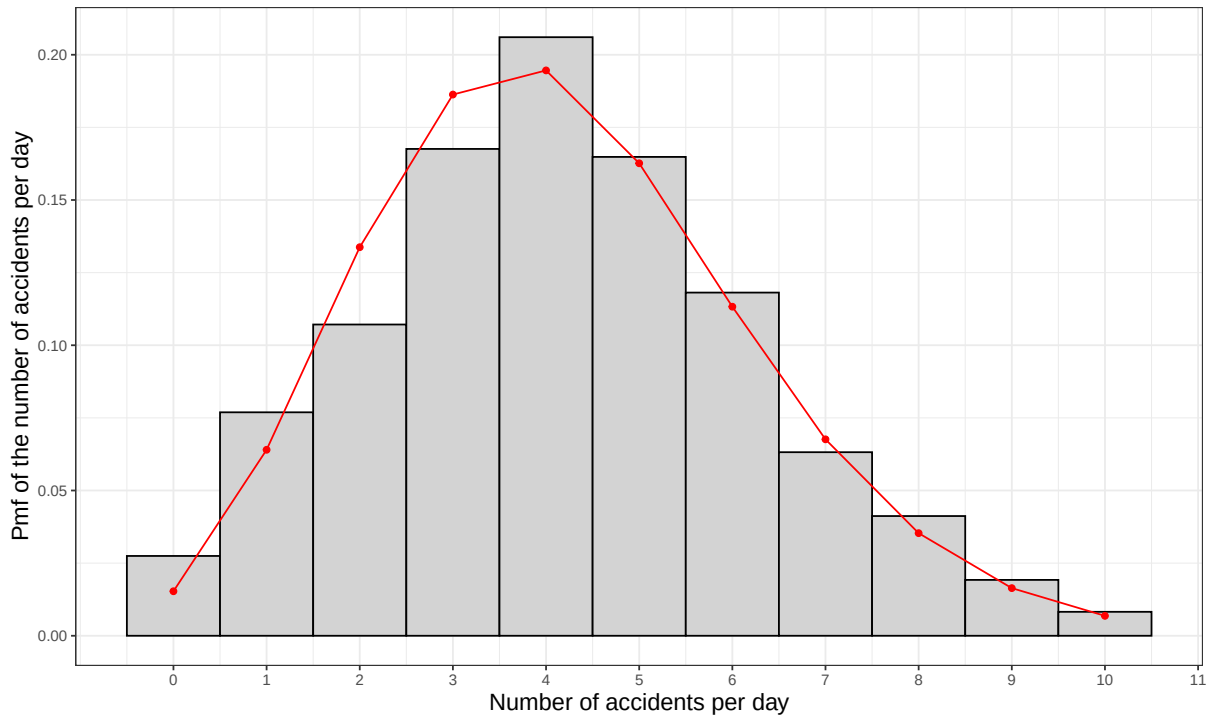


Figure 19: Pmf of $\text{Pois}(4.18)$ and scaled road accident counts per day in Westminster in 2019.

Example 4.14. The number of road accidents at a busy junction can be modelled by a Poisson distribution.

Figure 19 shows a **histogram** (grey bars) of the number of road accidents per day in Westminster¹² in the year 2019. The histogram shows how many days there have been in a year with a given number of accidents. To enable the comparison with a Poisson pmf, the day counts are divided by the total number of days, so that they sum to 1 like a pmf. The red curve shows the pmf of $\text{Pois}(4.18)$. Although the fit is not perfect, the Poisson model seems reasonable.

We do not go into the details of how the value $\lambda = 4.18$ was found (*spoiler alert*: wait until the second semester), but once we have it, we can conclude that on average, just over 4 road accidents per day occur in Westminster.

4.4 Joint distributions and independence of discrete random variables

4.4.1 Joint and marginal distributions

Suppose that X and Y are two discrete random variables defined on the same sample space Ω . Then each $\omega \in \Omega$ yields a value $X(\omega) = x$ for X and $Y(\omega) = y$ for Y so that we can regard the outcome to be a pair (x, y) . The probability of X and Y taking values x and y is

$$\mathbb{P}(X = x, Y = y) = \mathbb{P}(\{\omega \in \Omega : X(\omega) = x, Y(\omega) = y\}).$$

Example 4.15. We roll two fair dice. Let X be the sum of the two rolls, and let Y be the product of the two rolls. Then

$$\Omega = \{(a, b) : a, b \in \{1, 2, 3, 4, 5, 6\}\}$$

¹²Data can be found from (accessed on 5 Aug 2022): <https://data.gov.uk/dataset/cb7ae6f0-4be6-4935-9277-47e5ce24a11f/road-safety-data>.

and, for example,

$$\mathbb{P}(X = 4, Y = 3) = \mathbb{P}(\omega = (1, 3) \text{ or } \omega = (3, 1)) = \frac{2}{36} = \frac{1}{18}.$$

Definition 4.10 (Joint probability mass function). Let X and Y be discrete random variables defined on the sample space Ω with sets of possible values S_X and S_Y , respectively. The *joint probability mass function* of X and Y is defined for all $x, y \in \mathbb{R}$ as

$$f_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y).$$

Thus

$$\sum_{x \in S_X} \sum_{y \in S_Y} f_{X,Y}(x, y) = \sum_{x \in S_X} \sum_{y \in S_Y} \mathbb{P}(X = x, Y = y) = 1$$

and $f_{X,Y}(x, y) = 0$ for all (x, y) such that either $x \notin S_X$ or $y \notin S_Y$.

If $S_X = \{x_1, x_2, \dots, x_n\}$ and $S_Y = \{y_1, y_2, \dots, y_m\}$ then the double summation above can be written as

$$\begin{aligned} \sum_{i=1}^n \sum_{j=1}^m \mathbb{P}(X = x_i, Y = y_j) &= \mathbb{P}(X = x_1, Y = y_1) + \dots + \mathbb{P}(X = x_1, Y = y_m) \\ &\quad + \mathbb{P}(X = x_2, Y = y_1) + \dots + \mathbb{P}(X = x_1, Y = y_m) + \dots \\ &\quad + \mathbb{P}(X = x_n, Y = y_1) + \dots + \mathbb{P}(X = x_n, Y = y_m), \end{aligned}$$

and this can be extended to countably infinite S_X and S_Y .

Notice that the events $\{Y = y_1\}, \{Y = y_2\}, \dots, \{Y = y_m\}$ form a partition of Ω , and thus, by the law of total probability, we can find the **marginal distribution** (or marginal pmf) of X by summing $f_{X,Y}(x, y)$ over each possible value of Y :

$$\begin{aligned} f_X(x) = \mathbb{P}(X = x) &= \sum_{y \in S_Y} \mathbb{P}(\{X = x\} \mid \{Y = y\})\mathbb{P}(\{Y = y\}) \\ &= \sum_{y \in S_Y} \mathbb{P}(\{X = x\} \cap \{Y = y\}) \\ &= \sum_{y \in S_Y} \mathbb{P}(X = x, Y = y) \\ &= \sum_{y \in S_Y} f_{X,Y}(x, y) \end{aligned}$$

The marginal pmf of Y is found analogously by summing over X .

We don't give a formal definition of *marginal distribution* because there isn't one - the *marginal distribution* of X is just the distribution of X , but using the word *marginal* implies that X was introduced via its joint distribution with some other random variable(s).

Example 4.16. Suppose that the discrete random variables X and Y have a joint pmf

$$\begin{aligned} \mathbb{P}(X = 2, Y = 0) &= 1/8 \\ \mathbb{P}(X = 1, Y = 1) &= 1/4 \\ \mathbb{P}(X = 0, Y = 2) &= 1/4 \\ \mathbb{P}(X = 0, Y = 1) &= 1/4 \\ \mathbb{P}(X = 0, Y = 0) &= 1/8, \end{aligned}$$

with $\mathbb{P}(X = x, Y = y) = 0$ for all other values of x and y . Find the marginal distributions of X and Y .

We have

$$\begin{aligned} \mathbb{P}(X = 2) &= \mathbb{P}(X = 2, Y = 0) = 1/8, \\ \mathbb{P}(X = 1) &= \mathbb{P}(X = 1, Y = 1) = 1/4 \end{aligned}$$

and

$$\mathbb{P}(X = 0) = \mathbb{P}(X = 0, Y = 0) + \mathbb{P}(X = 0, Y = 1) + \mathbb{P}(X = 0, Y = 2) = \frac{1}{8} + \frac{1}{4} + \frac{1}{4} = \frac{5}{8}.$$

We also have

$$\mathbb{P}(Y = 2) = \mathbb{P}(X = 0, Y = 2) = 1/4,$$

$$\mathbb{P}(Y = 1) = \mathbb{P}(X = 1, Y = 1) + \mathbb{P}(X = 0, Y = 1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

and

$$\mathbb{P}(Y = 0) = \mathbb{P}(X = 2, Y = 0) + \mathbb{P}(X = 0, Y = 0) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}.$$

One possible way of generating random variables X and Y with this joint distribution is to toss a fair coin three times, and let X be the number of times we see two heads in a row, and Y be the number of times we see either HT or TH .

Example 4.17. Suppose that the discrete random variables X and Y have a joint pmf

$$\mathbb{P}(X = x, Y = y) = \begin{cases} \frac{2x+y}{36} & \text{if } x \in \{0, 1, 2\} \text{ and } y \in \{1, 2, 3\}, \\ 0 & \text{otherwise.} \end{cases}$$

Find the marginal distributions of X and Y .

The marginal distribution for X is, for each $x \in \{0, 1, 2\}$,

$$\begin{aligned} \mathbb{P}(X = x) &= \sum_{y=1}^3 \mathbb{P}(X = x, Y = y) \\ &= \sum_{y=1}^3 \frac{2x+y}{36} \\ &= \frac{1}{36} \left(6x + \sum_{y=1}^3 y \right) \\ &= \frac{6x + 1 + 2 + 3}{36} = \frac{x+1}{6}. \end{aligned}$$

The marginal distribution for Y is, for each $y \in \{1, 2, 3\}$,

$$\begin{aligned} \mathbb{P}(Y = y) &= \sum_{x=0}^2 \mathbb{P}(X = x, Y = y) \\ &= \sum_{x=0}^2 \frac{2x+y}{36} \\ &= \frac{1}{36} \sum_{x=0}^2 (2x+y) \\ &= \frac{0+2+4+3y}{36} = \frac{2+y}{12}. \end{aligned}$$

We can represent our values in a **joint distribution table**:

		Y			$\mathbb{P}(X = x)$
		1	2	3	
X	0	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{6}{36}$
	1	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{12}{36}$
	2	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{7}{36}$	$\frac{18}{36}$
		$\frac{9}{36}$	$\frac{12}{36}$	$\frac{15}{36}$	1
$\mathbb{P}(Y = y)$		$\frac{9}{36}$	$\frac{12}{36}$	$\frac{15}{36}$	1

Besides the joint pmf, we can also define the joint cdf, which will be used in the definition of independence for random variables in the next section; and useful also for working with continuous distributions.

Definition 4.11 (Joint cumulative distribution function). Let $X : \Omega \rightarrow S_X$ and $Y : \Omega \rightarrow S_Y$ be random variables defined on the same probability space. The function

$$F_{X,Y}(x, y) = \mathbb{P}(X \leq x, Y \leq y)$$

is the joint cumulative distribution function of X and Y .

Remark:

The joint cdf has similar properties to the standard cdf:

- a. $F_{X,Y}$ is non-decreasing in each co-ordinate, i.e.,

$$F_{X,Y}(x+a, y+b) \geq F_{X,Y}(x, y)$$

for any $a, b \geq 0$.

- b. $F_{X,Y}(+\infty, +\infty) = 1$.

- c. $F_{X,Y}(-\infty, y) = F_{X,Y}(x, -\infty) = 0$ for any x and y .

(Be careful with b and c. For example, it is *not* true that $F_{X,Y}(+\infty, y) = F_{X,Y}(x, +\infty) = 1$ for any x and y . Why not?)

4.4.2 Independence of random variables

The independence of events can be extended to define independence between random variables.

Definition 4.12 (Independent random variables). Let $X : \Omega \rightarrow S_X$ and $Y : \Omega \rightarrow S_Y$ be random variables on the same sample space Ω . We say that X and Y are *independent* if

$$\mathbb{P}(X \leq x, Y \leq y) = \mathbb{P}(X \leq x) \mathbb{P}(Y \leq y) \quad \text{for all } x, y \in \mathbb{R}. \quad (47)$$

Remark: We can write independence in terms of cdfs. Suppose that X and Y have cdfs F_X and F_Y respectively, and joint cdf $F_{X,Y}$. That is,

$$F_X(x) = \mathbb{P}(X \leq x), \quad F_Y(y) = \mathbb{P}(Y \leq y), \quad \text{and} \quad F_{X,Y}(x, y) = \mathbb{P}(X \leq x, Y \leq y).$$

Then X and Y are independent if

$$F_{X,Y}(x, y) = F_X(x)F_Y(y) \quad \text{for all } x, y \in \mathbb{R}.$$

For discrete random variables, we have the pmf available as well as the cdf, and it is often easier to work with the pmf. The good news is that the definition of independence for discrete random variables can also be written easily in terms of the pmf.

Theorem 4.5 (Independent discrete random variables). Let $X : \Omega \rightarrow S_X$ and $Y : \Omega \rightarrow S_Y$ be discrete random variables. Then X and Y are independent if and only if

$$\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y) \quad \text{for all } x \in S_X, y \in S_Y. \quad (48)$$

We can also write this in terms of pmfs: if the pmfs of X and Y are f_X and f_Y , and they have joint pmf $f_{X,Y}$, then X and Y are independent if and only if

$$f_{X,Y}(x, y) = f_X(x)f_Y(y) \quad \text{for all } x, y \in \mathbb{R}.$$

Note that if $S_{X,Y} = \{(x, y) : x \in S_X, y \in S_Y\}$ then $f_{X,Y}(x, y) = 0$ for all $(x, y) \notin S_{X,Y}$ and when this occurs at least one of $\mathbb{P}(X = x)$ and $\mathbb{P}(Y = y)$ is zero (as at least one of x, y is not possible). This means that we can restrict our attention to the set of possible values of X and Y only, i.e. S_X and S_Y , as in the first part of the theorem.

The proof of this theorem is a little bit longer than the proofs we have seen so far. Before we carry out the proof, we will look at some examples.

Example 4.18. Roll a fair dice twice. Let X be the result of the first roll and Y be the result of the second roll. Show that X and Y are independent.

This is a case of showing the obvious, but letting $\Omega = \{(a, b) : a, b \in \{1, 2, 3, 4, 5, 6\}\}$, for any $x, y \in \{1, 2, 3, 4, 5, 6\}$ we have

$$\mathbb{P}(X = x, Y = y) = \mathbb{P}(\omega = (x, y)) = \frac{1}{|\Omega|} = \frac{1}{36},$$

whereas

$$\mathbb{P}(X = x) = \sum_{y=1}^6 \mathbb{P}(X = x, Y = y) = \sum_{y=1}^6 \frac{1}{36} = \frac{1}{6}$$

and similarly $\mathbb{P}(Y = y) = 1/6$. Thus

$$\mathbb{P}(X = x, Y = y) = \frac{1}{36} = \frac{1}{6} \times \frac{1}{6} = \mathbb{P}(X = x)\mathbb{P}(Y = y).$$

By Theorem 4.5 this tells us that X and Y are independent.

Let Z be the sum of the two rolls. Show that X and Z are not independent.

Again this seems clear, since if X is small, Z is likely to be small. To show it rigorously we need to find x and z such that $\mathbb{P}(X = x, Z = z) \neq \mathbb{P}(X = x)\mathbb{P}(Z = z)$. There are plenty of possibilities, but one trivial one is that if $X = 1$, then Z can't equal (for example) 12; thus

$$\mathbb{P}(X = 1, Z = 12) = 0$$

but $\mathbb{P}(Z = 12) = \mathbb{P}(\omega = (6, 6)) = 1/36$ so

$$\mathbb{P}(X = 1)\mathbb{P}(Z = 12) = \frac{1}{6} \times \frac{1}{36} = \frac{1}{216} \neq 0.$$

So indeed, by Theorem 4.5, X and Z are not independent.

Example 4.19. *Let X, Y be discrete random variables taking values in $\{1, 2, 3, \dots\}$ with a joint pmf*

$$\mathbb{P}(X = x, Y = y) = \frac{3^x}{4^{x+y}}.$$

Show that X and Y are independent.

Notice that

$$\begin{aligned} \mathbb{P}(X = x) &= \sum_{y=1}^{\infty} \frac{3^x}{4^{x+y}} = \left(\frac{3}{4}\right)^x \sum_{y=1}^{\infty} \left(\frac{1}{4}\right)^y \\ &= \left(\left(\frac{3}{4}\right)^{x-1} \left(\frac{1}{4}\right)\right) \left(\sum_{y=1}^{\infty} \left(\frac{1}{4}\right)^{y-1} \left(\frac{3}{4}\right)\right) \\ &= \left(\frac{3}{4}\right)^{x-1} \frac{1}{4}, \end{aligned}$$

where the final equality holds because the sum is over the pmf of $\text{Geom}(3/4)$ and hence equal to 1. Thus, $X \sim \text{Geom}(1/4)$. Similarly we can show that

$$\mathbb{P}(Y = y) = \left(\frac{1}{4}\right)^{y-1} \frac{3}{4},$$

so that $Y \sim \text{Geom}(3/4)$.

For all $x, y \in \{1, 2, 3, \dots\}$,

$$\begin{aligned} \mathbb{P}(X = x, Y = y) &= \frac{3^x}{4^{x+y}} = \left(\frac{3}{4}\right)^x \left(\frac{1}{4}\right)^y \\ &= \left(\frac{3}{4}\right)^{x-1} \frac{3}{4} \left(\frac{1}{4}\right)^{y-1} \frac{1}{4} \\ &= \left(\left(\frac{3}{4}\right)^{x-1} \frac{1}{4}\right) \left(\left(\frac{1}{4}\right)^{y-1} \frac{3}{4}\right) \\ &= \mathbb{P}(X = x) \mathbb{P}(Y = y), \end{aligned}$$

so X and Y are independent by Theorem 4.5.

Before we prove Theorem 4.5, it will be useful to have yet another alternative formulation of independence.

Theorem 4.6 (Alternative formulation of independence for random variables). *Random variables $X : \Omega \rightarrow S_X$ and $Y : \Omega \rightarrow S_Y$ are independent if and only if*

$$\mathbb{P}(a < X \leq b, c < Y \leq d) = \mathbb{P}(a < X \leq b) \mathbb{P}(c < Y \leq d) \text{ for all } a < b \in \mathbb{R} \text{ and all } c < d \in \mathbb{R}. \quad (49)$$

We prove this in the case of discrete random variables with finite support. The proof in full generality is not much more difficult, but requires a small amount of analysis.

Proof. Idea: for any random variable Z and $p < q \in \mathbb{R}$,

$$\{p < Z \leq q\} = \{Z \leq q\} \setminus \{Z \leq p\},$$

and since $p < q$ we have $\{Z \leq p\} \subset \{Z \leq q\}$, so

$$\mathbb{P}(p < Z \leq q) = \mathbb{P}(Z \leq q) - \mathbb{P}(Z \leq p).$$

We will use this many times to change from probabilities like $\mathbb{P}(a < X \leq b)$ to probabilities like $\mathbb{P}(X \leq b)$ and $\mathbb{P}(X \leq a)$.

First we show that if X and Y are independent, then (49) holds. Using the idea above several times,

$$\begin{aligned} \mathbb{P}(a < X \leq b, c < Y \leq d) &= \mathbb{P}(X \leq b, c < Y \leq d) - \mathbb{P}(X \leq a, c < Y \leq d) && [\text{idea above for } X] \\ &= (\mathbb{P}(X \leq b, Y \leq d) - \mathbb{P}(X \leq b, Y \leq c)) \\ &\quad - (\mathbb{P}(X \leq a, Y \leq d) - \mathbb{P}(X \leq a, Y \leq c)) && [\text{idea above for } Y, \text{ twice}] \\ &= (\mathbb{P}(X \leq b)\mathbb{P}(Y \leq d) - \mathbb{P}(X \leq b)\mathbb{P}(Y \leq c)) \\ &\quad - (\mathbb{P}(X \leq a)\mathbb{P}(Y \leq d) - \mathbb{P}(X \leq a)\mathbb{P}(Y \leq c)) && [\text{independence, four times}] \\ &= \mathbb{P}(X \leq b)\mathbb{P}(c < Y \leq d) - \mathbb{P}(X \leq a)\mathbb{P}(c < Y \leq d) && [\text{idea above for } Y, \text{ twice}] \\ &= \mathbb{P}(a < X \leq b)\mathbb{P}(c < Y \leq d). && [\text{idea above for } X] \end{aligned}$$

This proves (49) (and note that it works for any random variables, not only those with finite support).

Now to show that (49) implies independence, if X and Y are discrete with finite support then we can just take the smallest values $a \in S_X$ and $b \in S_Y$. Then we know for sure that $X > a - 1$ and $Y > b - 1$. Thus for any $x, y \in \mathbb{R}$, by (49) we have

$$\begin{aligned} \mathbb{P}(X \leq x, Y \leq y) &= \mathbb{P}(a - 1 < X \leq x, b - 1 < Y \leq y) && [\text{since } X > a - 1 \text{ and } Y > b - 1] \\ &= \mathbb{P}(a - 1 < X \leq x)\mathbb{P}(b - 1 < Y \leq y) \\ &= \mathbb{P}(X \leq x)\mathbb{P}(Y \leq y). && [\text{since } X > a - 1 \text{ and } Y > b - 1] \end{aligned}$$

So X and Y are independent.

(If X and Y do not have finite support, then we can't necessarily find a smallest a and b as above, so we have to do a limiting argument. It's still fairly straightforward - just outside the scope of this course. If we have time at the end of the course we will learn a little bit about how probabilities interact with limits.) $\square$

We can now prove Theorem 4.5.

Proof of equivalence of pmf formulation of independence. Again, we will carry out the proof only in the case that X and Y are discrete random variables with finite support S_X and S_Y . Again, the proof for general X and Y is not much more complicated, but requires limits.

Let us write $S_X = \{x_1, \dots, x_n\}$ and $S_Y = \{y_1, \dots, y_m\}$, where $x_1 < x_2 < \dots < x_n$ and $y_1 < y_2 < \dots < y_m$.

1. First suppose that X and Y are independent. We want to show that (48) holds.

Idea: if we choose a and b appropriately, then we can rewrite $\{X = x_i\}$ as $\{a < X \leq b\}$.

We know from Theorem 4.6 that for any $a < b \in \mathbb{R}$ and $c < d \in \mathbb{R}$,

$$\mathbb{P}(a < X \leq b, c < Y \leq d) = \mathbb{P}(a < X \leq b) \mathbb{P}(c < Y \leq d). \quad (50)$$

Note that if we choose $a = x_{i-1}$ and $b = x_i$ for some $i = 2, \dots, n$, then

$$\{x_{i-1} < X \leq x_i\} = \{X = x_i\},$$

and similarly for $j = 2, \dots, m$,

$$\{y_{j-1} < Y \leq y_j\} = \{Y = y_j\}.$$

Thus, for any $i = 2, \dots, n$ and $j = 2, \dots, m$, by (50) we have

$$\begin{aligned} \mathbb{P}(X = x_i, Y = y_j) &= \mathbb{P}(x_{i-1} < X \leq x_i, y_{j-1} < Y \leq y_j) \\ &= \mathbb{P}(x_{i-1} < X \leq x_i) \mathbb{P}(y_{j-1} < Y \leq y_j) \\ &= \mathbb{P}(X = x_i) \mathbb{P}(Y = y_j). \end{aligned}$$

We can do the same for $i = 1$ and/or $j = 1$ by choosing $a = x_1 - 1$ and/or $c = y_1 - 1$. This completes the proof that independence implies (48).

2. Now suppose that (48) holds. We want to show that X and Y are independent.

Idea: Write $\mathbb{P}(X \leq x)$ as a sum of $\mathbb{P}(X = x_i)$ over the possible values of x_i that are smaller than x . If $x < x_1$ or $y < y_1$, then both sides of Equation (47) are zero and $\mathbb{P}(X \leq x, Y \leq y) = \mathbb{P}(X \leq x) \mathbb{P}(Y \leq y)$ is trivially true. Otherwise, let $p \in \{1, \dots, n\}$ and $q \in \{1, \dots, m\}$ be the largest values for which $x_p \leq x$ and $y_q \leq y$.

In this case, by (48),

$$\begin{aligned} \mathbb{P}(X \leq x, Y \leq y) &= \sum_{i=1}^p \sum_{j=1}^q \mathbb{P}(X = x_i, Y = y_j) \\ &= \sum_{i=1}^p \sum_{j=1}^q \mathbb{P}(X = x_i) \mathbb{P}(Y = y_j) \\ &= \left(\sum_{i=1}^p \mathbb{P}(X = x_i) \right) \left(\sum_{j=1}^q \mathbb{P}(Y = y_j) \right) \quad [\mathbb{P}(X = x_i) \text{ doesn't depend on } j, \text{ we can take it out of the sum}] \\ &= \mathbb{P}(X \leq x) \mathbb{P}(Y \leq y). \end{aligned}$$

This completes the proof that (48) implies independence. □

4.4.3 Sums of independent random variables

We know that the sum of n independent and identical Bernoulli random variables with success probability p has the Binomial distribution with parameters n and p , but what if the random variables that we add together are not Bernoulli trials but something else, say rolls of a 6-sided dice.

Consider independent $X : \Omega \rightarrow S_X = \{x_1, \dots, x_n\}$ and $Y : \Omega \rightarrow S_Y = \{y_1, \dots, y_m\}$. Then for any $k \in \mathbb{R}$ we have

$$\begin{aligned} \mathbb{P}(X + Y = k) &= \mathbb{P}(\{X = x_1 \cap Y = k - x_1\} \cup \dots \cup \{X = x_n \cap Y = k - x_n\}) \\ &= \sum_{i=1}^n \mathbb{P}(X = x_i, Y = k - x_i) && [\text{disjoint events}] \\ &= \sum_{i=1}^n \mathbb{P}(X = x_i) \mathbb{P}(Y = k - x_i), && [\text{independence}] \end{aligned}$$

which leads to the following general result:

Theorem 4.7 (Sum of independent discrete random variables). *Let $X : \Omega \rightarrow S_X$ and $Y : \Omega \rightarrow S_Y$ be independent discrete random variables. For all $k \in \mathbb{R}$ we have*

$$\mathbb{P}(X + Y = k) = \sum_{x \in S_X} \mathbb{P}(X = x) \mathbb{P}(Y = k - x).$$

Example 4.20. Let $X \sim \text{Pois}(\lambda)$ and $Y \sim \text{Pois}(\mu)$, for some $\lambda, \mu > 0$. Find the distribution of $X + Y$.

Let $k \geq 0$. We know that X and Y are both non-negative, so to have $X + Y = k$, the value of X must certainly be at most k . Thus

$$\begin{aligned}\mathbb{P}(X + Y = k) &= \sum_{i=0}^k \mathbb{P}(X = i, Y = k - i) \\ &= \sum_{i=0}^k \mathbb{P}(X = i) \mathbb{P}(Y = k - i) && \text{[independence]} \\ &= \sum_{i=0}^k e^{-\lambda} \frac{\lambda^i}{i!} e^{-\mu} \frac{\mu^{k-i}}{(k-i)!} && \text{[pmf of Poisson rvs]} \\ &= \frac{e^{-(\lambda+\mu)}}{k!} \sum_{i=0}^k \binom{k}{i} \lambda^i \mu^{k-i}.\end{aligned}$$

We now spot that the sum in the last line is equal to $(\lambda + \mu)^k$, so that

$$\mathbb{P}(X + Y = k) = \frac{e^{-(\lambda+\mu)} (\lambda + \mu)^k}{k!},$$

the pmf of a $\text{Pois}(\lambda + \mu)$ random variable. So $X + Y \sim \text{Pois}(\lambda + \mu)$.

This is not surprising if we recall our interpretation of a Poisson random variable as the number of events that are happening at a certain rate over a time period. If we have events happening at rate λ , and independently add on events happening at rate μ , then it intuitively makes sense that the overall rate should be $\lambda + \mu$.

Example 4.21. Let $X \sim \text{Bin}(n, p)$ and $Y \sim \text{Bin}(m, p)$, with $n < m$, and X, Y independent. Show that the distribution of $X + Y$ is a $\text{Bin}(n + m, p)$.

Proof by interpretation:

Recall the interpretation of a Binomial distribution: X represents the number of successes in n independent trials, each of which results in a success with probability p ; similarly, Y represents the number of successes in m independent trials, each trial being a success with probability p . Hence, as X and Y are assumed independent, it follows that $X + Y$ represents the number of successes in $n + m$ independent trials when each trial has a probability p of being a success. Therefore, $X + Y \sim \text{Bin}(n + m, p)$.

Calculations with pmfs:

What if we try to calculate the pmf of $X + Y$?

For $k \in \{0, 1, \dots, n + m\}$,

$$\mathbb{P}(X + Y = k) = \sum_{x=0}^k \mathbb{P}(X = x) \mathbb{P}(Y = k - x).$$

Note that if $x > n$ or $k - x > m$, the probabilities on the right are 0. To make our calculations easier to write down, we agree on a convention that for $r > N$, $\binom{N}{r} = 0$. This makes sense; if $r > N$ then the number of ways of choosing r items from N without replacement is zero.

Then we can write

$$\begin{aligned}\mathbb{P}(X + Y = k) &= \sum_{x=0}^k \binom{n}{x} p^x (1-p)^{n-x} \binom{m}{k-x} p^{k-x} (1-p)^{m-k+x} \\ &= p^k (1-p)^{n+m-k} \sum_{x=0}^k \binom{n}{x} \binom{m}{k-x}.\end{aligned}$$

But we know that, if $X + Y \sim \text{Bin}(n + m, p)$ (which we convinced ourselves of above), then

$$\mathbb{P}(X + Y = k) = \begin{cases} \binom{n+m}{k} p^k (1-p)^{n+m-k} & \text{for } k \in \{0, 1, \dots, n + m\} \\ 0 & \text{otherwise.} \end{cases}$$

It must therefore be true that

$$\sum_{x=0}^k \binom{n}{x} \binom{m}{k-x} = \binom{n+m}{k}.$$

We have thus proved a purely algebraic expression, known as Vandermonde's identity, using only probability.

There are other proofs of Vandermonde's identity (**non-examinable**):

- *Algebraic proof*

For any $z \in \mathbb{R}$,

$$\begin{aligned} \sum_{k=0}^{n+m} \binom{n+m}{k} z^k &= (1+z)^{n+m} \quad [\text{binomial expansion}] \\ &= (1+z)^n (1+z)^m \\ &= \left(\sum_{i=0}^n \binom{n}{i} z^i \right) \left(\sum_{j=0}^m \binom{m}{j} z^j \right) \quad [\text{binomial expansion}] \\ &= \left(\binom{n}{0} z^0 + \binom{n}{1} z^1 + \binom{n}{2} z^2 + \dots + \binom{n}{n} z^n \right) \left(\binom{m}{0} z^0 + \binom{m}{1} z^1 + \binom{m}{2} z^2 + \dots + \binom{m}{m} z^m \right) \\ &= \binom{n}{0} \binom{m}{0} z^0 + \left(\binom{n}{0} \binom{m}{1} + \binom{n}{1} \binom{m}{0} \right) z^1 \\ &\quad + \left(\binom{n}{0} \binom{m}{2} + \binom{n}{1} \binom{m}{1} + \binom{n}{2} \binom{m}{0} \right) z^2 + \dots \end{aligned}$$

Looking at the coefficient of z^k in the above expansion, we see that it can be written as

$$\binom{n}{0} \binom{m}{k} + \binom{n}{1} \binom{m}{k-1} + \binom{n}{2} \binom{m}{k-2} + \dots + \binom{n}{k} \binom{m}{0} = \sum_{x=0}^k \binom{n}{x} \binom{m}{k-x}.$$

Since this holds for all values of z , it must be true that

$$\binom{n+m}{k} = \sum_{x=0}^k \binom{n}{x} \binom{m}{k-x}.$$

- *Combinatorial proof*

Suppose there are m black balls and n red balls in an urn, and you want to pick k balls out of these $n+m$ to put in another urn, without replacement and with $k \in \{0, 1, \dots, n+m\}$.

- You can extract them in $\binom{n+m}{k}$ ways (unordered sampling without replacement).
- Another strategy would be to first choose x black balls out of m , with $x \leq k$, and $k-x$ red balls out of n red balls, and you have $\binom{n}{x} \binom{m}{k-x}$ ways of doing this. But then you still have the choice of what x is, so there are $\sum_{x=0}^k \binom{n}{x} \binom{m}{k-x}$ options in total.

This is two ways of counting the same thing, and therefore

$$\binom{n+m}{k} = \sum_{x=0}^k \binom{n}{x} \binom{m}{k-x}.$$

5 Continuous random variables

In the previous section, we saw that pmfs are a natural way to describe discrete random variables; but they are not useful at all for continuous random variables. Indeed, if X is a continuous random variable, then usually $\mathbb{P}(X = x) = 0$ for all x , so the pmf is identically zero. However, we also saw, in Theorems 4.1, 4.2 and 4.3 that we can characterise *any* random variable by simply stating its cumulative distribution function (cdf). We will now give a review of some of the most common distributions of continuous random variables via their cdfs.

5.1 Common continuous random variables

5.1.1 Uniform distribution

The uniform distribution can be regarded as a continuous analogue of the equally likely outcomes in finite sample spaces that we saw in Section 2, such as tossing a coin, rolling a fair die, or drawing a card from a well shuffled deck. In the discrete case, the probability of an event was proportional to the number of points in the event; similarly, the probability that a uniform random variable lies within an interval is proportional to the length of the interval. We define it as follows (see Figure 20 for illustration):

Definition 5.1 (Uniform distribution). Suppose that $a, b \in \mathbb{R}$ satisfy $a < b$. We say that the random variable X has a *Uniform* distribution on the interval $[a, b]$, $X \sim \text{Unif}([a, b])$, if its cumulative distribution function is

$$F_X(x) = \mathbb{P}(X \leq x) = \begin{cases} 0, & x \leq a, \\ \frac{x-a}{(b-a)}, & a < x \leq b, \\ 1, & x > b. \end{cases}$$

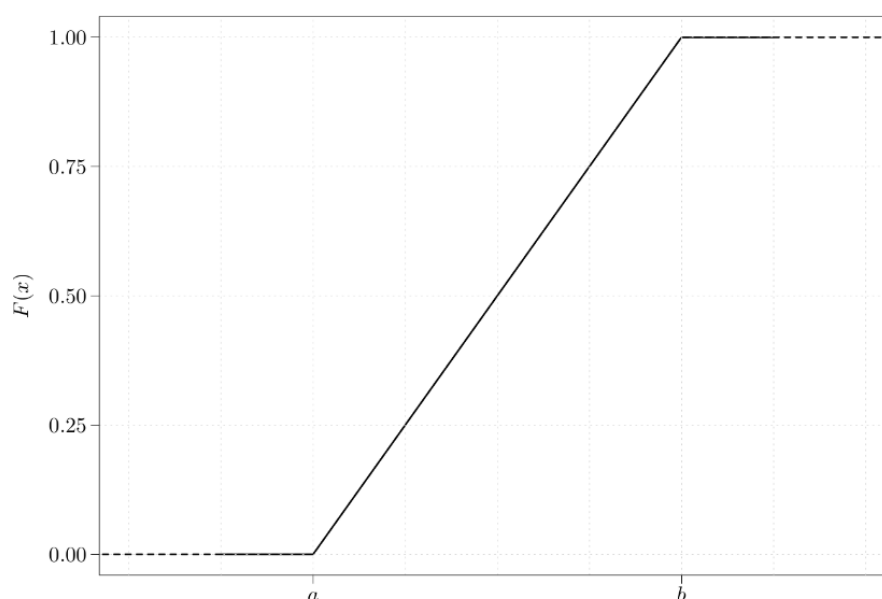


Figure 20: The cdf of a $\text{Unif}([a, b])$.

To verify that Definition 5.1 gives a valid cdf (recalling Theorem 4.1), we note that F_X as defined above is continuous (we will not check this rigorously in the sense of epsilons and deltas, but it is constant on $x \leq a$ and $x \geq b$, linear on $x \in [a, b]$, and continuous at $x = a$ and $x = b$). Moreover, for any $x, y \in [a, b]$, such that $x < y$, we have

$$0 \leq \frac{x-a}{b-a} < \frac{y-a}{b-a} \leq 1,$$

so F_X is non-decreasing. Finally, we observe that $F_X(+\infty) = F_X(b) = 1$ and $F_X(-\infty) = F_X(a) = 0$, so F_X is a valid distribution function.

Remark: Let $u, v \in \mathbb{R}$ satisfy $a \leq u < v \leq b$. Then, by Theorem 4.2,

$$\begin{aligned} \mathbb{P}(X \in (u, v]) &= F_X(v) - F_X(u) \\ &= \frac{v-a}{b-a} - \frac{u-a}{b-a} \\ &= \frac{v-u}{b-a}, \end{aligned}$$

so we see that $\mathbb{P}(X \in (u, v])$ depends only on the length $v - u$ of the interval, and not on the location of the interval. In other words, all intervals $[u, v]$ of equal length are equally likely and in this sense the distribution is **uniform**.

Example 5.1. An important special case of uniform distribution is $X \sim \text{Unif}([0, 1])$ which can be used for constructing more complicated random variables. The cdf of $\text{Unif}([0, 1])$ is

$$F_X(x) = \begin{cases} 0, & x \leq 0 \\ x, & 0 < x \leq 1 \\ 1, & x > 1. \end{cases}$$

Example 5.2. Let $X \sim \text{Unif}([0, 1])$. Find the distribution of $Y = 360X$.

$$\mathbb{P}(Y \leq y) = \mathbb{P}(360X \leq y) = \mathbb{P}\left(X \leq \frac{y}{360}\right) = \begin{cases} 0, & y \leq 0, \\ \frac{y}{360}, & 0 < y \leq 360, \\ 1, & y > 360, \end{cases}$$

so we see that $Y \sim \text{Unif}([0, 360])$.

5.1.2 Exponential distribution

Exponential random variables are often used for modelling the time one has to wait until the occurrence of a specific event. Examples of physical phenomena that follow closely an exponential distribution are the time between the decay of atoms in a radioactive sample, the time between the arrival of two consecutive customers arriving at a till, or the time between phone calls to a customer service line. It can loosely be thought of as a continuous counterpart to the geometric distribution. We define the exponential distribution as follows (see Figure 21 for illustration).

Definition 5.2 (Exponential distribution). A random variable X has an *Exponential* distribution with parameter $\lambda > 0$, $X \sim \text{Exp}(\lambda)$, if its cumulative distribution function is

$$F_X(x) = \mathbb{P}(X \leq x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-\lambda x}, & x \geq 0. \end{cases}$$

As e^x is a continuous function, and $F_X(0) = 1 - e^0 = 1 - 1 = 0$, we conclude that F_X in Definition 5.2 is continuous as well. Moreover, as $e^{-\lambda x}$ is decreasing in x when $\lambda > 0$, we see that F_X is non-decreasing. Finally, $F_X(-\infty) = F_X(0) = 0$ and $F_X(+\infty) = 1 - \lim_{x \rightarrow \infty} e^{-\lambda x} = 1$, so F_X is a valid cumulative distribution function.

Example 5.3. Max the Mechanic has noticed that the number N_t of customers that have visited her garage by time t after opening satisfies $N_t \sim \text{Pois}(\mu t)$. Find the distribution for the (random) time T that Max has to wait until the arrival of the first customer of the day.

We can write

$$\begin{aligned} \{T > t\} &= \{\text{No customers have arrived within } t \text{ minutes}\} \\ &= \{N_t = 0\} \end{aligned}$$

and therefore, for all $t \geq 0$ we have

$$\mathbb{P}(T > t) = \mathbb{P}(N_t = 0) = e^{-\mu t},$$

where the final equality follows from the pmf of a Poisson distribution of parameter μt . Thus $\mathbb{P}(T \leq t) = 1 - e^{-\mu t}$, and therefore $T \sim \text{Exp}(\mu)$.

Example 5.4. Suppose that $X \sim \text{Exp}(\lambda)$ and $Y \sim \text{Exp}(\mu)$ are independent. Let $Z = \min\{X, Y\}$. What is the distribution of Z ?

$\mathbb{P}(Z > z) = \mathbb{P}(\min\{X, Y\} > z) = \mathbb{P}(X > z, Y > z) = \mathbb{P}(X > z)\mathbb{P}(Y > z) = e^{-\lambda z}e^{-\mu z} = e^{-(\lambda+\mu)z}$,
so $\mathbb{P}(Z \leq z) = 1 - e^{-(\lambda+\mu)z}$, and therefore $Z \sim \text{Exp}(\lambda + \mu)$.

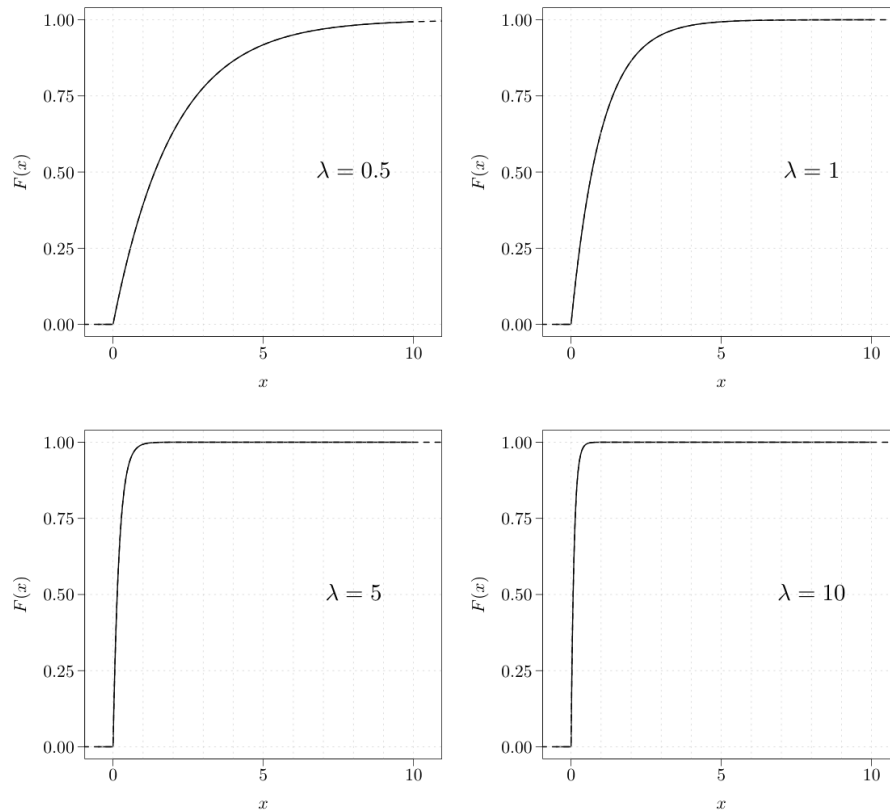


Figure 21: The cdf of an $\text{Exp}(\lambda)$ for $\lambda = 1/2, 1, 5, 10$.

Example 5.5. Suppose that $X \sim \text{Exp}(\lambda)$. Let $x, y > 0$. Calculate $\mathbb{P}(X \leq x + y \mid X > x)$.

$$\mathbb{P}(X \leq x+y \mid X > x) = \frac{\mathbb{P}(x < X \leq x+y)}{\mathbb{P}(X > x)} = \frac{F_X(x+y) - F_X(x)}{1 - F_X(x)} = \frac{1 - e^{-\lambda(x+y)} - (1 - e^{-\lambda x})}{e^{-\lambda x}} = 1 - e^{-\lambda y} = \mathbb{P}(X \leq y).$$

This is called the *memoryless* property of the exponential distribution: if X is exponentially distributed with parameter λ , and we know that X is larger than x , then $X - x$ is again exponentially distributed with parameter λ .

5.1.3 Normal distribution

One of the most frequently-used continuous random variables is the *normal* or *Gaussian* distribution¹³ which we define as follows (see Figures 22, 23 and 24 for illustration).

Definition 5.3 (Normal distribution). For any $\mu \in \mathbb{R}$ and $\sigma^2 \in (0, +\infty)$, a random variable X has a *Normal* or *Gaussian* distribution with parameters μ and σ^2 , $X \sim N(\mu, \sigma^2)$, if its cumulative distribution function is

$$F_X(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(u-\mu)^2} du.$$

As an integral of a continuous function, F_X in Definition 5.3 is continuous. Moreover, because the

¹³Johann Carl Friedrich Gauss (1777-1855).

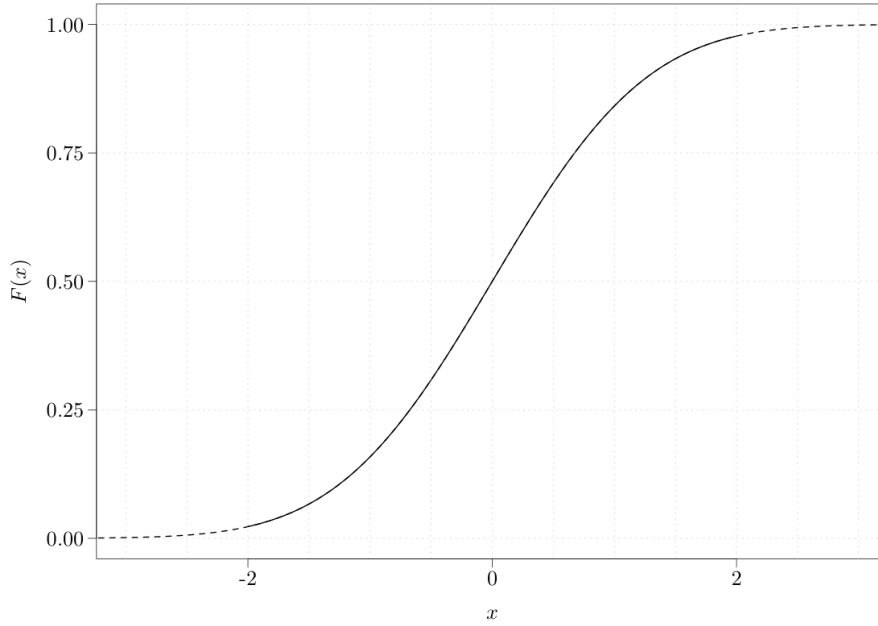


Figure 22: The cdf of an $N(\mu, \sigma^2)$ for $\mu = 0$ and $\sigma^2 = 1$.

integrand is positive, we have for any $x, y \in \mathbb{R}$ such that $x < y$

$$\begin{aligned} F_X(y) &= \int_{-\infty}^y \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(u-\mu)^2} du \\ &= \int_{-\infty}^x \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(u-\mu)^2} du + \int_x^y \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(u-\mu)^2} du \\ &\geq \int_{-\infty}^x \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(u-\mu)^2} du = F_X(x), \end{aligned}$$

so F_X is non-decreasing. The proof of

$$F_X(+\infty) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(u-\mu)^2} du = 1, \quad (51)$$

is technical and requires skills beyond this course and hence the proof is omitted. However, from this it can also be shown that $F_X(-\infty) = 0$ (the proof is omitted) and hence F_X is a valid cumulative distribution function.

Remarks:

1. Although the cumulative distribution function of a normal distribution cannot be written in terms of common functions, it is perhaps the most commonly encountered distribution in applications, because of its *universality* as a limit; you will see this in the form of the Central Limit Theorem in semester 2.
2. The distribution $N(0, 1)$ is typically referred to as the **standard normal distribution** and its cumulative distribution function is typically denoted by Φ .
3. Accurate and efficient numerical approximations of Φ are implemented in all major computing software. By appropriate transformations, the distribution function for any normal distribution can be evaluated.

5.2 Probability density functions

Cumulative distribution functions (cdfs) offer a consistent characterisation of distributions and independence across discrete and continuous random variables. However, they are not well-suited to visualising distributions, and are often less intuitive to use than probability mass functions (pmfs) for discrete random variables. We would, therefore, like a suitable continuous analogue to the pmf. This is the role of the **probability density function (pdf)**.

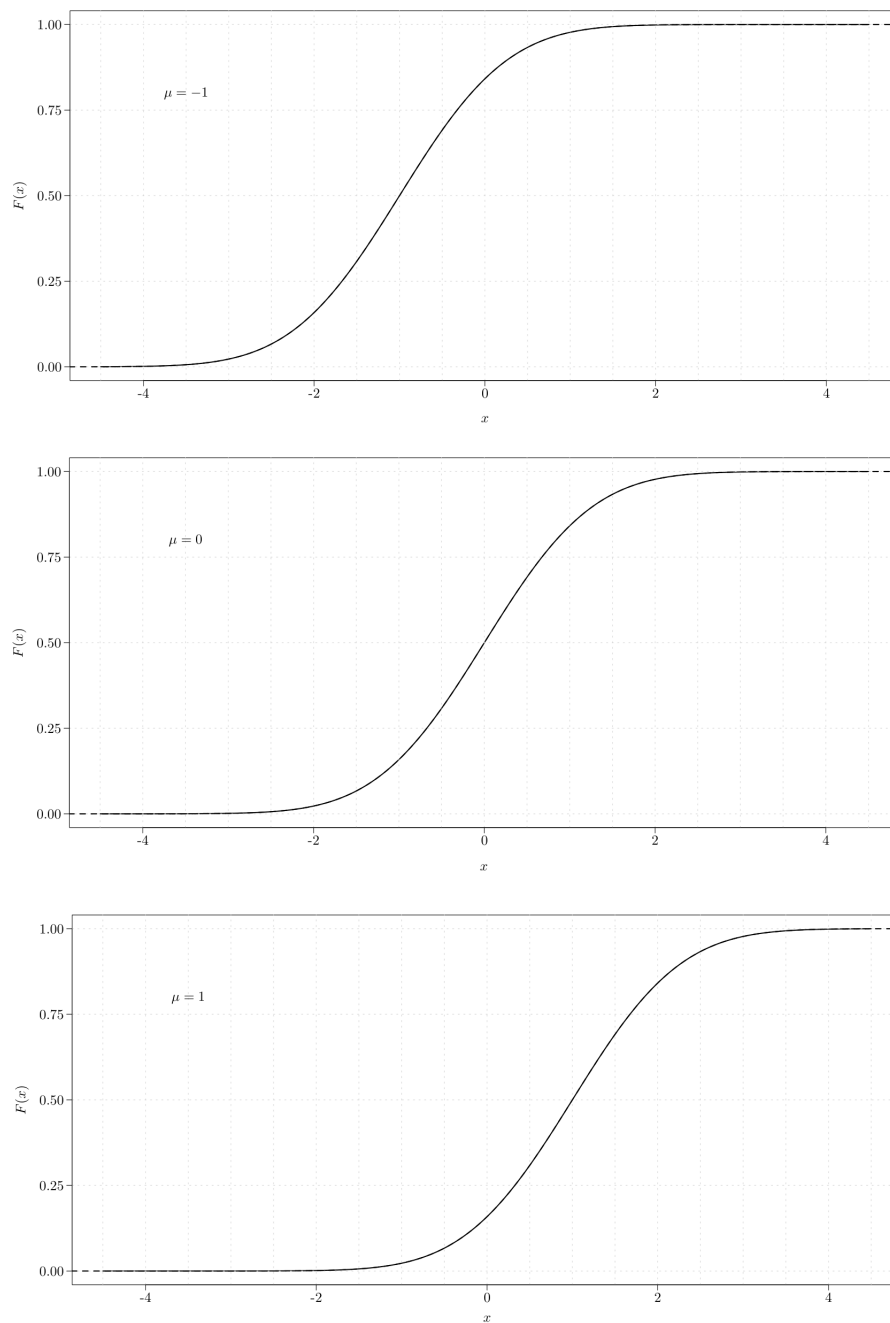


Figure 23: The cdf of an $N(\mu, \sigma^2)$ for $\mu = -1, 0, 1$ and $\sigma^2 = 1$.

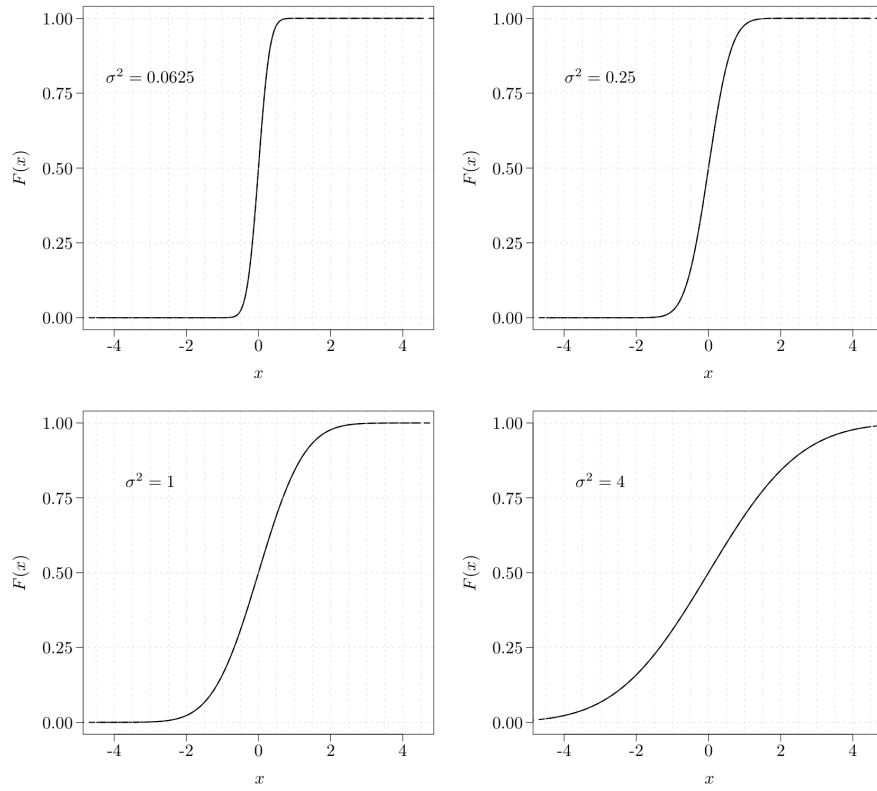


Figure 24: The cdf of an $N(\mu, \sigma^2)$ for $\mu = 0$ and $\sigma^2 = 1/16$ (0.0625), $1/4$ (0.25), 1, 4.

Definition 5.4 (Probability density function). Let X be a continuous random variable with support S and cumulative distribution function F_X . If there exists a function $f : \mathbb{R} \rightarrow [0, +\infty)$ such that $f(x) = 0$ for all $x \notin S$ and

$$F_X(x) = \int_{-\infty}^x f(u) \, du, \quad (52)$$

for all $x \in \mathbb{R}$, then f is said to be a *probability density function (pdf)* for X . We sometimes write f_X for the pdf of a continuous random variable X , in line with the notation for the pmf of a discrete random variable.

Remarks:

1. By Definition 5.4, the probability density function is not unique because individual points of f can be changed without changing the value of the integral in Equation (52). Typically, the probability density function is assumed to refer to the right-continuous function f that satisfies Equation (52).
2. By Theorem 4.2, we see that the probability of the event $X \in (a, b]$ can be obtained by integrating the probability density function of X over the interval $(a, b]$, i.e.,

$$\mathbb{P}(X \in (a, b]) = F_X(b) - F_X(a) = \int_{-\infty}^b f_X(u) \, du - \int_{-\infty}^a f_X(u) \, du = \int_a^b f_X(u) \, du. \quad (53)$$

This means that the probability of the event $(a, b]$ is the **area under the probability density function** across $(a, b]$, as illustrated in Figure 25.

3. We also see that

$$1 = F_X(+\infty) = \int_{-\infty}^{\infty} f(u) \, du,$$

i.e. the total area under the probability density function is always equal to 1.

4. In all the cases (of continuous random variables) that we have met so far, f_X is *more or less* the derivative of F_X ; that is, the pdf is the derivative of the cdf. This is not a coincidence; the Fundamental Theorem of Calculus says that under certain conditions, integration and differentiation are inverse operations. But since we haven't met the Fundamental Theorem of Calculus yet, we

don't know what those conditions are. And even in the simple case of the uniform distribution, F_X is not differentiable at either a or b , so it is not *quite* true to say that f_X is the derivative of F_X (though it is true at all but those two points).

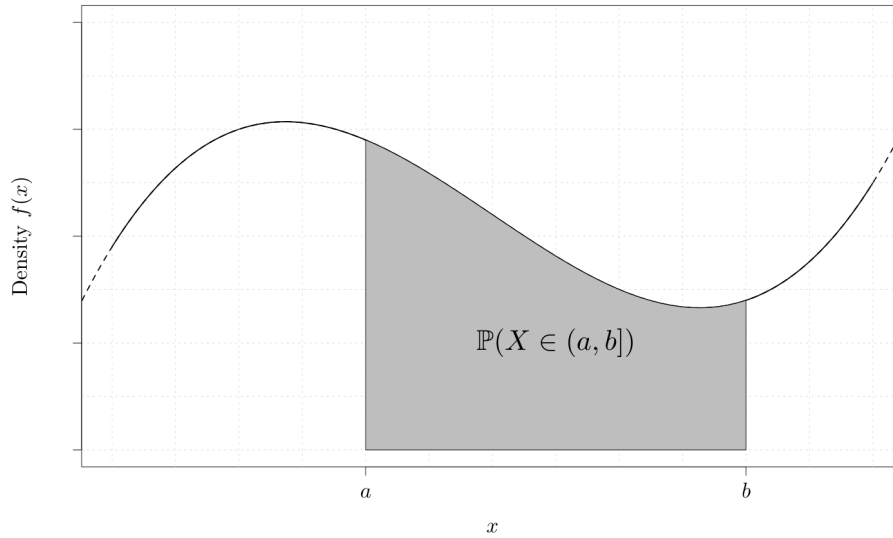


Figure 25: Integrating the probability density function over the interval $(a, b]$.

Following these observations, and with the convention that the pdf is the right-continuous function f_X satisfying Equation (52), we find the following densities for uniform, exponential and normal distributions.

Theorem 5.1 (PDF of a uniform distribution). *If $X \sim \text{Unif}([a, b])$, then*

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leq x < b \\ 0, & \text{otherwise.} \end{cases}$$

is the probability density function (pdf) of X .

Proof. Let F_X be the cumulative distribution function of $\text{Unif}([a, b])$. We need to consider three cases:

1. $x < a$,
2. $a \leq x < b$, and
3. $x \geq b$.

Case 1: for $x < a$,

$$\int_{-\infty}^x f_X(u) \, du = \int_{-\infty}^x 0 \, du = 0 = F_X(x).$$

Case 2: for $a \leq x < b$,

$$\int_{-\infty}^x f_X(u) \, du = \int_{-\infty}^a 0 \, du + \int_a^x \frac{1}{b-a} \, du = 0 + \frac{x-a}{b-a} = F_X(x),$$

Case 3: for $x \geq b$,

$$\int_{-\infty}^x f_X(u) \, du = \int_{-\infty}^a 0 \, du + \int_a^b \frac{1}{b-a} \, du + \int_b^x 0 \, du = 0 + \frac{b-a}{b-a} + 0 = 1 = F_X(x),$$

from which we conclude that f_X satisfies Equation (52) for the uniform distribution function F_X . □

Theorem 5.2 (PDF of an exponential distribution). *If $X \sim \text{Exp}(\lambda)$, then*

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

is the probability density function (pdf) of X .

Proof. Let as an exercise. □

Theorem 5.3 (PDF of a normal distribution). *If $X \sim N(\mu, \sigma^2)$, then*

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} \quad \forall x \in \mathbb{R}.$$

is the probability density function (pdf) of X .

Proof. Obvious from the definition of the distribution function for X . □

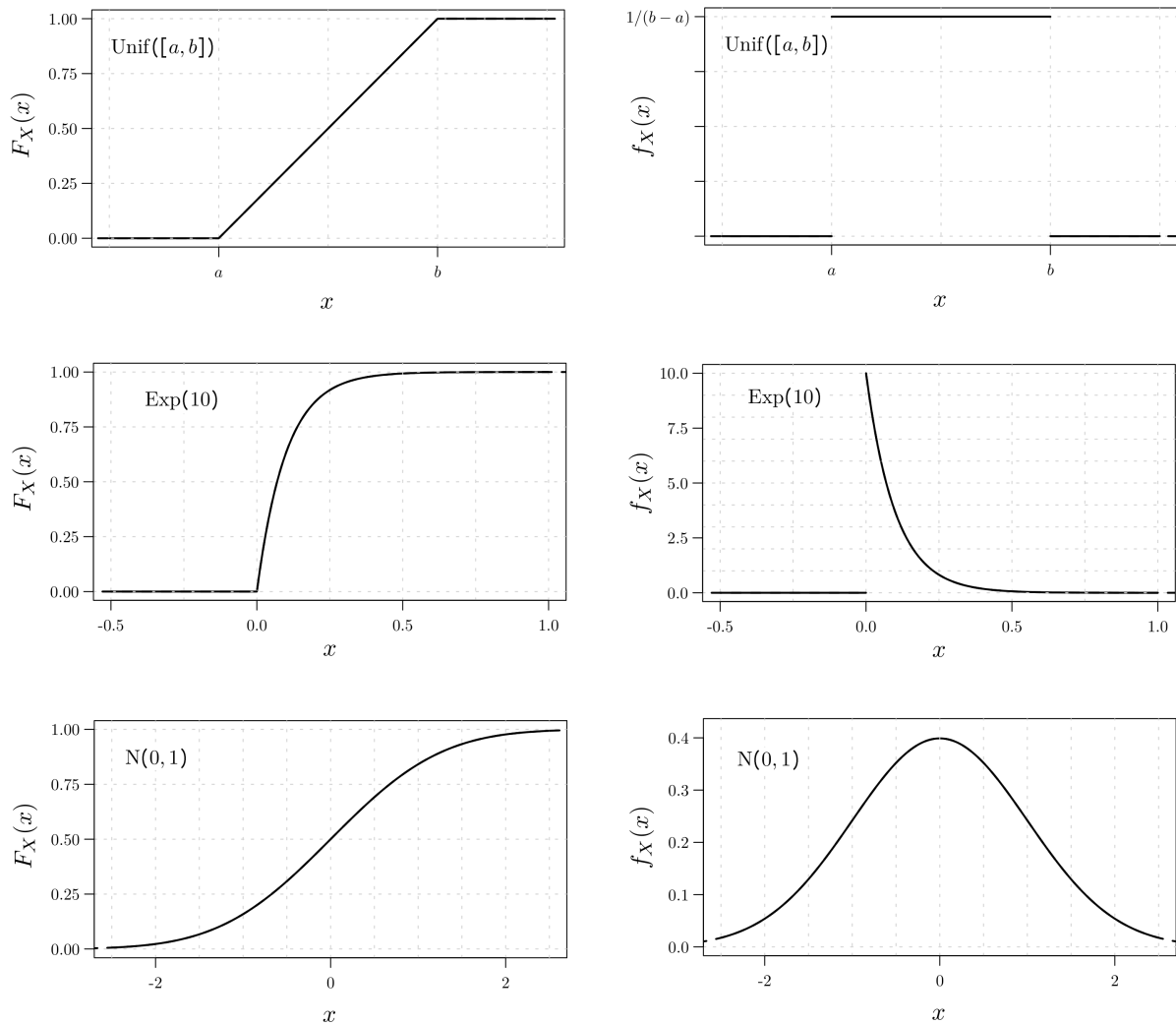


Figure 26: Cumulative distribution functions and probability density functions of $\text{Unif}([a, b])$, $\text{Exp}(10)$ and $N(0, 1)$ distributions.

Example 5.6 (Examples of cumulative distribution and probability density functions). Consider distributions $\text{Unif}([a, b])$, $\text{Exp}(\lambda)$ and $N(\mu, \sigma^2)$, where $\lambda = 10$, $\mu = 0$, and $\sigma^2 = 1$. The cumulative distribution and probability density functions for each of these distributions are shown in Figure 26.

Example 5.7. Let $X \sim \text{Exp}(\lambda)$, where $\lambda = 1$. Find $\mathbb{P}(X \in [1, 2])$.

By Equation (53),

$$\mathbb{P}(X \in [1, 2]) = \int_1^2 e^{-u} du = F_X(2) - F_X(1) = e^{-1} - e^{-2} \approx 0.2325 \text{ (4 dp),}$$

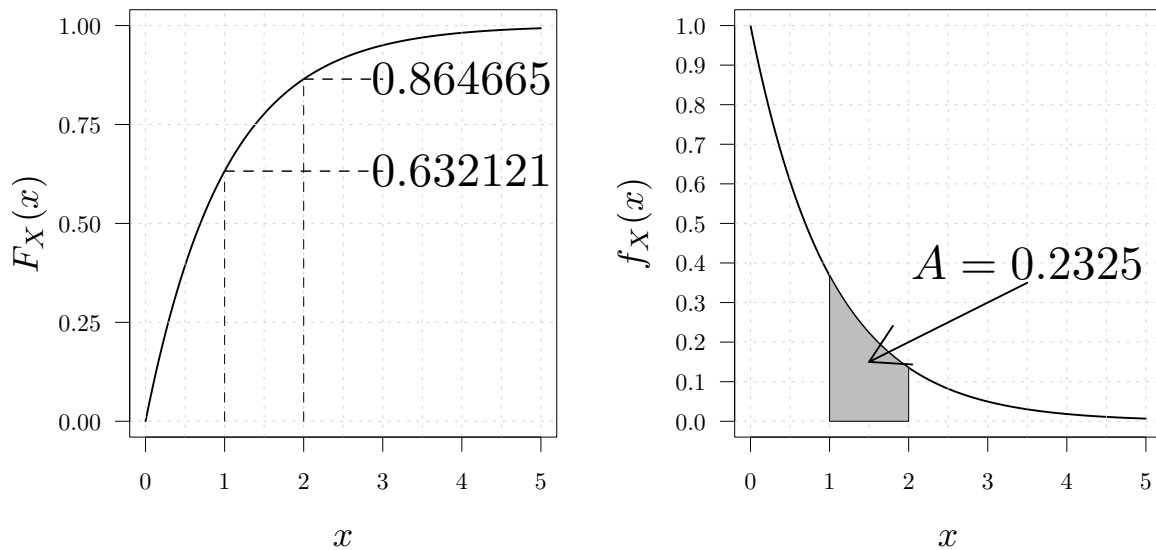


Figure 27: Probability of an event $X \in [1, 2]$ for an exponential random variable of parameter 1.

as illustrated in Figure 27.

Example 5.8 (Example 4.14 revisited). The time between consecutive road accidents can be modelled as an exponentially distributed random variable $T \sim \text{Exp}(\lambda)$ for some λ .

Figure 28 shows a histogram of the times between consecutive road accidents of Example 4.14. The histogram is scaled to match probability density (i.e., the area under the histogram is equal to 1). The picture also shows the probability density function of an exponential distribution whose parameter has been chosen appropriately so that the density matches the histogram as well as possible¹⁴. The exponential distribution appears to be a good model for the time between road accidents.

5.3 Joint distributions and independence of continuous random variables

5.3.1 Joint distributions

As with discrete random variables, it may be the case that we have two continuous random variables X and Y defined on the same probability space Ω and we need to calculate probabilities such as

$$\mathbb{P}(X \in (a, b], Y \in (c, d]). \quad (54)$$

We can do this using the joint cdf for X and Y , defined in Definition 4.11:

$$F_{X,Y}(x, y) = \mathbb{P}(X \leq x, Y \leq y).$$

Recall that the Definition 5.4 of a probability density function for a single random variable relied on integration. Similarly, the definition of a **joint** probability density function relies on integration over a two dimensional domain.

Definition 5.5 (Joint probability density function). Let $X : \Omega \rightarrow S_X$ and $Y : \Omega \rightarrow S_Y$ be random variables defined on the same probability space. A function f which satisfies

$$F_{X,Y}(x, y) = \mathbb{P}(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f(u, v) \, dv \, du, \quad (55)$$

¹⁴Finding this parameter value is a topic in more advanced statistics.

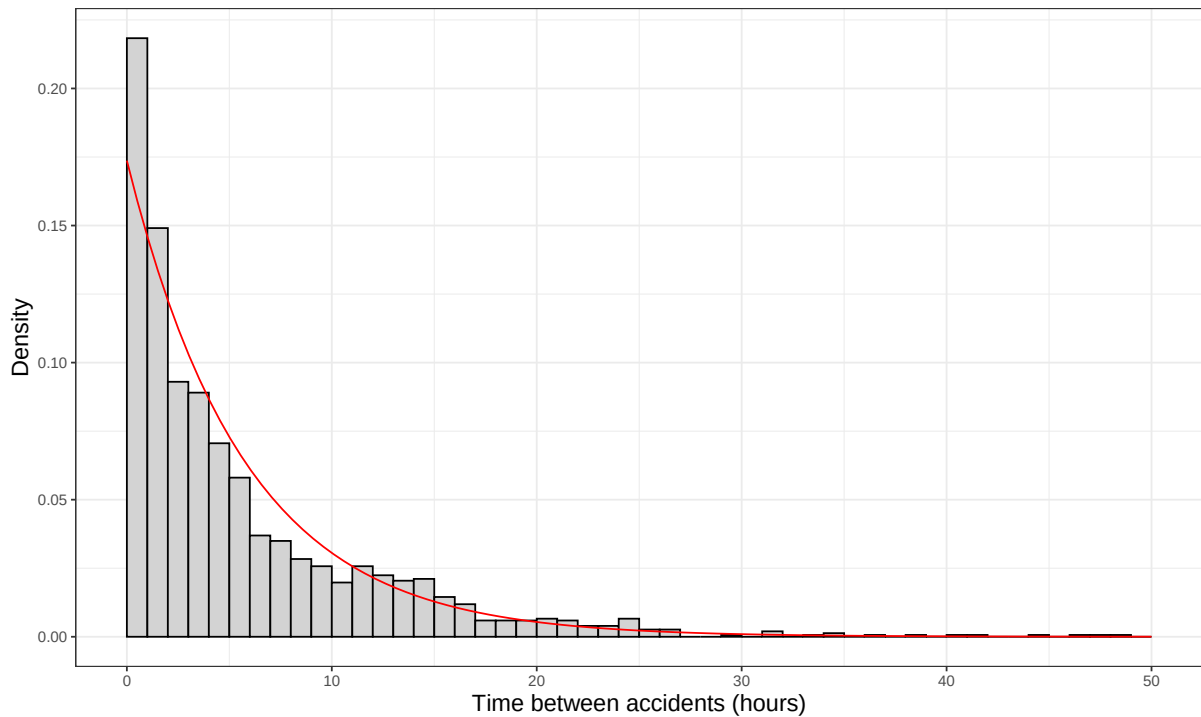


Figure 28: Histogram for the times between consecutive road accidents, and the probability density function of an exponential distribution with well-chosen parameter.

is a joint probability density function of X and Y . We sometimes write $f_{X,Y}$ for the joint pdf of X and Y , in line with the notation for the joint pmf of two discrete random variables.

One subtle issue we face is about the definition of the **double integral** in (55). In particular, it is not at all clear whether the order of integration matters. The way it is written in (55) says that we should take the function $f(u, v)$, integrate over v (holding u fixed, i.e. imagining that it is a constant), and then integrate the result over u . But does it matter if we first integrate $f(u, v)$ over u , holding v fixed, and then integrate the result over v ? That is, do we have

$$\int_{-\infty}^x \left(\int_{-\infty}^y f(u, v) \, dv \right) du = \int_{-\infty}^y \left(\int_{-\infty}^x f(u, v) \, du \right) dv ??$$

It turns out that the answer is, essentially, yes **when f is non-negative**, but the proof of this would require some advanced tools. In this course we won't have to worry, but you should be wary in general about swapping the order of integration. Integration involves taking limits, so swapping the order of integration involves exchanging limits, which can go horribly wrong (you may have encountered this in analysis). Some of the most useful and celebrated results of *measure theory* involve giving nice conditions under which limits and integrals can be exchanged.

Example 5.9. Suppose that $a = 0$, $b = 2$, $c = 0$, $d = 1$, and

$$g(x, y) = \frac{3}{16} x^2 + \frac{1}{2} y.$$

Find

$$\int_a^b \int_c^d g(x, y) \, dy dx.$$

We integrate first with respect to y :

$$\int_0^1 \left(\frac{3}{16} x^2 + \frac{1}{2} y \right) dy = \left[\frac{3}{16} x^2 y + \frac{1}{4} y^2 \right]_0^1 = \frac{3}{16} x^2 + \frac{1}{4}.$$

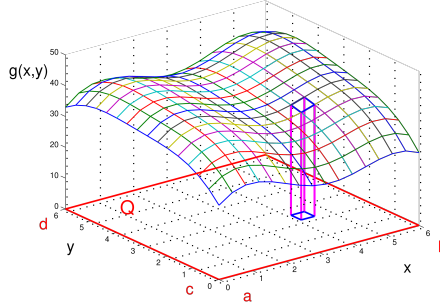


Figure 29: The intuition behind double integrals: as long as $g(x, y)$ is smooth and non-negative, we can think of the double integral of g as approximately the sum of the volumes of all the blocks under the surface; and then it shouldn't matter whether we sum in the x -direction first and then the y -direction, or vice versa.

Then we integrate with respect to x :

$$\int_0^2 \left(\frac{3}{16} x^2 + \frac{1}{4} \right) dx = \left[\frac{1}{16} x^3 + \frac{1}{4} x \right]_0^2 = \frac{1}{2} + \frac{1}{2} = 1.$$

If we do the integration in different order, we have first

$$\int_0^2 \left(\frac{3}{16} x^2 + \frac{1}{2} y \right) dx = \left[\frac{1}{16} x^3 + \frac{1}{2} xy \right]_0^2 = \frac{1}{2} + y$$

and then

$$\int_0^1 \left(\frac{1}{2} + y \right) dy = \left[\frac{1}{2} y + \frac{1}{2} y^2 \right]_0^1 = 1.$$

In either case the answer is 1.

Recall from the proof of Theorem 4.6 that

$$\begin{aligned} \mathbb{P}(a < X \leq b, c < Y \leq d) &= \mathbb{P}(X \leq b, c < Y \leq d) - \mathbb{P}(X \leq a, c < Y \leq d) \\ &= (\mathbb{P}(X \leq b, Y \leq d) - \mathbb{P}(X \leq b, Y \leq c)) \\ &\quad - (\mathbb{P}(X \leq a, Y \leq d) - \mathbb{P}(X \leq a, Y \leq c)) \\ &= F_{X,Y}(b, d) - F_{X,Y}(b, c) - F_{X,Y}(a, d) + F_{X,Y}(a, c) \end{aligned} \quad (56)$$

Moreover,

$$\begin{aligned} F_{X,Y}(b, d) - F_{X,Y}(b, c) &= \int_{-\infty}^b \int_{-\infty}^d f_{X,Y}(x, y) dy dx - \int_{-\infty}^b \int_{-\infty}^c f_{X,Y}(x, y) dy dx \\ &= \int_{-\infty}^b \left(\int_{-\infty}^d f_{X,Y}(x, y) dy - \int_{-\infty}^c f_{X,Y}(x, y) dy \right) dx \\ &= \int_{-\infty}^b \int_c^d f_{X,Y}(x, y) dy dx \end{aligned} \quad (57)$$

and similarly we have

$$-F_{X,Y}(a, d) + F_{X,Y}(a, c) = - \int_{-\infty}^a \int_c^d f_{X,Y}(x, y) dy dx. \quad (58)$$

By combining Equations (56) – (58) we have

$$\begin{aligned} \mathbb{P}(X \in (a, b], Y \in (c, d]) &= \int_{-\infty}^b \int_c^d f_{X,Y}(x, y) dy dx - \int_{-\infty}^a \int_c^d f_{X,Y}(x, y) dy dx \\ &= \int_a^b \int_c^d f_{X,Y}(x, y) dy dx, \end{aligned}$$

so we can calculate the probability that our two random variables fall in a pair of intervals by integrating the joint probability density function over those intervals.

Remarks:

1. We have immediately that the total volume under the joint probability density function is

$$F_{X,Y}(+\infty, +\infty) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f_{X,Y}(x, y) \, dy dx = 1.$$

2. If X and Y have joint probability density function $f_{X,Y}$, then we can find the **marginal distribution** of X by integrating just over y . That is,

$$\mathbb{P}(X \in (a, b]) = \mathbb{P}(X \in (a, b], Y \in (-\infty, \infty)) = \int_a^b \int_{-\infty}^{\infty} f_{X,Y}(x, y) \, dy \, dx$$

or we can calculate the **marginal probability density function** of X via

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) \, dy. \quad (59)$$

3. The discussion above on the double integrals applies to **rectangular** events, such as

$$\{x, y \in \mathbb{R}^2 : x \in (a, b], y \in (c, d]\}$$

only – see Figure 30. Non-rectangular events, such as

$$\{x, y \in \mathbb{R}^2 : x \in [-1, 1], y \in [-\sqrt{1-x^2}, \sqrt{1-x^2}]\},$$

illustrated in Figure 31, will be discussed in more detail in Semester 2.

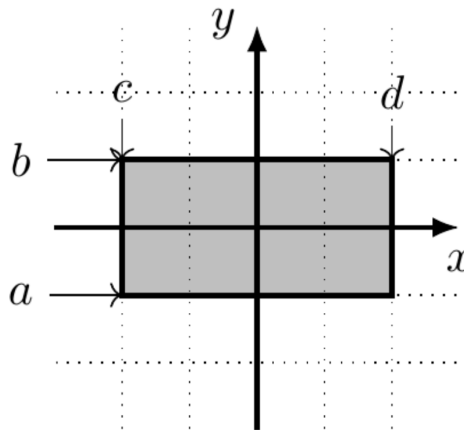


Figure 30: A rectangular event.

5.3.2 Independence

Recall the definition of independence of random variables from Definition 4.12: random variables X and Y are *independent* if

$$\mathbb{P}(X \leq x, Y \leq y) = \mathbb{P}(X \leq x) \mathbb{P}(Y \leq y) \text{ for all } x, y \in \mathbb{R}.$$

This definition applies to both continuous and discrete random variables. It only uses the cdf, which there is no problem defining for any random variable.

However, Theorem 4.5 gave us an alternative characterisation of independence for discrete random variables, in terms of their pmfs. The natural question for continuous random variables is: is there a similar characterisation of independence in terms of pdfs? The answer is yes.

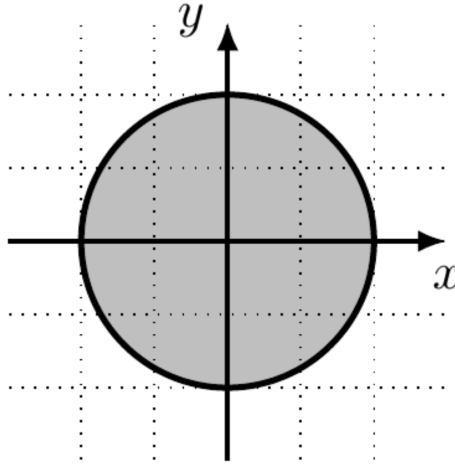


Figure 31: A non-rectangular event.

Theorem 5.4. Suppose that random variables $X : \Omega \rightarrow S_X$ and $Y : \Omega \rightarrow S_Y$ have densities f_X and f_Y respectively, and a joint probability density function $f_{X,Y}$. If

$$f_{X,Y}(x, y) = f_X(x)f_Y(y) \text{ for all } x, y \in \mathbb{R},$$

then X and Y are independent.

Conversely, if we have independent random variables X and Y with probability density functions f_X and f_Y respectively, then $f(x, y) = f_X(x)f_Y(y)$ is a joint probability density function for X and Y .

Proof. Suppose that $f_{X,Y}(x, y) = f_X(x)f_Y(y)$ for all $x, y \in \mathbb{R}$. Then we have

$$\begin{aligned} \mathbb{P}(X \leq x, Y \leq y) &= \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(u, v) \, dv \, du \\ &= \int_{-\infty}^x \int_{-\infty}^y f_X(u)f_Y(v) \, dv \, du \\ &= \int_{-\infty}^x f_X(u) \left(\int_{-\infty}^y f_Y(v) \, dv \right) \, du \\ &= \int_{-\infty}^x f_X(u) \mathbb{P}(Y \leq y) \, du \\ &= \mathbb{P}(X \leq x) \mathbb{P}(Y \leq y) \end{aligned}$$

so X and Y are independent.

On the other hand, suppose that X and Y are independent. Then

$$\begin{aligned} \int_{-\infty}^x \int_{-\infty}^y f_X(u)f_Y(v) \, dv \, du &= \int_{-\infty}^x f_X(u) \left(\int_{-\infty}^y f_Y(v) \, dv \right) \, du \\ &= F_Y(y) \int_{-\infty}^x f_X(u) \, du \\ &= F_X(x)F_Y(y) \\ &= F_{X,Y}(x, y) \quad \text{[by independence]} \end{aligned}$$

so $f_X(u)f_Y(v)$ is a joint pdf for X and Y . □

Example 5.10. Suppose that $X \sim \text{Exp}(1)$ and $Y \sim \text{Exp}(2)$ are independent. Find the joint probability density function of X and Y .

We know the densities of X and Y , so by Theorem 5.4

$$f_{X,Y}(x, y) = f_X(x)f_Y(y) = e^{-x} \times 2e^{-2y} = 2e^{-x-2y},$$

for all $x, y \geq 0$, and 0 otherwise.

Example 5.11. Suppose that $X \sim \text{Exp}(1)$ and $Y \sim \text{Exp}(2)$ are independent. Find $\mathbb{P}(X \in (1, 2], Y \in (1, 2])$.

Because X and Y are independent, we have

$$\begin{aligned}\mathbb{P}(X \in (1, 2], Y \in (1, 2]) &= \mathbb{P}(X \in (1, 2]) \mathbb{P}(Y \in (1, 2]) \\ &= \int_1^2 f_X(x) dx \times \int_1^2 f_Y(y) dy \\ &= [F_X(2) - F_X(1)] \times [F_Y(2) - F_Y(1)] \\ &= (e^{-2} - e^{-1})(e^{-4} - e^{-2}) = 0.0272 \text{ (4 dp)}.\end{aligned}$$

Theorem 5.5. Suppose that X and Y are random variables on the same probability space.

- (a) If X and Y are discrete with joint pmf $f_{X,Y}$, then X and Y are independent if and only if $f_{X,Y}(x, y)$ can be written as a product of functions $g(x)$ and $h(y)$ depending only on x and y respectively.
- (b) Suppose that X and Y are continuous and have a joint pdf $f_{X,Y}$, that can be written as a product of functions $g(x)$ and $h(y)$ depending only on x and y respectively. Then X and Y are independent.

Proof. (a) We know from Theorem 4.5 that if X and Y are independent, then $f_{X,Y}(x, y) = f_X(x)f_Y(y)$, so we are done by setting $g = f_X$ and $h = f_Y$. Now suppose conversely that $f_{X,Y}(x, y) = g(x)h(y)$ for all x and y . Then we can calculate the marginal distributions:

$$f_X(x) = \sum_{j \in S_Y} f_{X,Y}(x, j) = \sum_{j \in S_Y} g(x)h(j) = g(x) \sum_{j \in S_Y} h(j)$$

and similarly

$$f_Y(y) = h(y) \sum_{i \in S_X} g(i).$$

Multiplying these together, we get

$$f_X(x)f_Y(y) = g(x)h(y) \sum_{i \in S_X} g(i) \sum_{j \in S_Y} h(j) = f_{X,Y}(x, y) \sum_{i \in S_X} \sum_{j \in S_Y} f_{X,Y}(i, j)$$

but the last double sum must equal 1 by the definition of joint pmf. So

$$f_X(x)f_Y(y) = f_{X,Y}(x, y)$$

and we deduce that X and Y are independent.

- (b) The same as above but with integrals in place of sums:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, v) dv = \int_{-\infty}^{\infty} g(x)h(v) dv = g(x) \int_{-\infty}^{\infty} h(v) dv$$

and similarly

$$f_Y(y) = h(y) \int_{-\infty}^{\infty} g(u) du.$$

Multiplying these together, we get

$$f_X(x)f_Y(y) = g(x)h(y) \int_{-\infty}^{\infty} g(u) du \int_{-\infty}^{\infty} h(v) dv = f_{X,Y}(x, y) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(u, v) du dv$$

but the last double integral must equal 1 by the definition of joint pdf. So

$$f_X(x)f_Y(y) = f_{X,Y}(x, y)$$

and we deduce from Theorem 5.4 that X and Y are independent. □

As a final note in this section, we point out that any random variable X with a bounded probability density function satisfies $\mathbb{P}(X = x) = 0$ for all $x \in \mathbb{R}$.

Theorem 5.6. *Suppose that X is a random variable with a bounded probability density function f_X ; that is, there exists $C > 0$ such that $f_X(x) \leq C$ for all $x \in \mathbb{R}$. Then $\mathbb{P}(X = x) = 0$ for all $x \in \mathbb{R}$.*

Proof. This proof is **non-examinable**, not because it is difficult, but because it is an analysis proof rather than a probability proof.

Suppose that $f_X(x) \leq C$ for all $x \in \mathbb{R}$. Fix $\varepsilon > 0$. Then for any $x \in \mathbb{R}$,

$$\mathbb{P}(X = x) \leq \mathbb{P}(X \in (x - \varepsilon/C, x]) = F_X(x) - F_X(x - \varepsilon/C) = \int_{x - \varepsilon/C}^x f_X(u) du \leq \frac{\varepsilon}{C} \cdot C = \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, and we showed that $\mathbb{P}(X = x) \leq \varepsilon$, we must have $\mathbb{P}(X = x) = 0$. □

We recall that not every continuous random variable has a probability density function, and there absolutely are (many) continuous random variables with $\mathbb{P}(X = x) > 0$ for some value(s) of x . See question 6 on problem sheet 9 for an example. But the most common continuous random variables, such as uniform, exponential and normal random variables, have bounded probability density functions and therefore satisfy $\mathbb{P}(X = x) = 0$.

6 Expectation and variance

The expectation $\mathbb{E}[X]$ of a random variable X is a measure of where we expect X to be “on average”, according to its distribution. You might think of $\mathbb{E}[X]$ as your best guess for the value of X before you perform the experiment; but how realistic this is depends on the situation (for example, if X is the outcome of a single roll of a dice, then $\mathbb{E}[X] = 7/2$, but the probability that X actually equals $7/2$ is, of course, zero).

The variance $\text{Var}(X)$ of a random variable X is a measure of how “spread out” the distribution of X is, relative to its expectation.

6.1 Expectation of a discrete random variable

Definition 6.1 (Expectation of a discrete random variable). Let X be a discrete random variable with support S . If $\sum_{x \in S} |x| \mathbb{P}(X = x) < \infty$ then the expectation (or expected value) of X is

$$\mathbb{E}[X] = \sum_{x \in S} x \mathbb{P}(X = x). \quad (60)$$

Remark: The requirement $\sum_{x \in S} |x| \mathbb{P}(X = x) < \infty$ ensures that the sum in (60) makes sense. You might have seen in analysis lectures that we need to be careful with infinite sums: in some cases, adding up terms in different orders can change the value of the sum. The requirement $\sum_{x \in S} |x| \mathbb{P}(X = x) < \infty$ ensures this can’t happen. Note that if S is finite, say $S = \{x_1, \dots, x_n\}$, then

$$\mathbb{E}[X] = \sum_{i=1}^n x_i \mathbb{P}(X = x_i)$$

contains only a finite number of terms and so the sum is automatically finite.

Example 6.1. Let X be the score obtained on a single roll of a fair dice. That is,

$$\mathbb{P}(X = x) = \frac{1}{6} \text{ for } x \in \{1, 2, 3, 4, 5, 6\}$$

and $\mathbb{P}(X = x) = 0$ otherwise. Find $\mathbb{E}[X]$.

$$\begin{aligned}
\mathbb{E}[X] &= \sum_{x=1}^6 x \mathbb{P}(X = x) \\
&= 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 5\left(\frac{1}{6}\right) + 6\left(\frac{1}{6}\right) \\
&= \frac{21}{6} = \frac{7}{2}.
\end{aligned}$$

Example 6.2. Suppose that $X \sim \text{Geom}(p)$. Show that $\mathbb{E}[X] = \frac{1}{p}$.

If $X \sim \text{Geom}(p)$ then $\mathbb{P}(X = x) = (1-p)^{x-1}p$ for $x \in \{1, 2, \dots\}$. Thus,

$$\begin{aligned}
\mathbb{E}[X] &= \sum_{x=1}^{\infty} x(1-p)^{x-1}p \\
&= p(1 + 2(1-p) + 3(1-p)^2 + \dots) \\
&= -p\left(\frac{d}{dp}1 + \frac{d}{dp}(1-p) + \frac{d}{dp}(1-p)^2 + \frac{d}{dp}(1-p)^3 + \dots\right) \\
&= -p\frac{d}{dp}\left(\sum_{i=0}^{\infty}(1-p)^i\right) && [\text{see note below}] \\
&= -p\frac{d}{dp}\frac{1}{p} && [\text{as } |1-p| < 1 \text{ for } p \in (0, 1)] \\
&= \frac{1}{p}.
\end{aligned}$$

Note: making the fourth equality rigorous requires some analysis which is beyond the scope of this course, but it is not too difficult: we just show that the infinite sum on the second line is within ε of the sum of the first n terms when n is large, and use linearity of the derivative on the finite sum. The fourth line then involves the derivative of a finite geometric sum, which we can calculate, take the derivative, and then take the limit as $n \rightarrow \infty$. We won't do the details here!

6.2 Expectation of a continuous random variable

The interpretation of the expectation of a continuous random variable is precisely the same as for discrete random variables, but since the definition of $\mathbb{E}[X]$ for discrete X relies on the pmf, a different (but analogous) definition for continuous X is needed.

Definition 6.2 (Expectation of a continuous random variable). Let X be a continuous random variable with a probability density function $f_X : \mathbb{R} \rightarrow [0, +\infty)$. If $\int_{-\infty}^{+\infty} |x| f_X(x) dx < +\infty$, then

$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} x f_X(x) dx. \quad (61)$$

Remarks:

1. Absolute convergence $\int_{-\infty}^{+\infty} |x| f_X(x) dx < +\infty$ is needed for the same reason as in the discrete case: we need the integral $\int_{-\infty}^{+\infty} x f_X(x) dx$ to be well-defined..
2. The definition is similar to Definition 6.1, but the sum over S is replaced by an integral, and the pmf $\mathbb{P}(X = x)$ is replaced with a probability density function $f_X(x)$.

Example 6.3. Suppose that $X \sim \text{Exp}(\lambda)$. Find $\mathbb{E}[X]$.

By Definition 6.2 we have

$$\mathbb{E}[X] = \int_0^{\infty} x \lambda e^{-\lambda x} dx.$$

Integration by parts¹⁵ gives

$$\mathbb{E}[X] = \left[-\frac{(\lambda x + 1)e^{-\lambda x}}{\lambda} \right]_0^{\infty} = \frac{1}{\lambda}.$$

We often think of exponential random variables as “waiting times” and call the parameter λ the “rate”. Example 5.3 gave a clue as to why: the time to the first arrival in a Poisson queue with rate λ is exponentially distributed with parameter λ . It turns out that also the times between any pair of arrivals are also exponentially distributed with parameter λ . For example, if we assume that customers arrive at Fresh randomly at rate (say) 12 per minute, then the time in minutes between one customer and the next joining the queue in Fresh is exponentially distributed with parameter 12, so we expect it to take 1/12 minutes (or 5 seconds) for the next customer to arrive. “Higher rate means less time to wait.”

6.3 Properties of expectation

6.3.1 General results

Theorem 6.1 (Non-negative RVs have non-negative expectation). *If X is a non-negative random variable, i.e. it has support $S \subset [0, +\infty)$, then $\mathbb{E}[X] \geq 0$.*

Proof. In the discrete case, we have $x \geq 0$ for all $x \in S$, so $x\mathbb{P}(X = x) \geq 0$ for all $x \in S$, so

$$\mathbb{E}[X] = \sum_{x \in S} x\mathbb{P}(X = x) \geq 0.$$

Similarly, if X is continuous and has a probability density function f_X , then since $S \subset [0, +\infty)$, we have (by the definition of pdf) $f_X(x) = 0$ for all $x < 0$. Thus

$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} x f_X(x) dx = \int_0^{+\infty} x f_X(x) dx \geq 0.$$

□

Theorem 6.2 (The expectation of a constant is that constant). *If $\mathbb{P}(X = a) = 1$ for some $a \in \mathbb{R}$, then $\mathbb{E}[X] = a$. Thus, $\mathbb{E}[a] = a$.*

Proof. This is trivial: $\mathbb{E}[X] = a\mathbb{P}(X = a) = a$. □

Now we come to a much less trivial property, with an appropriate name. The key question is: for a random variable X and a function $g : \mathbb{R} \rightarrow \mathbb{R}$, how do we calculate $\mathbb{E}[g(X)]$? For example, how do we calculate $\mathbb{E}[X^2]$? It is worth pointing out first of all that we absolutely do **NOT** have $\mathbb{E}[g(X)] = g(\mathbb{E}[X])$ (this could perhaps be called the law of the incompetent statistician). But the next simplest possibility *is* true.

Theorem 6.3 (The law of the unconscious statistician). *1. Suppose that X is a discrete random variable with support S . For any function $g : \mathbb{R} \rightarrow \mathbb{R}$ such that $\sum_{x \in S} |g(x)|\mathbb{P}(X = x) < \infty$, we have*

$$\mathbb{E}[g(X)] = \sum_{x \in S} g(x)\mathbb{P}(X = x)$$

.

2. Suppose that X is a continuous random variable, with a probability density function f_X . If $g : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $\int_{-\infty}^{+\infty} |g(x)|f_X(x) dx < \infty$, then

$$\mathbb{E}[g(X)] = \int_{-\infty}^{+\infty} g(x)f_X(x) dx.$$

¹⁵You will not need to do any difficult integrals in the exam. The exam is aiming to test your knowledge and skill in probability, not integration. Any non-trivial integrals required will be stated as part of the question.

3. Suppose that $X_1, \dots, X_n$ are discrete random variables with support $S_1, \dots, S_n$ respectively. For any function $g : \mathbb{R}^n \rightarrow \mathbb{R}$ with

$$\sum_{x_1 \in S_1} \cdots \sum_{x_n \in S_n} |g(x_1, \dots, x_n)| \mathbb{P}(X_1 = x_1, \dots, X_n = x_n) < \infty$$

we have

$$\mathbb{E}[g(X_1, \dots, X_n)] = \sum_{x_1 \in S_1} \cdots \sum_{x_n \in S_n} g(x_1, \dots, x_n) \mathbb{P}(X_1 = x_1, \dots, X_n = x_n).$$

4. Suppose that $X_1, \dots, X_n$ are continuous random variables with joint probability density function $f_{X_1, \dots, X_n}$. If $g : \mathbb{R}^n \rightarrow \mathbb{R}$ satisfies

$$\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} |g(x_1, \dots, x_n)| f_{X_1, \dots, X_n}(x_1, \dots, x_n) dx_1 \cdots dx_n < \infty,$$

then

$$\mathbb{E}[g(X_1, \dots, X_n)] = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} g(x_1, \dots, x_n) f_{X_1, \dots, X_n}(x_1, \dots, x_n) dx_1 \cdots dx_n.$$

Proof. We will only prove claim 1. Claim 2 will be discussed in the second semester. Claim 3 is a natural extension of claim 1 and its proof is the same but with more notation. Similarly for claim 4 as an extension of claim 2.

Claim 1.

Idea: Let $Y = g(X)$, write out the definition of $\mathbb{E}[Y]$ and convert anything about Y into something about X . Will need to construct a partition of S according to the possible values of $g(x)$.

Let $Y = g(X)$. Then since X is a discrete rv, Y is also a discrete random variable. Let S_Y be the support of Y . Now,

$$\mathbb{E}[g(X)] = \mathbb{E}[Y] = \sum_{y \in S_Y} y \mathbb{P}(Y = y).$$

We want to change from Y to X on the right-hand side, so for each $y \in S_Y$ we let A_y be the set of $x \in S$ that are mapped to y by g , i.e.

$$A_y = \{x \in S : g(x) = y\}.$$

Then

$$\mathbb{P}(Y = y) = \mathbb{P}(X \in A_y) = \sum_{x \in A_y} \mathbb{P}(X = x).$$

Thus

$$\mathbb{E}[g(X)] = \sum_{y \in S_Y} y \sum_{x \in A_y} \mathbb{P}(X = x) = \sum_{y \in S_Y} \sum_{x \in A_y} y \mathbb{P}(X = x).$$

Now, continuing the theme of changing y to x on the right-hand side, notice that for every $x \in A_y$, we have $g(x) = y$, so

$$\mathbb{E}[g(X)] = \sum_{y \in S_Y} \sum_{x \in A_y} g(x) \mathbb{P}(X = x).$$

We almost have the result we want; it only remains to notice that each $x \in S$ belongs to A_y for exactly one y , so that the double sum on the right-hand side above is actually summing over all $x \in S$. $\square$

Example 6.4. Suppose that X and Y are random variables with support S_X and S_Y respectively. If X and Y are both discrete, then

$$\begin{aligned} \mathbb{E}[X^2] &= \sum_{x \in S_X} x^2 \mathbb{P}(X = x) \\ \mathbb{E}[XY] &= \sum_{x \in S_X} \sum_{y \in S_Y} xy \mathbb{P}(X = x, Y = y). \end{aligned}$$

If X is continuous with probability density function f_X , then

$$\mathbb{E}[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) \, dx$$

and if Y is also continuous and X and Y have joint probability density function $f_{X,Y}$, then

$$\mathbb{E}[XY] = \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} xy f_{X,Y}(x, y) \, dy \right) dx.$$

Theorem 6.4 (Linearity of expectation). *If X and Y are random variables and $a, b \in \mathbb{R}$, then*

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

Proof. We can only prove this when *both* X and Y are discrete, or when X and Y are continuous and have a joint probability density function $f_{X,Y}$ and marginal densities f_X and f_Y . It is true more generally, but we need more advanced tools to prove it.

Idea: Apply the law of the unconscious statistician, and use linearity of sums / integrals.

Let S_X be the support of X , and S_Y the support of Y . For discrete random variables, we have by Theorem 6.3

$$\begin{aligned} \mathbb{E}[aX + bY] &= \sum_{x \in S_X} \sum_{y \in S_Y} (ax + by) \mathbb{P}(X = x, Y = y) \\ &= a \sum_{x \in S_X} x \left(\sum_{y \in S_Y} \mathbb{P}(X = x, Y = y) \right) + b \sum_{y \in S_Y} y \left(\sum_{x \in S_X} \mathbb{P}(X = x, Y = y) \right) \\ &= a \sum_{x \in S_X} x \mathbb{P}(X = x) + b \sum_{y \in S_Y} y \mathbb{P}(Y = y) \\ &= a\mathbb{E}[X] + b\mathbb{E}[Y]. \end{aligned}$$

The proof for continuous random variables is similar, but with sums replaced by integrals. We also need to use the formula for calculating marginals, (59). We have

$$\begin{aligned} \mathbb{E}[aX + bY] &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} (ax + by) f_{X,Y}(x, y) \, dx \, dy \\ &= a \int_{-\infty}^{+\infty} x \left(\int_{-\infty}^{+\infty} f_{X,Y}(x, y) \, dy \right) dx + b \int_{-\infty}^{+\infty} y \left(\int_{-\infty}^{+\infty} f_{X,Y}(x, y) \, dx \right) dy \\ &= a \int_{-\infty}^{+\infty} x f_X(x) \, dx + b \int_{-\infty}^{+\infty} y f_Y(y) \, dy \\ &= a\mathbb{E}[X] + b\mathbb{E}[Y]. \end{aligned}$$

□

Remark:

We can chain Theorem 6.4 together to get linearity of a sum of many random variables:

$$\mathbb{E}[a_1X_1 + a_2X_2 + \dots + a_nX_n] = a_1\mathbb{E}[X_1] + a_2\mathbb{E}[X_2] + \dots + a_n\mathbb{E}[X_n],$$

or written another way,

$$\mathbb{E} \left[\sum_{i=1}^n a_i X_i \right] = \sum_{i=1}^n a_i \mathbb{E}[X_i].$$

Note that this **does not** allow us to exchange expectations with *infinite* sums. That is a much more delicate matter.

Example 6.5. Suppose that $X \sim \text{Geom}(1/4)$ and $Y \sim \text{Exp}(1/2)$. Find $\mathbb{E}[3X - Y + 5]$.

In Example 6.2, we saw that if $X \sim \text{Geom}(p)$, then $\mathbb{E}[X] = 1/p$. And in Example 6.3, we saw that if $Y \sim \text{Exp}(\lambda)$ then $\mathbb{E}[Y] = 1/\lambda$. So when $p = 1/4$ and $\lambda = 1/2$, we have $\mathbb{E}[X] = 4$ and $\mathbb{E}[Y] = 2$. We also saw that the expectation of a constant is that constant, so $\mathbb{E}[5] = 5$. Using the linearity of expectation, Theorem 6.4, we have

$$\mathbb{E}[3X - Y + 5] = 3\mathbb{E}[X] - \mathbb{E}[Y] + \mathbb{E}[5] = (3 \times 4) - 2 + 5 = 15.$$

Example 6.6. Show that if $Y \sim \text{Binom}(n, p)$, then $\mathbb{E}[Y] = np$.

We can do this two ways. The easier option is to write $Y = \sum_{i=1}^n X_i$, where $X_i \sim \text{Ber}(p)$ is the outcome of the i th trial. Now

$$\mathbb{E}[X_i] = 0 \times \mathbb{P}(X_i = 0) + 1 \times \mathbb{P}(X_i = 1) = \mathbb{P}(X_i = 1) = p.$$

By the linearity of expectation, Theorem 6.4, we have

$$\mathbb{E}[Y] = \mathbb{E}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \mathbb{E}[X_i] = np.$$

The alternative is to do the calculation directly:

$$\mathbb{E}[Y] = \sum_{j=0}^n j \mathbb{P}(Y = j) = \sum_{j=0}^n j \binom{n}{j} p^j (1-p)^{n-j}$$

and then we have to notice that for each $j = 1, \dots, n$,

$$j \binom{n}{j} p^j (1-p)^{n-j} = np \binom{n-1}{j-1} p^{j-1} (1-p)^{n-j}$$

so that

$$\begin{aligned} \mathbb{E}[Y] &= \sum_{j=0}^n j \binom{n}{j} p^j (1-p)^{n-j} = \sum_{j=1}^n j \binom{n}{j} p^j (1-p)^{n-j} \\ &= np \sum_{j=1}^n \binom{n-1}{j-1} p^{j-1} (1-p)^{n-j} \\ &= np \sum_{i=0}^{n-1} \binom{n-1}{i} p^i (1-p)^{n-1-i} \\ &= np(p + (1-p))^{n-1} = np. \end{aligned}$$

Using linearity of expectation is much quicker and easier!

Example 6.7. Let $X \sim N(\mu, \sigma^2)$. Find $\mathbb{E}[X]$.

By linearity of expectation,

$$\mathbb{E}[X] = \mathbb{E}[X - \mu] + \mu,$$

so it suffices to show that $\mathbb{E}[X - \mu] = 0$. By the law of the unconscious statistician,

$$\mathbb{E}[X - \mu] = \int_{-\infty}^{\infty} \frac{(x - \mu)}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx.$$

Letting $y = x - \mu$ gives

$$\mathbb{E}[X - \mu] = \int_{-\infty}^{\infty} \frac{y}{\sqrt{2\pi\sigma^2}} e^{-\frac{y^2}{2\sigma^2}} dy = \int_{-\infty}^0 \frac{y}{\sqrt{2\pi\sigma^2}} e^{-\frac{y^2}{2\sigma^2}} dy + \int_0^{+\infty} \frac{y}{\sqrt{2\pi\sigma^2}} e^{-\frac{y^2}{2\sigma^2}} dy.$$

We now spot that on the right-hand side, the integral from $-\infty$ to 0 is the negative of the integral from 0 to ∞ (e.g. by substituting $y = -z$ in one of them), from which we conclude that $\mathbb{E}[X - \mu] = 0$. This then implies that $\mathbb{E}[X] = \mu$.

Corollary 6.1 (expectation preserves non-strict inequalities). *If X and Y are random variables on the same sample space such that $X(\omega) \geq Y(\omega)$ for all ω , then $\mathbb{E}[X] \geq \mathbb{E}[Y]$.*

Proof. Let $Z = X - Y$. Then since $X(\omega) \geq Y(\omega)$ for all ω , we have $Z(\omega) \geq 0$ for all ω , i.e. Z is a non-negative random variable. So by Theorem 6.1, $\mathbb{E}[X - Y] = \mathbb{E}[Z] \geq 0$. But by linearity of expectation, $\mathbb{E}[X - Y] = \mathbb{E}[X] - \mathbb{E}[Y]$. So we have shown that $\mathbb{E}[X] - \mathbb{E}[Y] \geq 0$, i.e. $\mathbb{E}[X] \geq \mathbb{E}[Y]$. $\square$

Corollary 6.2 (Expectation of the modulus $\geq$ the modulus of the expectation). *If $\mathbb{E}[X]$ exists then $\mathbb{E}[|X|] \geq |\mathbb{E}[X]|$.*

Proof. We consider two cases: $\mathbb{E}[X] \geq 0$ and $\mathbb{E}[X] < 0$. If $\mathbb{E}[X] \geq 0$, then

$$\begin{aligned} |X| &\geq X \\ &= \mathbb{E}[X] + X - \mathbb{E}[X] \\ &= |\mathbb{E}[X]| + X - \mathbb{E}[X], \quad [\text{since } \mathbb{E}[X] \geq 0 \text{ implies } \mathbb{E}[X] = |\mathbb{E}[X]|] \end{aligned}$$

and by taking the expectation throughout and using Corollary 6.1 and linearity of expectation, we have

$$\mathbb{E}[|X|] \geq |\mathbb{E}[X]| + \mathbb{E}[X] - \mathbb{E}[X] = |\mathbb{E}[X]|.$$

For the case $\mathbb{E}[X] < 0$ we proceed similarly:

$$|X| \geq -X = -\mathbb{E}[X] - X + \mathbb{E}[X] = |\mathbb{E}[X]| - X + \mathbb{E}[X],$$

which, by taking the expectation throughout, yields the claim by Corollary 6.1 and linearity of expectation. $\square$

6.3.2 Expectation of the product of independent random variables

We have seen that $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$. What about $\mathbb{E}[XY]$? It is **not true** (!!!) that $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ in general. But this does hold when X and Y are independent.

Theorem 6.5 (Expected value of the product of independent random variables). *If X and Y are independent random variables and functions $g : \mathbb{R} \rightarrow \mathbb{R}$ and $h : \mathbb{R} \rightarrow \mathbb{R}$ are such that $\mathbb{E}[|g(X)|] < \infty$ and $\mathbb{E}[|h(Y)|] < \infty$, then*

$$\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)].$$

Proof. We do the proof only when X and Y are both discrete. The case when X and Y are both continuous with a joint probability density function is similar. The theorem is true more generally but we would need more advanced tools to prove it.

Idea: Use the law of the unconscious statistician and then apply independence in the form of probability mass functions (i.e. Theorem 4.5).

Suppose that X and Y are discrete random variables with support S_X and S_Y respectively. Theorem 4.5 says that if X and Y are independent, then for all $x \in S_X$, $y \in S_Y$, we have

$$\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y).$$

Then, by the law of the unconscious statistician, Theorem 6.3,

$$\begin{aligned} \mathbb{E}[g(X)h(Y)] &= \sum_{x \in S_X} \sum_{y \in S_Y} g(x)h(y)\mathbb{P}(X = x, Y = y) \\ &= \sum_{x \in S_X} \sum_{y \in S_Y} g(x)h(y)\mathbb{P}(X = x)\mathbb{P}(Y = y) \\ &= \left(\sum_{x \in S_X} g(x)\mathbb{P}(X = x) \right) \left(\sum_{y \in S_Y} h(y)\mathbb{P}(Y = y) \right) \\ &= \mathbb{E}[g(X)]\mathbb{E}[h(Y)]. \end{aligned}$$

$\square$

Remarks:

1. This illustrates the intuition that “independence means multiplication”: if X and Y are independent then

$$\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)],$$

which is not true in general.

2. This result can be extended to n independent random variables in the obvious fashion.
3. An important application of this corollary is when $g(X) = X$ and $h(Y) = Y$ yielding

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$$

for independent X and Y . This will be especially useful later when we consider the *covariance* of two random variables.

6.4 Variance

Definition 6.3 (Variance). If X is a random variable with $\mathbb{E}[X^2] < \infty$ ¹⁶, then the *variance* of X is defined as

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2].$$

The *standard deviation* of X is the (positive) square root of $\text{Var}(X)$.

Thus, for a discrete random variable X with support S_X and expectation

$$\mu_X = \mathbb{E}[X] = \sum_{x \in S_X} x\mathbb{P}(X = x),$$

we have

$$\text{Var}(X) = \sum_{x \in S_X} (x - \mu_X)^2 \mathbb{P}(X = x).$$

Note that in this calculation, $\mu_X = \mathbb{E}[X]$ is just a real number, not a random variable. Similarly, for continuous X with pdf f_X we have

$$\text{Var}(X) = \int_{S_X} (x - \mu_X)^2 f_X(x) \, dx, \text{ where } \mu_X = \mathbb{E}[X] = \int_{S_X} x f_X(x) \, dx.$$

Corollary 6.3 (The variance is non-negative). *For any random variable X , $\text{Var}(X) \geq 0$.*

Proof. Let $Y = (X - \mathbb{E}[X])^2$. Then $Y \geq 0$ so, from Theorem 6.1, $\text{Var}(X) = \mathbb{E}[Y] \geq 0$. □

Theorem 6.6 (Calculating the variance). *If X is a random variable with $\mathbb{E}[X^2] < \infty$, then*

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2.$$

Proof.

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[(X - \mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2 - 2X\mathbb{E}[X] + \mathbb{E}[X]^2] \\ &= \mathbb{E}[X^2] - 2\mathbb{E}[X]\mathbb{E}[X] + \mathbb{E}[X]^2 && \text{[by linearity of expectation]} \\ &= \mathbb{E}[X^2] - \mathbb{E}[X]^2. \end{aligned}$$

□

¹⁶Note that if $\mathbb{E}[X^2] < \infty$, then $\mathbb{E}[|X|] < \infty$ so that the expectation $\mathbb{E}[X]$ is well defined.

Remark: Sometimes $\mathbb{E}[X]$ is called the *first moment* of X and $\mathbb{E}[X^2]$ is called the *second moment* (in general $\mathbb{E}[X^n]$ is called the *nth moment*). The variance formula of Theorem 6.6 is sometimes called the **moment form** for the variance.

Hence, if X is discrete then we can calculate $\text{Var}(X)$ as

$$\text{Var}(X) = \sum_{x \in S_X} x^2 \mathbb{P}(X = x) - \left(\sum_{x \in S_X} x \mathbb{P}(X = x) \right)^2.$$

For continuous X this becomes

$$\text{Var}(X) = \int_{S_X} x^2 f_X(x) \, dx - \left(\int_{S_X} x f_X(x) \, dx \right)^2.$$

Often this is the easiest way to compute $\text{Var}(X)$ from the pmf or pdf.

Example 6.8. Suppose that $X \sim \text{Geom}(p)$. Show that $\text{Var}(X) = \frac{1-p}{p^2}$.

We will use an observation similar to Example 6.2. Let $q = 1 - p$.

$$\begin{aligned} \mathbb{E}[X^2] &= \sum_{x=1}^{\infty} x^2 (1-p)^{x-1} p \\ &= p(1^2 + 2^2 q + 3^2 q^2 + \dots) && [\text{substituting } q = 1 - p] \\ &= p(1 + 2q + 3q^2 + \dots) + p(2(2-1)q + 3(3-1)q^2 + \dots) \\ &= \mathbb{E}[X] + pq \left(\frac{d^2}{dq^2} 1 + \frac{d^2}{dq^2} q + \frac{d^2}{dq^2} q^2 + \frac{d^2}{dq^2} q^3 + \dots \right) \\ &= \mathbb{E}[X] + pq \frac{d^2}{dq^2} \left(\sum_{i=0}^{\infty} q^i \right) \\ &= \mathbb{E}[X] + pq \frac{d^2}{dq^2} \frac{1}{1-q} \\ &= \frac{1}{p} + pq \frac{2}{(1-q)^3} = \frac{1}{p} + \frac{2(1-p)}{p^2} = \frac{2-p}{p^2}. \end{aligned}$$

Thus,

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{2-p}{p^2} - \left(\frac{1}{p} \right)^2 = \frac{1-p}{p^2}.$$

As in Example 6.2, changing the sum of derivatives to the derivative of a sum can (in this case) be made rigorous without too much work.

Example 6.9. Suppose that $X \sim \text{Unif}([0, 1])$. Find $\text{Var}(X)$.

Note that $f_X(x) = 1$ for $x \in [0, 1]$ and 0 otherwise. First we find $\mathbb{E}[X]$:

$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} x f_X(x) \, dx = \int_0^1 x \, dx = \left[\frac{1}{2} x^2 \right]_0^1 = \frac{1}{2} \times 1^2 - \frac{1}{2} \times 0^2 = \frac{1}{2}.$$

Then we find $\mathbb{E}[X^2]$:

$$\mathbb{E}[X^2] = \int_{-\infty}^{+\infty} x^2 f_X(x) \, dx = \int_0^1 x^2 \, dx = \left[\frac{1}{3} x^3 \right]_0^1 = \frac{1}{3} \times 1^3 - \frac{1}{3} \times 0^3 = \frac{1}{3}.$$

Thus

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}.$$

Theorem 6.7 (Variance of a constant is zero). $\text{Var}(X) = 0$ if and only if $\mathbb{P}(X = a) = 1$ for some constant $a \in \mathbb{R}$. Thus, $\text{Var}(a) = 0$.

Proof. Suppose that $\mathbb{P}(X = a) = 1$. Then from Theorem 6.2, $\mathbb{E}[X] = a$, and also, by the law of the unconscious statistician,

$$\mathbb{E}[X^2] = a^2 \mathbb{P}(X = a) = a^2$$

so

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = a^2 - a^2 = 0.$$

So the variance of a constant is zero. We now want to show that the variance of any non-constant random variable is non-zero. We can do this very easily when the random variable is discrete. However, the general case is a bit more subtle.

It does not use any tools that you don't already know, so I will include it here. However, it puts a number of ingredients together and has a more advanced feel than many of the proofs in this course, so you can consider it **non-examinable**.

Suppose that we do not have $\mathbb{P}(X = a) = 1$ for any constant a . Let $a = \mathbb{E}[X]$; then since $\mathbb{P}(X = a) < 1$, there must exist $\delta > 0$ such that either $\mathbb{P}(X \geq a + \delta) > 0$ or $\mathbb{P}(X \leq a - \delta) > 0$. (In fact, it must be possible to find $\delta > 0$ such that both these hold; why? But we won't need this.)

Let Y be a random variable on the same sample space as X , such that $Y(\omega) = 1$ if $X(\omega) \geq a + \delta$ or $X(\omega) \leq a - \delta$, and $Y(\omega) = 0$ otherwise. Then $Y \leq 1$, so $(X - a)^2 Y \leq (X - a)^2$. Thus

$$\text{Var}(X) = \mathbb{E}[(X - a)^2] \geq \mathbb{E}[(X - a)^2 Y]$$

. We also claim that $(X - a)^2 Y \geq \delta^2 Y$. Indeed, this inequality holds if $Y = 1$, since $Y = 1$ exactly when $(X - a)^2 \geq \delta^2$; but it also holds if $Y = 0$, since then both sides are 0. Thus

$$\text{Var}(X) \geq \mathbb{E}[(X - a)^2 Y] \geq \mathbb{E}[\delta^2 Y] = \delta^2 \mathbb{E}[Y].$$

Finally, by our choice of δ we have

$$\mathbb{E}[Y] = 1\mathbb{P}(Y = 1) + 0\mathbb{P}(Y = 0) = \mathbb{P}(Y = 1) = \mathbb{P}(\{X \geq a + \delta\} \cup \{X \leq a - \delta\}) > 0$$

and therefore we have shown that $\text{Var}(X) > 0$. □

Theorem 6.8 (Variance of a linear function). If X is a random variable with $\mathbb{E}[X^2] < \infty$ and $a, b \in \mathbb{R}$ then

$$\text{Var}(aX + b) = a^2 \text{Var}(X).$$

Proof. Note that, by the linearity of expectation, $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$ so that, from Definition 6.3,

$$\begin{aligned} \text{Var}(aX + b) &= \mathbb{E}[(aX + b - \mathbb{E}[aX + b])^2] \\ &= \mathbb{E}[(aX + b - a\mathbb{E}[X] - b)^2] \\ &= \mathbb{E}[a^2(X - \mathbb{E}[X])^2] = a^2 \text{Var}(X). \end{aligned}$$

□

Example 6.10. For $a = -1$ and $b = 0$ we have immediately that $\text{Var}(-X) = \text{Var}(X)$.

If $Y \sim \text{Geom}(1/2)$, then by Example 6.8 we have

$$\text{Var}(Y) = \frac{1 - \frac{1}{2}}{\left(\frac{1}{2}\right)^2} = 2$$

and therefore

$$\text{Var}(2Y - 5) = 2^2 \text{Var}(Y) = 8.$$

6.4.1 Covariance

The **covariance** of two random variables is a measure of their joint variability, or their degree of association.

Definition 6.4 (Covariance and correlation). For random variables X and Y with $\mathbb{E}[X^2] < \infty$ and $\mathbb{E}[Y^2] < \infty$, the *covariance* of X and Y is defined as

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])].$$

If $\text{Var}(X) > 0$ and $\text{Var}(Y) > 0$ then the *correlation* of X and Y is defined as

$$\rho_{XY} = \text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}.$$

Remarks:

1. Notice that $\text{Cov}(X, Y) = \text{Cov}(Y, X)$ and $\text{Cov}(X, X) = \text{Var}(X)$.
2. The correlation is the covariance divided by the product of the standard deviations.
3. The covariance is the average value of the product of the deviation of X from its expectation $\mathbb{E}[X]$ and Y from its expectation $\mathbb{E}[Y]$.
 - a. If X is larger than $\mathbb{E}[X]$ whenever Y is larger than $\mathbb{E}[Y]$, and smaller than $\mathbb{E}[X]$ whenever Y is smaller than $\mathbb{E}[Y]$, then $\text{Cov}(X, Y)$ is positive.
 - b. If X is smaller than $\mathbb{E}[X]$ whenever Y is larger than $\mathbb{E}[Y]$, and larger than $\mathbb{E}[X]$ whenever Y is smaller than $\mathbb{E}[Y]$, then $\text{Cov}(X, Y)$ is negative.

Theorem 6.9 (Calculating the covariance). *The covariance of X and Y can be calculated as*

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

Proof. Idea: Multiply out the definition and use linearity of expectation.

$$\begin{aligned}\text{Cov}(X, Y) &= \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] \\ &= \mathbb{E}[XY - \mathbb{E}[X]Y - \mathbb{E}[Y]X + \mathbb{E}[X]\mathbb{E}[Y]] \\ &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] - \mathbb{E}[Y]\mathbb{E}[X] + \mathbb{E}[X]\mathbb{E}[Y] \quad [\text{linearity of expectation}] \\ &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].\end{aligned}$$

□

Thus, if X and Y are discrete then we can calculate $\text{Cov}(X, Y)$ as

$$\text{Cov}(X, Y) = \sum_{x \in S_X} \sum_{y \in S_Y} xy \mathbb{P}(X = x, Y = y) - \left(\sum_{x \in S_X} x \mathbb{P}(X = x) \right) \left(\sum_{y \in S_Y} y \mathbb{P}(Y = y) \right).$$

For continuous X and Y we have

$$\text{Cov}(X, Y) = \int_{S_X} \int_{S_Y} xy f_{X,Y}(x, y) \, dx dy - \left(\int_{S_X} x f_X(x) \, dx \right) \left(\int_{S_Y} y f_Y(y) \, dy \right).$$

Corollary 6.4 (Independent random variables have zero covariance). *If X and Y are independent, then $\text{Cov}(X, Y) = 0$.*

Proof. From Theorem 6.5, if X and Y are independent, then $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ and so, from Theorem 6.9, $\text{Cov}(X, Y) = 0$. □

It will be useful later to note that covariance is linear in each of its arguments. Since covariance is symmetric, it suffices to check one of the arguments.

Corollary 6.5 (Covariance is linear in each argument). *For any random variables X, Y and Z satisfying $\mathbb{E}[X^2] < \infty$, $\mathbb{E}[Y^2] < \infty$ and $\mathbb{E}[Z^2] < \infty$, and any $a, b \in \mathbb{R}$, we have*

$$\text{Cov}(aX + bY, Z) = a\text{Cov}(X, Z) + b\text{Cov}(Y, Z).$$

Proof. Idea: Write out the definition and use linearity of expectation.

$$\begin{aligned} \text{Cov}(aX + bY, Z) &= \mathbb{E}[(aX + bY - \mathbb{E}[aX + bY])(Z - \mathbb{E}[Z])] \\ &= \mathbb{E}[a(X - \mathbb{E}[X])(Z - \mathbb{E}[Z]) + b(Y - \mathbb{E}[Y])(Z - \mathbb{E}[Z])] && \text{[linearity of expectation]} \\ &= a\mathbb{E}[(X - \mathbb{E}[X])(Z - \mathbb{E}[Z])] + b\mathbb{E}[(Y - \mathbb{E}[Y])(Z - \mathbb{E}[Z])] && \text{[linearity of expectation]} \\ &= a\text{Cov}(X, Z) + b\text{Cov}(Y, Z). \end{aligned}$$

□

Remark:

(Non-examinable but might be helpful!) Covariance acts *almost* like an *inner product* (if you have met inner products) - it is symmetric and bilinear. It's not positive definite, so it's not actually an inner product, but it is *positive semi-definite* - $\text{Cov}(X, X) \geq 0$ and (by Theorem 6.7) if $\text{Cov}(X, X) = 0$ then X is a constant.

6.4.2 The variance of a sum of random variables

We have seen that expectation is linear, and covariance is linear in each of its arguments. But variance is **not** linear.

Theorem 6.10 (Variance of a sum of two RVs). *For random variables X and Y , with $\mathbb{E}[X^2] < \infty$ and $\mathbb{E}[Y^2] < \infty$, and constants $a, b \in \mathbb{R}$,*

$$\text{Var}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab\text{Cov}(X, Y).$$

Proof. We can do this two ways.

Idea 1: The direct way. Multiply out the definition, group appropriate terms together and apply linearity of expectation.

Let $\mathbb{E}[X] = \mu_X$ and $\mathbb{E}[Y] = \mu_Y$ then $\mathbb{E}[aX + bY] = a\mu_X + b\mu_Y$ so that

$$\begin{aligned} \text{Var}(aX + bY) &= \mathbb{E}[(aX + bY - \mathbb{E}[aX + bY])^2] \\ &= \mathbb{E}[(a(X - \mu_X) + b(Y - \mu_Y))^2] \\ &= \mathbb{E}[a^2(X - \mu_X)^2 + b^2(Y - \mu_Y)^2 + 2ab(X - \mu_X)(Y - \mu_Y)] \\ &= a^2\mathbb{E}[(X - \mu_X)^2] + b^2\mathbb{E}[(Y - \mu_Y)^2] + 2ab\mathbb{E}[(X - \mu_X)(Y - \mu_Y)] \\ &= a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab\text{Cov}(X, Y). \end{aligned}$$

Idea 2: Use linearity of covariance in each argument.

$$\begin{aligned} \text{Var}(aX + bY) &= \text{Cov}(aX + bY, aX + bY) \\ &= a\text{Cov}(X, aX + bY) + b\text{Cov}(Y, aX + bY) && \text{[linearity in the first argument]} \\ &= a^2\text{Cov}(X, X) + ab\text{Cov}(X, Y) + ba\text{Cov}(Y, X) + b^2\text{Cov}(Y, Y) && \text{[linearity in the second argument]} \\ &= a^2\text{Var}(X) + 2ab\text{Cov}(X, Y) + b^2\text{Var}(Y). \end{aligned}$$

□

Example 6.11. Two important special cases are:

$$\begin{aligned}\text{Var}(X + Y) &= \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y) && [\text{taking } a = b = 1] \\ \text{Var}(X - Y) &= \text{Var}(X) + \text{Var}(Y) - 2\text{Cov}(X, Y). && [\text{taking } a = 1 \text{ and } b = -1]\end{aligned}$$

Notice that if X and Y are independent, then $\text{Cov}(X, Y) = 0$ and so $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) = \text{Var}(X - Y)$.

In fact, essentially the same proof (at least if we use the second method) can be used to find the variance of a sum of random variables.

Corollary 6.6 (Variance of a sum of n random variables). *Suppose that $X_1, \dots, X_n$ are random variables satisfying $\mathbb{E}[X_i^2] < \infty$ for all $i = 1, \dots, n$. Let $Y = \sum_{i=1}^n a_i X_i$ for some constants $a_1, \dots, a_n \in \mathbb{R}$. Then*

$$\text{Var}(Y) = \sum_{i=1}^n \sum_{j=1}^n a_i a_j \text{Cov}(X_i, X_j) = \sum_{i=1}^n a_i^2 \text{Var}(X_i) + 2 \sum_{i=1}^n \sum_{j=i+1}^n a_i a_j \text{Cov}(X_i, X_j). \quad (62)$$

Proof. Idea: Again use linearity of covariance in each argument.

$$\begin{aligned}\text{Var}(Y) &= \text{Cov}\left(\sum_{i=1}^n a_i X_i, \sum_{j=1}^n a_j X_j\right) \\ &= \sum_{i=1}^n a_i \text{Cov}\left(X_i, \sum_{j=1}^n a_j X_j\right) && [\text{linearity in the first argument}] \\ &= \sum_{i=1}^n \sum_{j=1}^n a_i a_j \text{Cov}(X_i, X_j). && [\text{linearity in the second argument}]\end{aligned}$$

This proves the first form of the statement of the corollary. We now notice that the sum over j can be split into a sum over $j < i$, the term when $j = i$, and a sum over $j > i$. That is,

$$\text{Var}(Y) = \sum_{i=1}^n \sum_{j=1}^{i-1} a_i a_j \text{Cov}(X_i, X_j) + \sum_{i=1}^n a_i^2 \text{Var}(X_i) + \sum_{i=1}^n \sum_{j=i+1}^n a_i a_j \text{Cov}(X_i, X_j).$$

Finally, by symmetry of the covariance, we can swap the roles of i and j in the first double sum, and notice that it equals the second double sum, so we obtain

$$\text{Var}(Y) = \sum_{i=1}^n a_i^2 \text{Var}(X_i) + 2 \sum_{i=1}^n \sum_{j=i+1}^n a_i a_j \text{Cov}(X_i, X_j).$$

□

Corollary 6.7 (Sum of independent random variables). *If $X_1, \dots, X_n$ are independent, then*

$$\text{Var}(Y) = \text{Var}\left(\sum_{i=1}^n b_i X_i\right) = \sum_{i=1}^n b_i^2 \text{Var}(X_i). \quad (63)$$

Proof. If $X_1, \dots, X_n$ are independent then $\text{Cov}(X_i, X_j) = 0$ for all $i \neq j$ and so Equation (62) reduces to Equation (63). □

Remarks: Corollary 6.6 is sometimes known as Bienaymé's¹⁷ identity, and Corollary 6.7 is also known as Bienaymé's formula. We won't use these names in the exam.

¹⁷Irénée-Jules Bienaymé.

Example 6.12. Suppose that $Y \sim \text{Binom}(n, p)$. Show that $\text{Var}(Y) = np(1 - p)$.

In Example 6.6 we showed that if $Y \sim \text{Binom}(n, p)$ then $\mathbb{E}[Y] = np$. This was accomplished by writing $Y = \sum_{i=1}^n X_i$ where each $X_i \sim \text{Ber}(p)$ denotes the outcome of the i th Bernoulli trial. We also saw in Example 6.6 that $\mathbb{E}[X_i] = p$ and $\mathbb{E}[X_i^2] = p$ so that

$$\text{Var}(X_i) = \mathbb{E}[X_i^2] - \mathbb{E}[X_i]^2 = p - p^2 = p(1 - p).$$

As the trials, and hence $X_1, X_2, \dots$, are independent, the formula for the variance of a sum of independent random variables (Bienaymé's formula, Corollary 6.7) gives

$$\text{Var}(Y) = \text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) = np(1 - p).$$

Example 6.13. We can use the linearity of covariance in each argument to find that

$$\begin{aligned} \text{Cov}(X + Y, X - Y) &= \text{Cov}(X, X - Y) + \text{Cov}(Y, X - Y) \\ &= \text{Cov}(X, X) - \text{Cov}(X, Y) + \text{Cov}(Y, X) - \text{Cov}(Y, Y) \\ &= \text{Cov}(X, X) - \text{Cov}(Y, Y) = \text{Var}(X) - \text{Var}(Y). \end{aligned}$$

To summarise the formulas for sums of random variables,

$$\begin{aligned} \mathbb{E}\left[\sum_{i=1}^n a_i X_i\right] &= \sum_{i=1}^n a_i \mathbb{E}[X_i], \\ \text{Cov}\left(\sum_{i=1}^n a_i X_i, Y\right) &= \sum_{i=1}^n a_i \text{Cov}(X_i, Y), \\ \text{Var}\left(\sum_{i=1}^n a_i X_i\right) &= \sum_{i=1}^n a_i^2 \text{Var}(X_i) + 2 \sum_{i=1}^n \sum_{j=i+1}^n a_i a_j \text{Cov}(X_i, X_j), \end{aligned}$$

and if $X_1, X_2, \dots, X_n$ are independent then

$$\text{Var}\left(\sum_{i=1}^n a_i X_i\right) = \sum_{i=1}^n a_i^2 \text{Var}(X_i).$$

6.5 The Law of Large Numbers

Consider a sequence of independent random variables $X_1, X_2, \dots$ that all have the same distribution, with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2$ for all $i = 1, 2, \dots$. The **(sample) average** $\bar{X}_n$ of the first n of these random variables is defined as

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

By linearity of expectation, we immediately have

$$\mathbb{E}[\bar{X}_n] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n} n\mu = \mu,$$

so the expected value of the sample average is equal to the expectation of each individual X_i . For the variance of the sample average, we use

$$\text{Var}(\bar{X}_n) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}.$$

Thus, if we let n increase, then we clearly see that $\text{Var}(\bar{X}_n)$ will tend to zero. As the variance is a measure of the spread of a random variable, we can interpret this as suggesting that as n increases, $\bar{X}_n$ will get closer and closer to its expected value, μ .

Example 6.14. Consider tossing a fair coin n times and assigning value 0 to heads and 1 to tails, so that a single toss constitutes a Bernoulli trial, which we denote by X_i for $i \in \{1, \dots, n\}$. As we know already,

$$\sum_{i=1}^n X_i \sim \text{Binom}\left(n, \frac{1}{2}\right).$$

Thus the pmf of the sample average of $X_1, \dots, X_n$ is easily obtained as

$$\mathbb{P}(\bar{X}_n = x) = \mathbb{P}\left(\frac{1}{n} \sum_{i=1}^n X_i = x\right) = \mathbb{P}\left(\sum_{i=1}^n X_i = nx\right),$$

where the final probability is equal to the pmf of $\text{Binom}(n, p)$ evaluated at nx . Figure 32 shows the pmfs, expectations and variances of $\bar{X}_n$ for $n \in \{1, \dots, 14\}$. The distribution of $\bar{X}_n$ concentrates around $p = \mathbb{E}[X_i] = 0.5$ as expected.

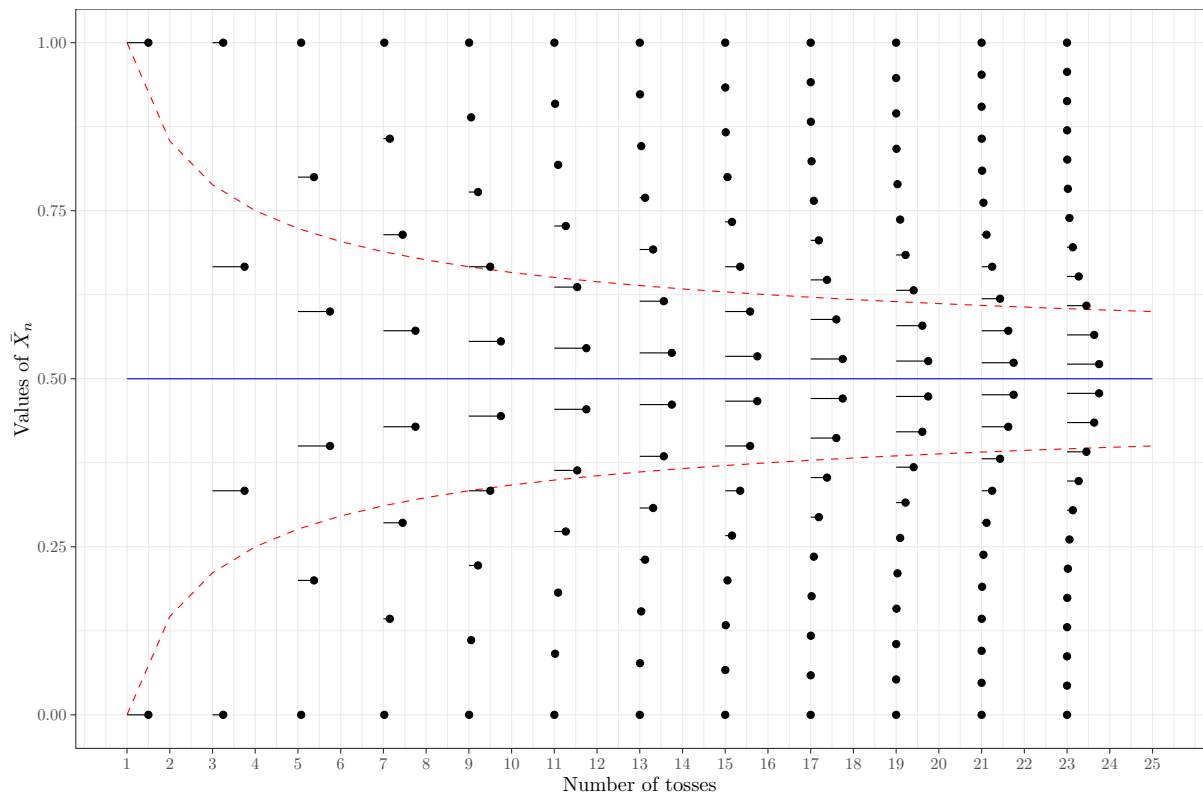


Figure 32: Probability mass function (black lollipops), expectation (solid blue line), and expectation $\pm$ standard deviation (dotted red curves) of $\bar{X}_n$ in Example refexm:LLN1 for $n = 1, \dots, 14$.

These observations can be made rigorous to give the following result, which we state without a proof.

Theorem 6.11 ((Weak) Law of Large Numbers, (W)LLN). *Suppose that $X_1, X_2, \dots$ are independent and identically distributed random variables with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2$ for all $i = 1, 2, \dots$. Then for any $\varepsilon > 0$,*

$$\lim_{n \rightarrow +\infty} \mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n X_i - \mu\right| > \varepsilon\right) = 0.$$

We have not covered the formal definition of limits, but Theorem 6.11 should be understood as follows. No matter how small a margin of error ($\varepsilon > 0$) we choose, by taking a sufficiently large sample (i.e. n large enough) we can make the probability that the sample average $\bar{X}_n$ is within distance ε of μ , arbitrarily close to one.

Example 6.15 (Example 6.14 revisited). A typical use of the law of large numbers is the estimation of unknown quantities. Suppose that we were not sure whether heads and tails on a coin toss were really equally likely¹⁸. Figure 33 shows a realisation of the sample average $\bar{X}_n$ for $n \in \{1, \dots, 2000\}$ tosses¹⁹.

Due to the law of large numbers, we know that for large n the distribution of $\bar{X}_n$ will be concentrated around the true probability, p , of heads and thus it is unlikely to lie far from $\bar{X}_n$. In this experiment, we have $\bar{X}_{2000} = 0.4885$ which, although not quite 0.5, is a reasonable estimate of the true probability of heads²⁰.

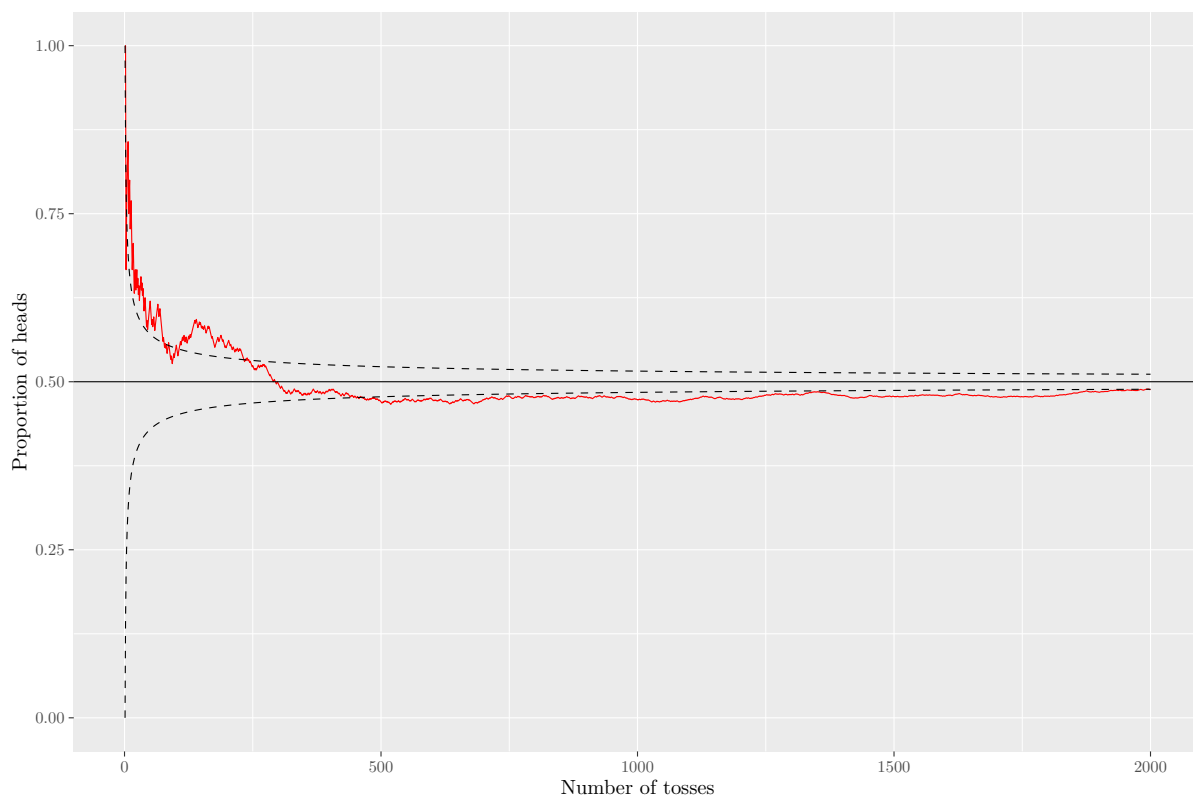


Figure 33: Realisation of $\bar{X}_n$ for $n \in \{1, \dots\}$ (red), the true probability of heads (solid), and the standard deviation from the true p (dashed).

7 Indicator functions and applications

In this section, we introduce a new class of simple but extremely useful discrete random variables, called *indicator functions*. We will then use these to prove some simple but extremely useful results: the Markov and Chebyshev inequalities, and another formula for the expectation of a random variable.

This section was written after the exam, so there will be no material from this section in the exam. View it as bonus material that might be interesting and good preparation if you are thinking of taking probability courses in later years. If your aim is just to get through the exam with the minimal amount of work, you can ignore this section.

¹⁸Persi Diaconis, a professor of Statistics and Mathematics at Stanford University, and a former magician, claims to have trained himself to toss a coin so that it falls the desired way almost all the time. He has also studied the fairness of coin tossing.

¹⁹For this example, a fair Bernoulli random variable was simulated 2000 times on a computer in a fraction of a second, but the same experiment has been conducted with a real coin by many statisticians, such as Karl Pearson (1857-1936), who famously tossed a coin 24000 times and recorded 12012 heads.

²⁰In more advanced statistics courses you will learn to analyse this kind of estimates rigorously.

7.1 Indicator functions

Recall that a random variable is a function from our sample space to $\mathbb{R}$.

Definition 7.1 (Indicator function). Suppose that we have a sample space Ω and a σ -algebra $\mathcal{F}$ on Ω . For an event $A \in \mathcal{F}$, the *indicator function* of A is the discrete random variable $\mathbf{1}_A : \Omega \rightarrow \{0, 1\}$ such that

$$\mathbf{1}_A(\omega) = \begin{cases} 1 & \text{if } \omega \in A \\ 0 & \text{if } \omega \notin A. \end{cases}$$

Example 7.1. Toss a coin once. Let A be the event that we see a head. Then $\mathbf{1}_A = 1$ if we see a head, and $\mathbf{1}_A = 0$ if we see a tail.

(What is the cdf or pmf of $\mathbf{1}_A$? What is the cdf or pmf of $\mathbf{1}_{A^c}$?)

Since an indicator function can only take two values, 0 and 1, it is very easy to calculate its expectation.

Lemma 7.1. For any event A , $\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A)$.

Proof. Indeed, since the indicator function can only take values 0 or 1, by the definition of expectation for discrete random variables,

$$\mathbb{E}[\mathbf{1}_A] = 0 \cdot \mathbb{P}(\mathbf{1}_A = 0) + 1 \cdot \mathbb{P}(\mathbf{1}_A = 1) = \mathbb{P}(\mathbf{1}_A = 1) = \mathbb{P}(\{\omega : \mathbf{1}_A(\omega) = 1\}).$$

But

$$\{\omega : \mathbf{1}_A(\omega) = 1\} = \{\omega : \omega \in A\} = A.$$

So $\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A)$. □

7.2 The Markov and Chebyshev inequalities

We now come to one of the simplest, but most powerful and most-used theorems in all of probability.

Theorem 7.1 (Markov's inequality). For any non-negative random variable X , and any $x > 0$,

$$\mathbb{P}(X \geq x) \leq \frac{\mathbb{E}[X]}{x}.$$

Proof. Idea: Note that $\mathbf{1}_{\{X \geq x\}} \leq X/x$ and take expectations.

We claim that $\mathbf{1}_{\{X \geq x\}} \leq X/x$. Indeed, if $X \geq x$, the left-hand side is 1 and the right-hand side is at least 1; whereas if $X < x$, then the left-hand side is 0, but the right-hand side is at least 0, since X is non-negative.

Taking expectations of both sides,

$$\mathbb{E}[\mathbf{1}_{\{X \geq x\}}] \leq \mathbb{E}[X/x].$$

We have just seen in Lemma 7.1 that $\mathbb{E}[\mathbf{1}_{\{X \geq x\}}] = \mathbb{P}(X \geq x)$, and by linearity of expectation, $\mathbb{E}[X/x] = \mathbb{E}[X]/x$, which completes the proof. □

Example 7.2. Let $X \sim \text{Bin}(1000, 0.01)$. Estimate $\mathbb{P}(X \geq 35)$ using Markov's inequality.

To calculate $\mathbb{P}(X \geq 35)$ by hand would take a long time: we would have to calculate

$$1 - \sum_{n=0}^{34} \binom{1000}{n} \times 0.01^n \times 0.99^{1000-n}.$$

Markov's inequality gives us a bound with almost no work at all: it says that

$$\mathbb{P}(X \geq 35) \leq \frac{\mathbb{E}[X]}{35} = \frac{1000 \times 0.01}{35} = \frac{2}{7}.$$

It is reasonable to ask whether this bound is any good. In the case of the binomial random variable above, it is not very good at all; a computer can tell us the real answer is approximately 0.0000000004. But the idea of using the expectation - which is often easy to calculate - to upper bound probabilities is a powerful

one, and using Markov's inequality in the right way is the key to proving many theorems in probability, even in cutting-edge research.

Markov's inequality gives us a way of bounding the probability that a non-negative random variable is much bigger than its expectation. Three questions immediately spring to mind:

1. Can we improve on this bound (particularly in the binomial example above)?
2. What about the probability that a random variable is much *smaller* than its expectation?
3. What if our random variable isn't non-negative?

One way of answering all three questions (if our random variable has a finite second moment) is to use *Chebyshev's inequality*, which is really just Markov's inequality in disguise.

Corollary 7.1 (Chebyshev's inequality). *For any random variable X with $\mathbb{E}[X^2] < \infty$, and any $x > 0$,*

$$\mathbb{P}\left(|X - \mathbb{E}[X]| \geq x\right) \leq \frac{\text{Var}(X)}{x^2}.$$

Proof. Idea: Apply Markov's inequality to $|X - \mathbb{E}[X]|^2$.

Note that

$$\mathbb{P}\left(|X - \mathbb{E}[X]| \geq x\right) = \mathbb{P}\left(|X - \mathbb{E}[X]|^2 \geq x^2\right).$$

But $|X - \mathbb{E}[X]|^2$ is a non-negative random variable, and $x^2 > 0$, so by Markov's inequality,

$$\mathbb{P}\left(|X - \mathbb{E}[X]|^2 \geq x^2\right) \leq \frac{1}{x^2} \mathbb{E}\left[|X - \mathbb{E}[X]|^2\right] = \frac{1}{x^2} \text{Var}(X).$$

□

Example 7.3. Let $X \sim \text{Bin}(1000, 0.01)$. Estimate $\mathbb{P}(X \geq 36)$ using Chebyshev's inequality.

Since $\mathbb{E}[X] = 1000 \times 0.01 = 10$, we have

$$\mathbb{P}(X \geq 36) \leq \mathbb{P}\left(|X - \mathbb{E}[X]| \geq 26\right) \leq \frac{\text{Var}(X)}{26^2} = \frac{9.9}{676} \approx 0.015.$$

This estimate is about twice as good as our previous estimate, when we used Markov's inequality. One might then ask if we can improve even further by looking at higher moments (i.e. using X^k , or $|X - \mathbb{E}[X]|^k$, for $k > 2$), and the answer is yes, in some cases this can yield even better results, and taking exponential moments even more so (this is the basis of the theory of *large deviations*), but this is a story for another course. Instead we apply Chebyshev's inequality to another example.

Example 7.4. Suppose that $X_1, X_2, \dots$ are independent and identically distributed random variables with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2$ for all $i = 1, 2, \dots$. Let

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Then

$$\mathbb{E}[\bar{X}_n] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n} n\mu = \mu,$$

and since the X_i are independent,

$$\text{Var}(\bar{X}_n) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} n\sigma^2 = \frac{\sigma^2}{n}.$$

Thus, for any $\varepsilon > 0$, by Chebyshev's inequality,

$$\mathbb{P}\left(|\bar{X}_n - \mu| > \varepsilon\right) \leq \frac{\text{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sigma^2}{\varepsilon n}$$

which tends to 0 as $n \rightarrow \infty$.

We have just proved the Weak Law of Large Numbers!

7.3 Another formula for calculating the expectation

Sometimes the pmf or pdf is simpler or easier to work with than the cdf; but sometimes the cdf is simpler or easier to work with (or easier to estimate). Thus the following formula for the expectation of a non-negative random variable is sometimes useful.

Theorem 7.2. *If X is a non-negative random variable, then*

$$\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X \geq x) dx.$$

Proof. Idea: Write $X = \int_0^X 1 dx = \int_0^\infty \mathbf{1}_{\{x \leq X\}} dx$ and take expectations.

As the idea suggests, we simply observe that since X is non-negative,

$$X = \int_0^X 1 dx = \int_0^X 1 dx + \int_X^\infty 0 dx = \int_0^\infty \mathbf{1}_{\{x \leq X\}} dx.$$

Taking expectations,

$$\mathbb{E}[X] = \mathbb{E}\left[\int_0^\infty \mathbf{1}_{\{x \leq X\}} dx\right].$$

We now swap the expectation and the integral (we have not shown that this is allowed; it is not always allowed; but it turns out that it is allowed as long as the integrand is non-negative, which is true in our case) to get

$$\mathbb{E}[X] = \int_0^\infty \mathbb{E}[\mathbf{1}_{\{x \leq X\}}] dx = \int_0^\infty \mathbb{P}(x \leq X) dx,$$

which completes the proof. $\square$

This formula can in fact be extended to give a formula for random variables that aren't necessarily non-negative, by writing $X = X\mathbf{1}_{\{X \geq 0\}} + X\mathbf{1}_{\{X < 0\}}$, but we leave this as an exercise.

Example 7.5. Let $X \sim \text{Exp}(\lambda)$ for $\lambda > 0$. Calculate $\mathbb{E}[X]$.

Using Theorem 7.2,

$$\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X \geq x) dx = \int_0^\infty e^{-\lambda x} dx = \frac{1}{\lambda}.$$

This is significantly easier than using the definition of expectation, for which we have to integrate $xe^{-\lambda x}$.

The formula in 7.2 works for any random variable, but if our random variable is discrete, then calculating the integral is actually equivalent to calculating a sum, as we show in the following corollary.

Corollary 7.2. *If X is a discrete random variable with support $S_X = \{0, 1, 2, \dots\}$, then*

$$\mathbb{E}[X] = \sum_{j=1}^\infty \mathbb{P}(X \geq j).$$

Proof. Two options: we can apply Theorem 7.2, noting that $\mathbb{P}(X \geq x) = \mathbb{P}(X \geq \lceil x \rceil)$. Or we can mimic the proof of Theorem 7.2 in the discrete setting:

$$X = \sum_{j=1}^X 1 = \sum_{j=1}^\infty \mathbf{1}_{\{j \leq X\}}$$

so taking expectations and swapping the sum and the limit (again without worrying about whether we are actually allowed to do this),

$$\mathbb{E}[X] = \mathbb{E}\left[\sum_{j=1}^\infty \mathbf{1}_{\{j \leq X\}}\right] = \sum_{j=1}^\infty \mathbb{E}[\mathbf{1}_{\{j \leq X\}}] = \sum_{j=1}^\infty \mathbb{P}(j \leq X).$$

$\square$

Example 7.6. Let $X \sim \text{Geom}(p)$ for $p \in (0, 1)$. Calculate $\mathbb{E}[X]$.

Using Corollary 7.2,

$$\mathbb{E}[X] = \sum_{j=1}^{\infty} \mathbb{P}(X \geq j) = \sum_{j=1}^{\infty} (1-p)^{j-1} = \sum_{i=0}^{\infty} (1-p)^i = \frac{1}{p}.$$

This is significantly easier than using the definition of expectation, for which we need to sum terms involving $j(1-p)^{j-1}$.

7.4 The expected return time to 0 for simple symmetric random walk

This section is more difficult, and you may find it easier to skip some of the details, but it gives us a nice final application of our tools to the random walk. As with all of Section 7, it is non-examinable.

Recall our key example, a random walk $S_n = \sum_{i=1}^n X_i$ where each “step” X_i equals +1 with probability 1/2 and -1 with probability 1/2. That is, S_n starts from 0 and goes up with probability 1/2 and down with probability 1/2 at each step.

In fact, we can start S_n from any integer $k \in \mathbb{Z}$ by setting $S_n = k + \sum_{i=1}^n X_i$. Usually we will start from 0 but occasionally it is useful to start from somewhere else. We write $\mathbb{P}_k$ to signify that we are starting from k . (When we start from 0, we will sometimes just write $\mathbb{P}$ and sometimes write $\mathbb{P}_0$.)

Let $T = \min\{n \geq 1 : S_n = 0\}$, the first time at which S_n returns to 0. We showed in an earlier lecture (with a little bit of hand-waving) that S_n hits 0 infinitely many times. So $\mathbb{P}(T < \infty) = 1$.

What is $\mathbb{E}[T]$? We use the formula from Corollary 7.2. This tells us that

$$\mathbb{E}[T] = \sum_{j=1}^{\infty} \mathbb{P}(T \geq j).$$

Note that we always have $T \geq 2$, since our random walk cannot return to 0 before time 2, so the first two terms in our sum are both equal to 1. Also, T cannot be an odd number, since the random walk cannot hit 0 at odd times, so $\mathbb{P}(T \geq 2j-1) = \mathbb{P}(T \geq 2j)$ for each $j = 1, 2, \dots$. Thus

$$\mathbb{E}[T] = 2 + 2 \sum_{j=2}^{\infty} \mathbb{P}(T \geq 2j).$$

Now, at time 1, we must either step up or down, and the distribution of T does not depend on which choice we make, so without loss of generality suppose that we step up. The event that $T \geq 2j$ when our random walk starts from 0 is then equal to the event that $T \geq 2j-1$ when our random walk starts from 1. That is,

$$\begin{aligned} \mathbb{P}(T \geq 2j) &= \mathbb{P}_1(\text{random walk doesn't hit 0 by time } 2j-2) \\ &= \mathbb{P}_1(S_i > 0 \text{ for all } i \leq 2j-2) \\ &= \mathbb{P}_1(S_i > 0 \text{ for all } i \leq 2j-2, S_{2j-2} \geq 1). \end{aligned}$$

(For the last line above, we can add in the condition $S_{j-2} \geq 1$ for free, since if $S_i > 0$ for all $i \leq 2j-2$, then the random walk certainly has to be ≥ 1 at time $2j-2$.) We now look at the complementary event of the random walk staying positive; if it does *not* stay positive up to time $2j-2$, then there must be some time at which it hits 0. Thus

$$\mathbb{P}(T \geq 2j) = \mathbb{P}_1(S_{2j-2} \geq 1) - \mathbb{P}_1(S_{2j-2} \geq 1, \text{ there exists } i < 2j-2 : S_i = 0).$$

Now the *reflection principle* (we may have time to discuss this more in lectures) says that asking the random walk (starting from 1) to go down to zero and back up to 1 is the same as asking it to move at least distance 2 (it doesn't care whether it has to go down at least 1 and back up at least 1, or up at least 2, or down at least 2). That is,

$$\mathbb{P}_1(S_{2j-2} \geq 1, \text{ there exists } i < 2j-2 : S_i = 0) = \mathbb{P}_1(S_{2j-2} \geq 3).$$

So

$$\begin{aligned}
 \mathbb{P}(T \geq 2j) &= \mathbb{P}_1(S_{2j-2} \geq 1) - \mathbb{P}_1(S_{2j-2} \geq 3) && \text{[by the argument above]} \\
 &= \mathbb{P}_1(S_{2j-2} \in \{1, 2\}) && \text{[containment]} \\
 &= \mathbb{P}_0(S_{2j-2} \in \{0, 1\}) && \text{[shift everything down by 1]} \\
 &= \mathbb{P}_0(S_{2j-2} = 0). && \text{[can only be at even positions at even times]}
 \end{aligned}$$

(This in itself is a remarkable formula: the probability that we don't hit zero until time $2j$ is the same as the probability that we hit zero at time $2j - 2$!)

Returning to our formula for $\mathbb{E}[T]$, we see that

$$\mathbb{E}[T] = 2 + 2 \sum_{j=2}^{\infty} \mathbb{P}_0(S_{2j-2} = 0) = 2 + 2 \sum_{n=0}^{\infty} \mathbb{P}_0(S_{2n} = 0).$$

We already calculated and estimated $\mathbb{P}_0(S_{2n} = 0)$ in an earlier lecture: to be at 0 after $2n$ steps we have to take exactly n steps up and n steps down, which has probability

$$\mathbb{P}_0(S_{2n} = 0) = \binom{2n}{n} \frac{1}{2^{2n}}.$$

We then estimated this using *Stirling's formula* (which is not part of the course) to show that

$$\mathbb{P}_0(S_{2n} = 0) \approx \frac{1}{\sqrt{\pi n}}.$$

Thus

$$\mathbb{E}[T] \approx 2 + 2 \sum_{n=0}^{\infty} \frac{1}{\sqrt{\pi n}} \dots$$

but this sum does not converge, so $\mathbb{E}[T] = \infty$.

So the random walk always returns to 0, but you might be waiting a long time to see it!